

**UNIVERSITY OF THE PUNJAB**

**NOTIFICATION**

It is hereby notified that the Syndicate at its meeting held on 17-12-2022 has approved the recommendations of the Academic Council made at its meetings held on 11-03-2022 & 21-03-2022, respectively, regarding approval of revised curriculum of BS Economics (4-Years) Program at the Department of Economics, School of Economics with effect from the Academic Session, 2021 and onward.

The revised curriculum of BS Economics is enclosed herewith as Annexure-'A'.

Admin. Block,  
Quaid-i-Azam Campus,  
Lahore.

Sd/-  
REGISTRAR

No. D/ 1925 - /Acad.,

Dated: 3-3-2023.

Copy of the above is forwarded to the following for information and necessary action:-

1. Dean, Faculty of Business, Economics & Administrative Sciences
2. Head, School of Economics
3. Chairman, Department of Economics
4. Controller of Examinations
5. Director, IT for placement at website
6. Admin. Officer (Statutes)
7. Secretary to the Vice-Chancellor
8. Private Secretary to the Registrar
9. Assistant (Syllabus)

  
Assistant Registrar (Academic)  
for Registrar



## Revised Curriculum BS Economics Program

Department of Economics  
School of Economics  
University of the Punjab, Lahore.

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**Program Title: BS Economics**  
**Department: School of Economics**  
**Faculty: Business, Economics and Administrative Sciences**

## **1. Department Mission**

### **Vision**

The School of Economics is committed to be an institution of excellence, following international standards, producing quality leaders in the area of economics, business and public policy, stirring debate and guiding the policy discourse through advanced cutting-edge research produced by the department.

### **Mission**

Our mission is to:

- Provide a quality and affordable platform of learning and education for students from all over Pakistan, especially Punjab.
- Provide students with an outstanding academic training.
- Striking balance between research, teaching and service to the community.
- Provide teachers with stimulating and conducive environment to produce distinguished research.
- Advancing knowledge in the area of economics, governance and policy making to resolve issues and challenges faced by Pakistan.

## **2. Introduction**

Economics research and education at University of the Punjab has a rich and long history spanning over a century. The Department of Economics has played a vital role in providing valuable knowledge and enriching human resources within this field. The BS Economics program and curriculum is being followed by many affiliated colleges across Punjab.

In 2018, focus was laid on enhancing the study and research of economics at University of the Punjab. After numerous discussions and feedback from all walks of life, the Department of Economics was upgraded to the School of

Economics, comprising of four departments and a research center. The department plans to introduce an additional undergraduate and a graduate degree program, along with certificate courses to enhance teaching of Economics.

### 3. Program Introduction

This program is the flagship program of the School, focusing on the issues primarily in the domain of economics, including but not limited to, development, growth, International trade (Globalization), Labor Markets, Industrial organization, poverty alleviation, etc. BS Economic program is well taken in the market, and our first batch of this program just graduated in June 2021.

### 4. Program Objectives

Graduates from this program shall be able to:

1. Understand various aspects of domestic and global economy as well as business world.
2. Analyze problems and apply critical thinking to finding solutions.
3. Think logically, critically, and creatively.
4. Apply algebraic, graphical, and statistical tools for problem solving
5. Present ideas in any medium succinctly, with clarity and conviction.
6. Communicate with clarity verbally (in English or Urdu) as well as non-verbally

### 5. Market Need / Rationale of the Program

While there is always further room of improvement for which the whole exercise of revising the program was undertaken, it also incorporated the feedback received from the affiliated colleges.

Program need assessment may include feedback from multiple sources such as:

- a) **Potential Students for the program.** (career needs, subject interest etc.): those interested in economics, development, economic policy and finance.
- b) **Potential Employers** (Public, private, NGOs, required skill set, industry projections, employment opportunities/estimated market

demand/Number of job openings, Current and future prospects): Academia, Research Centers, Multi-national, Commercial Banks.

- c) **Academic Projections** (The national/ international universities that have launched the similar program): BS Program is a staple program of all the universities offering Social Science Schools, such as LUMS, IBA and NUST.
- d) **Faculty** (Faculty credentials, capacity, resources sufficiency etc.): Faculty of School of Economics will be employed, which has 11 permanent faculty members, expected to grow shortly as hiring process is underway.
- e) **Physical Facilities** (Lab and library facilities etc.): While the Department of Economics has its own Computer Lab and Library, Separate office block for Department of Economics is under construction at the new School of Economics building. Lab and Library facilities will be in common use at School of Economics, which has a library with over 10,000 books and a computer lab (currently with 30 computers due to space limitation, expected to be of over 150 computers in the new premises)

## 6. Admission Eligibility Criteria

- F.A/F.Sc./A-Levels or equivalent
- Basic Admission Criteria
- Additional Marks
  - Hafiz-e-Quran 20 Marks
  - Elective Subjects Marks
  - Economics 10 Marks
  - Statistics 05 Marks
  - Mathematics 05 Marks

## 7. Duration of the Program

8 (Eight) Semesters/ 04 (Four) Years/ 137 Credit hours



## 8. Categorization of Courses as per HEC Recommendation and Difference

| Semester              | Courses | Category(Credit Hours) |                         |                    |                        |           |               |
|-----------------------|---------|------------------------|-------------------------|--------------------|------------------------|-----------|---------------|
|                       |         | Core Courses           | Diversification Courses | Economic Electives | Non-Economics Optional | Any Other | Semester Load |
| 1                     | 7       | 11                     | 6                       | 0                  | -                      | -         | 17            |
| 2                     | 7       | 12                     | 6                       | 0                  | -                      | -         | 18            |
| 3                     | 7       | 9                      | 9                       | 0                  | -                      | -         | 18            |
| 4                     | 7       | 13                     | 6                       | 0                  | -                      | -         | 19            |
| 5                     | 7       | 18                     | 0                       | 0                  | -                      | -         | 18            |
| 6                     | 6       | 10                     | 0                       | 6                  | -                      | -         | 16            |
| 7                     | 6       | 9                      | 0                       | 6                  | -                      | -         | 15            |
| 8                     | 6       | 10                     | 0                       | 6                  | -                      | -         | 16            |
| PU                    | 53      | 92                     | 27                      | 18                 | -                      | -         | 137           |
| HEC Guidelines        | 52      | 89                     | 27                      | 18                 | -                      | -         | 134           |
| Difference (HEC &) PU | 1       | 3                      | 0                       | 0                  | -                      | -         | 3             |

## 9. Scheme of Studies / Semester-wise workload

| SR                 | Code      | Course Title                              | Course Type     | Prerequisite          | Credit hours |
|--------------------|-----------|-------------------------------------------|-----------------|-----------------------|--------------|
| <b>Semester I</b>  |           |                                           |                 |                       |              |
| 1.                 | ISE-111   | Islamiyat / Ethics                        | Core            |                       | 2            |
| 2.                 | ENG-111   | English-I (Language in Use)               | Core            |                       | 3            |
| 3.                 | COMP-111  | Computer (Introduction & Applications)    | Diversification |                       | 3            |
| 4.                 | STAT-101A | Elementary Statistics                     | Diversification |                       | 3            |
| 5.                 | ECON-101  | Principles of Microeconomics              | Core            |                       | 3            |
| 6.                 | MATH-111  | Elementary Mathematics-I (Algebra)        | Core            |                       | 3            |
| 7.                 | HQ-001    | Tadrees-e-Tarjuma Quran                   | Core            |                       | 0            |
| <b>Semester II</b> |           |                                           |                 |                       |              |
| 1.                 | PST-111   | Pakistan Studies                          | Core            |                       | 2            |
| 2.                 | ENG-112   | English-II (Academic Reading & Writing)   | Core            |                       | 3            |
| 3.                 | STAT-102  | Probability and Probability Distributions | Core            | Elementary Statistics | 3            |



| SR                  | Code     | Course Title                            | Course Type     | Prerequisite                                                  | Credit hours |
|---------------------|----------|-----------------------------------------|-----------------|---------------------------------------------------------------|--------------|
| 4.                  | ECON-102 | Principles of Macroeconomics            | Core            |                                                               | 3            |
| 5.                  | CAL-111  | Calculus-I                              | Diversification | Elementary Mathematics I (Algebra)                            | 3            |
| 6.                  | HQ-002   | Tadrees-e-Tarjuma Quran                 | Core            |                                                               | 1            |
| 7.                  |          | Optional-I (Non Economics)              | Diversification |                                                               | 3            |
| <b>Semester III</b> |          |                                         |                 |                                                               |              |
| 1.                  | ENG-211  | English-III (Communication Skills)      | Core            |                                                               | 3            |
| 2.                  | CAL-211  | Calculus-II                             | Core            | Calculus-I                                                    | 3            |
| 3.                  | ECON-201 | Intermediate Microeconomics             | Core            | Principles of Microeconomics, Calculus I                      | 3            |
| 4.                  | HQ-003   | Tadrees-e-Tarjuma Quran                 | Core            |                                                               | 0            |
| 5.                  |          | Optional-II (Non Economics)             | Diversification |                                                               | 3            |
| 6.                  |          | Optional-III (Non Economics)            | Diversification |                                                               | 3            |
| 7.                  |          | Optional-IV (Non Economics)             | Diversification |                                                               | 3            |
| <b>Semester IV</b>  |          |                                         |                 |                                                               |              |
| 1.                  | ENG-212  | English-IV (English for Practical Aims) | Core            |                                                               | 3            |
| 2.                  | ECON-202 | Intermediate Macroeconomics             | Core            | Principles of Macroeconomics                                  | 3            |
| 3.                  | ECON-203 | Research Methods-I                      | Core            | Elementary Statistics                                         | 3            |
| 4.                  | ECON-204 | Islamic Economics                       | Core            | Principles of Microeconomics and Principles of Macroeconomics | 3            |
| 5.                  | HQ-004   | Tadrees-e-Tarjuma Quran                 | Core            |                                                               | 1            |
| 6.                  |          | Optional-V (Non Economics)              | Diversification |                                                               | 3            |
| 7.                  |          | Optional-VI (Non Economics)             | Diversification |                                                               | 3            |
| <b>Semester V</b>   |          |                                         |                 |                                                               |              |
| 1.                  | ECON-301 | Advanced Microeconomics                 | Core            | Intermediate Microeconomics                                   | 3            |
| 2.                  | ECON-303 | Research Methods-II                     | Core            | Research Methods-I                                            | 3            |
| 3.                  | ECON-304 | History of Economic Thought             | Core            | Intermediate Microeconomics and Intermediate Macroeconomics   | 3            |

| SR                   | Code     | Course Title                     | Course Type | Prerequisite                                                     | Credit hours |
|----------------------|----------|----------------------------------|-------------|------------------------------------------------------------------|--------------|
| 4.                   | ECON-305 | Econometrics-I                   | Core        | Elementary Statistics, Probability and Probability Distributions | 3            |
| 5.                   | ECON-306 | Development Economics-I          | Core        | Intermediate Microeconomics and Intermediate Macroeconomics      | 3            |
| 6.                   | ECON-307 | Advanced Mathematical Economics  | Core        | Calculus -I & Calculus -II                                       | 3            |
| 7.                   | HQ-005   | Tadrees-e-Tarjuma Quran          | Core        |                                                                  | 0            |
| <b>Semester VI</b>   |          |                                  |             |                                                                  |              |
| 1.                   | ECON-302 | Advanced Macroeconomics          | Core        | Intermediate Macroeconomics                                      | 3            |
| 2.                   | ECON-308 | Monetary Economics               | Core        | Intermediate Macroeconomics                                      | 3            |
| 3.                   | ECON-405 | Econometrics-II                  | Core        | Econometrics-I                                                   | 3            |
| 4.                   | HQ-006   | Tadrees-e-Tarjuma Quran          | Core        |                                                                  | 1            |
| 5.                   |          | Elective-I                       | Elective    |                                                                  | 3            |
| 6.                   |          | Elective-II                      | Elective    |                                                                  | 3            |
| <b>Semester VII</b>  |          |                                  |             |                                                                  |              |
| 1.                   | ECON-401 | Public Sector Economics          | Core        | Intermediate Microeconomics and Intermediate Macroeconomics      | 3            |
| 2.                   | ECON-407 | Major Issues in Pakistan Economy | Core        | Intermediate Microeconomics and Intermediate Macroeconomics      | 3            |
| 3.                   | ECON-406 | Development Economics-II         | Core        | Development Economics-I                                          | 3            |
| 4.                   | HQ-007   | Tadrees-e-Tarjuma Quran          | Core        |                                                                  | 0            |
| 5.                   |          | Elective-III                     | Elective    |                                                                  | 3            |
| 6.                   |          | Elective-IV                      | Elective    |                                                                  | 3            |
| <b>Semester VIII</b> |          |                                  |             |                                                                  |              |
| 1.                   | ECON-402 | Growth Theories and Models       | Core        | Advanced Macroeconomics, Development Economics-II                | 3            |
| 2.                   | ECON-403 | Applied Economics                | Core        | Advanced Microeconomics, Advanced Macroeconomics,                | 3            |

| SR | Code     | Course Title            | Course Type | Prerequisite                                                | Credit hours |
|----|----------|-------------------------|-------------|-------------------------------------------------------------|--------------|
|    |          |                         |             | Mathematical Economics-I and Econometrics-II                |              |
| 3. | ECON-404 | International Trade     | Core        | Intermediate Microeconomics and Intermediate Macroeconomics | 3            |
| 4. | HQ-008   | Tadrees-e-Tarjuma Quran | Core        |                                                             | 1            |
| 5. |          | Elective-V              | Elective    |                                                             | 3            |
| 6. |          | Elective-VI             | Elective    |                                                             | 3            |

### Research Thesis

- Department has an option to offer the research thesis worth 06 (six) credit hours in lieu of two elective courses.
- The thesis should span over two semesters 7<sup>th</sup> and 8<sup>th</sup> Semester in lieu of one elective course in each of the two semesters.
- Eligibility Criteria:
  - Minimum of 3.0 CGPA
  - One faculty member should be willing to supervise the thesis of the student.

### List of Optional Courses: (Non-Economics)

| Sr. No. | Course Code | Course Name                             |
|---------|-------------|-----------------------------------------|
| 1       | SOC-211     | Introduction to Sociology               |
| 2       | EDU-111     | Introduction to Education               |
| 3       | APSY-111A   | Introduction to Psychology              |
| 4       | HIS-111     | Introduction to History                 |
| 5       | GEOG-211    | Introduction to Geography               |
| 6       | PHIL-211A   | Introduction to Philosophy              |
| 7       | URDU-111    | Functional Urdu                         |
| 8       | BLAW-111    | Introduction to Business Law            |
| 9       | BUS-111     | Introduction to Business Administration |
| 10      | ACC-111     | Principles of Accounting                |
| 11      | POL-111     | Introduction to Political Science       |
| 12      | IR-111      | International Relations                 |
| 13      | BSCS-111    | Mass Communication                      |

| 14 | Foreign Language |                         |
|----|------------------|-------------------------|
| 1  | ARB-111          | Arabic Language - I     |
| 2  | FRN-111          | French Language         |
| 3  | SPN-111          | Introduction to Spanish |
| 4  | CHI-111          | Introduction to Chinese |
| 5  | GER-111          | Introduction to German  |
| 6  | ITL-111          | Introduction to Italian |
| 7  | TUR-111          | Introduction to Turkish |
| 8  | PER-111          | Introduction to Persian |

- For all the common courses at the university level, the department will be following the course codes and the syllabus provided by the university.



List of Elective Courses:

| Sr. No. | Course Code | Course Name                                                 |
|---------|-------------|-------------------------------------------------------------|
| 1       | ECON-309    | Labor Economics                                             |
| 2       | ECON-310    | Urban Economics                                             |
| 3       | ECON-311    | Comparative Economic System                                 |
| 4       | ECON-312    | Managerial Economics                                        |
| 5       | ECON-313    | Introduction to Political Economy                           |
| 6       | ECON-314    | Agricultural Economics                                      |
| 7       | ECON-315    | Human Resource Development                                  |
| 8       | ECON-316    | Money, Banking and Financial Institutions                   |
| 9       | ECON-317    | Economic Principles, Strategies and Techniques of Marketing |
| 10      | ECON-318    | Population Economics                                        |
| 11      | ECON-408    | Institutional Economics                                     |
| 12      | ECON-409    | Health Economics                                            |
| 13      | ECON-410    | Project Appraisal and Investment Analysis                   |
| 14      | ECON-411    | Environmental and Natural Resource Economics                |
| 15      | ECON-412    | Economic Dynamics                                           |
| 16      | ECON-413    | International Finance                                       |
| 17      | ECON-414    | Development Policy                                          |
| 18      | ECON-415    | Experimental and Behavioral Economics                       |
| 19      | ECON-416    | Gender Analysis in Economics                                |
| 20      | ECON-417    | Poverty and Income Distribution                             |
| 21      | ECON-418    | Energy Economics                                            |
| 22      | ECON-419    | Islamic Banking and Finance                                 |
| 23      | ECON-420    | Impact Evaluation Methods                                   |
| 24      | ECON-421    | World Economic History                                      |
| 25      | ECON-422    | General Equilibrium and Welfare Economics                   |
| 26      | ECON-423    | Regional Economics                                          |

|    |          |                                                                                                                |
|----|----------|----------------------------------------------------------------------------------------------------------------|
| 27 | ECON-424 | Thesis (In lieu of two elective courses – one in 7 <sup>th</sup> semester and one in 8 <sup>th</sup> semester) |
|----|----------|----------------------------------------------------------------------------------------------------------------|

#### 10. Award of Degree

Degree awarding criteria stating:

Minimum CGPA of 2.00 is required to Qualify for degree

#### 11. Faculty Strength

| Category                         | Regular Faculty |                | SBP Chair    |                |
|----------------------------------|-----------------|----------------|--------------|----------------|
|                                  | No. of Males    | No. of Females | No. of Males | No. of Females |
| <b>Professor (PhD)</b>           | 01              | -              | 1            | -              |
| <b>Associate Professor (PhD)</b> | 02              | -              | -            | -              |
| <b>Assistant Professor (PhD)</b> | 02              | 04             | -            | -              |
| <b>Lecturer (Mphil)</b>          | 01              | 01             | -            | -              |

#### 12. Present Student Teacher Ratio in the Department

The School of Economics has 11 regular teaching staff as reported above and has total student body of 629. Thereby, student - teacher ratio is 57 to 1.

##### Student - Teacher Ratio

| Total No. of Students<br>(New Intake + Existing) | Total No. of Teachers<br>(Regular) |
|--------------------------------------------------|------------------------------------|
| 629                                              | 11                                 |
| It means Student – teacher ratio is 57:1         |                                    |

#### 13. Course Outlines: Course outlines separately for each course has been attached in the next section.

#### 14. Assessment Criterion: The assessment is common for all the courses and the standard criteria is followed:

| Sr. No. | Elements             | Weightage | Details                                                                                                                                                                                                                                           |
|---------|----------------------|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1.      | Midterm Assessment   | 35%       | It takes place at the mid-point of the semester.                                                                                                                                                                                                  |
| 2.      | Formative Assessment | 25%       | It is continuous assessment. It includes: classroom participation, attendance, assignments and presentations, homework, attitude and behavior, hands-on-activities, short tests, quizzes etc.                                                     |
| 3.      | Final Assessment     | 40%       | It takes place at the end of the semester. It is mostly in the form of a test, but owing to the nature of the course the teacher may assess their students based on term paper, research proposal development, field work and report writing etc. |





## Economics Courses

**Course Code: ECON-101**

**Title: Principles of Microeconomics**

**Credit Hours: 03**

### Course Objectives:

This course is meant to be an introduction to the decision-making process of households, firms, and government, and the resulting allocation of resources through markets. The course examines how individuals and firms make decisions by weighing up costs and benefits, and how the interaction of their decisions leads to market and social outcomes. The objective of the course is to provide students with a clear understanding of the subject of economics as it pertains to the behavior of consumers, firms, and markets.

### Learning Outcomes:

This course will help the students to relate the various concepts in economic theory. The main areas under this course are law of Demand and Supply, their elasticity, theory of consumer behavior and production.

### Course Contents:

|                                          |                                                                                                                                                                                                                                                                                                    |
|------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Preliminaries</b>                     | The Themes of Microeconomics, Scarcity and Economics, what is a Market? Real versus Nominal Prices, Why Study Microeconomics?                                                                                                                                                                      |
| <b>Consumer Choice</b>                   | Cardinal Approach/Utility Analysis: Marginal Utility, Law of Diminishing Marginal Utility, Law of Equi-Marginal Utility, consumer equilibrium. Ordinal Approach of Consumer Behavior: Indifference Curves, Features of Indifference Curves, Budget Line, Consumer Equilibrium. Comparative Statics |
| <b>The Basics of Demand &amp; Supply</b> | The Market Mechanism: Demand and Supply, factors effecting the market forces, Changes in Market Equilibrium.                                                                                                                                                                                       |
| <b>Elasticities</b>                      | Price Elasticity of Demand, Price Elasticity of Supply, Point versus Arc Elasticities, Computing Elasticities, Price Elasticity and Total Expenditure, Cross-Price Elasticity of Demand, Income Elasticity of Demand.                                                                              |
| <b>Price Changes and Welfare</b>         | Consumer Surplus and producer surplus, price effect, Income effect, and Substitution Effect                                                                                                                                                                                                        |
| <b>Technology and Production</b>         | The Technology of Production, Production with one Variable Input (Labour), Production with two Variables, Inputs, Returns to Scale                                                                                                                                                                 |
| <b>Cost and Revenue</b>                  | Measuring Cost: Which Costs Matter? Costs in the Short Run, Costs in the Long Run, Long Run versus Short Run Cost Curves. Average, total, marginal cost curves.                                                                                                                                    |

|                          |                                                                                                                                                                                                                                                                                          |
|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                          | Measuring revenue: total, average, and marginal revenue.                                                                                                                                                                                                                                 |
| <b>Market Structures</b> | Introduction to markets – Perfect competition, monopoly, monopolistic competition, oligopoly. Difference in revenue between perfect competition and other market types. Profit Maximization, derivation of supply curve under perfect competition.<br><br>Introduction to factor markets |

### Teaching Methodology:

- To deliver lectures on topics included in course outline
- To require each student to solve independent assignments on topics included in the course.

### Evaluation Criteria:

| Evaluation Method   | Total Percentage |
|---------------------|------------------|
| Quizzes/Assignments | 25%              |
| Mid-Term Exam       | 35%              |
| Final-Term Exam     | 40%              |

### Recommended Books:

- Gans, J.S., King, S.P. & Mankiw, N.G. (2012). *Principles of Microeconomics* (5<sup>th</sup> ed.), Cengage Learning.
- Colander, Principles of Microeconomics
- Bernanke, Principles of Microeconomics

### Supplementary Material:

- McConnell, C., & Bruce, S. (2006). *Principles of Economics* (17<sup>th</sup> ed.). McGraw-Hill.
- Fuller, N. (1997). *Principles of Microeconomics*. UK: Tudor Business Publisher.

**Course Code: ECON-102****Title: Principles of Macroeconomics****Credit Hours: 03****Course Objectives:**

This course is designed for the beginners with no formal background or little acquaintance with economics. The objective is to give the students an idea of the basic concepts, tools of analysis and terminologies used in macroeconomics. Emphasis will be on the use of graphs, diagrams and numerical tables/schedules for exposition.

**Learning Outcomes:**

Upon the completion of the course, students should be able to

- Understand the structure of the economy and have a thorough understanding of the key macroeconomic terminologies
- Apply these macro-economic theories to the real world situations.

**Course Contents:**

|                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Introduction</b>                            | The economy in aggregate, Complexities of the world of business, Scope of macroeconomics, Brief account of the development of macro-economic after the World War-II, Concept of business cycles: Boom and Depression, Concepts of Inflation and Unemployment, Price Indices and Inflation, the simple macroeconomic models, Lead and Lagged variables, Exogenous and Endogenous variables and their functional relationships, Macro-models as abstraction from the real economy.                                                                                                                                                                           |
| <b>Macroeconomics: The Classical Framework</b> | Ingredients of the classical model: The Says Law of Market, The Quantity theory of money, The marginal productivity theory, Equilibrium in the goods and money market, A critical appraisal of classical model, classical dichotomy.                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>National Income</b>                         | Definition and concept of national income, Computation of national income: Product, Income and Expenditure approaches, Circular flow of income, Nominal versus Real income, Per capita income and the standard of living. The circular flow of National Income, GDP and GNP, Measurement of GNP: Expenditure (demand), Product (supply) and Income (factor rewards) approach, Value added at factor cost and market prices, Personal and disposable income, Per capita income, GNP as measure of welfare, Injections into and Leakages from the economy: Saving and Investment, Exports and Imports, Domestic Absorption, Nominal and Real income, The GDP |

|                                    |                                                                                                                                                                                                                                                                                                                                                                                                                             |
|------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                    | deflator, Problems in computation of national income, The underground economy.                                                                                                                                                                                                                                                                                                                                              |
| <b>Money and Banking</b>           | Money: definition, forms and functions, The demand for money, The Liquidity Preference/ Portfolio Balance Approach, the supply of money: M1, M2, M3 concepts, the money supply multiplier and the role of banking system, Instruments of credit control and the role of central bank, Commercial banking, Monetary policy: brief introduction. Central Bank and its functions with reference to the State Bank of Pakistan. |
| <b>Public Finance and Taxation</b> | Sources of public revenue, Various forms of taxes: Direct and Indirect, Income and Commodity taxes, Sales, Excise, Customs, Non-tax revenues, Major heads of public expenditure, Revenue and Capital account, Fiscal policy: meaning and objectives.                                                                                                                                                                        |
| <b>International Trade</b>         | Global distribution of resources, Concept of imports and exports, Currency exchange rates, Balance of Payments: causes of deficits in BOP of Pakistan and remedial measures,                                                                                                                                                                                                                                                |

#### Teaching Methodology:

- To deliver lectures on topics included in course outline
- To require each student to solve independent assignments on topics included in the course.

#### Evaluation Criteria:

| Evaluation Method   | Total Percentage |
|---------------------|------------------|
| Quizzes/Assignments | 25%              |
| Mid-Term Exam       | 35%              |
| Final-Term Exam     | 40%              |

#### Recommended Books:

- Mankiw, G–Principles of Economics- Latest Edition- South- West Publishers.
- McConnell, Campbell & Stanley Brue, and Sean Flynn, Macroeconomics, Latest Edition. (McGraw-Hill Economics)
- Samulson and Nordhaus - Economics –Latest Edition - McGraw Hill. Inc.
- Parkin, Michael - Macroeconomics, Latest Edition - Prentice Hall.

## Course Code: ECON-201

**Title: Intermediate Microeconomics**

**Credit Hours: 03**

**Prerequisite: Principles of Microeconomics, Calculus I**

### Course Objectives:

The course seeks to provide an understanding of the core economic models for the analysis of decisions by consumers, firms, and markets. It would focus on illustrating the functionality of certain core microeconomic models. The objective of this course is to enable students to develop ability to use analytic reasoning to understand and apply basic models of microeconomic theory. Simple Calculus and Graphical methods will be used extensively.

### Learning Outcomes:

On completion of this course, students would be able to,

- Have complete knowledge of the different types of market structures
- Understand why some producers reap abnormal profits while other are driven out of the business.
- Have an insight of the different business strategies employed by the firms.

### Course Contents:

|                                                   |                                                                                                                                                                                                                       |
|---------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Consumer Theory</b>                            | Consumer Preferences, Well-Behaved Preferences, Indifference Curves, Budget Constraint. Utility Functions, Utility Maximization, Choice, Deriving the Demand Curve, Consumer Surplus. The idea of revealed preference |
| <b>Goods classifications</b>                      | Elasticity computation. Normal and Inferior Goods, and Giffen Goods, Substitutes and Complements, The Price and income Offer Curves, Engel curve, The Inverse Demand Function                                         |
| <b>Price Effects</b>                              | Substitution Effect, Income effect, Sign of Substitution Effect, Total Change in Demand, Rates of Change, Law of demand, Compensating Demand Curves.                                                                  |
| <b>Technology</b>                                 | Inputs and Outputs, Describing Technological Constraints, Examples of Technology, Properties of Technology, Long Run and Short Run                                                                                    |
| <b>Cost and Revenue</b>                           | Cost Minimization, returns to Scale and the Cost Function, Long Run and Short Run Costs, Fixed and Quasi-Fixed Costs, Sunk Costs.                                                                                     |
| <b>Profit under perfect and imperfect markets</b> | Profits: Short run and long run profit maximization. Comparative Statistic, Profit Maximization and Returns to Scale (with respect to production technology and profits)                                              |

|                          |                                                                                                                                                                                       |
|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Monopoly</b>          | Maximizing Profits, Linear Demand Curve and Monopoly, Markup Pricing, Inefficiency of Monopoly, Deadweight Loss of Monopoly, what causes Monopolies?                                  |
| <b>Monopoly Behavior</b> | Price Discrimination, First-Degree Price Discrimination, Second-Degree Price Discrimination, Third-Degree Price Discrimination, Bundling, Two-Part Tariffs, Monopolistic Competition. |
| <b>Factors Market</b>    | Introduction to the market of factors of production, demand and supply curves, derivation of derived demand curve, market equilibrium, monopsony                                      |

### Teaching Methodology:

- To deliver lectures on topics included in course outline
- To require each student to solve independent assignments on topics included in the course.

### Evaluation Criteria:

| <b>Evaluation Method</b> | <b>Total Percentage</b> |
|--------------------------|-------------------------|
| Quizzes/Assignments      | 25%                     |
| Mid-Term Exam            | 35%                     |
| Final-Term Exam          | 40%                     |

### Recommended Books:

- Varian, H. R. (2009). Intermediate Microeconomics: A Modern Approach (8th ed.). W.W. Norton & Company.
- Nicholson, W. (2000). Intermediate microeconomics and its applications. Dryden Press: New York.
- Maddala, G. S. (2004), Microeconomics: theory & applications. Tata McGraw-Hill Education: New York.

### Course Code: ECON-202

**Title: Intermediate Macroeconomics**

**Credit Hours: 03**

**Prerequisite: Principles of Macroeconomics**

### Course Objectives:

This course is built on the macroeconomic concepts developed in the course on Principles of Macroeconomics. It will enable students to use various macroeconomic concepts to

study the dynamics of an economy at a more advanced level. Students will be able to think critically and to formulate their own perspective on various macroeconomic issues

### Learning Outcomes:

This course would help the students to:

- develop a consistent and logical method of identifying economic variables.
- comprehend the concepts related to macro variables and their relationship with each other.
- to resolve macroeconomic problems them in both a closed and open economy through alternative policies.

### Course Contents:

|                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Introduction</b>                       | The Keynesian revolution and growth of Macroeconomics, Actual vs. Potential Output, Analytical approach to macroeconomics, Okun's Law and unemployment, Price Indices and Inflation, the simple macroeconomic models, Lead and Lagged variables, Exogenous and Endogenous variables and their functional relationships, Introduction to the Classical and Keynesian Schools of Thought.                                                                                                                          |
| <b>Theories of Consumption</b>            | Consumption as a function of income, Theories of Consumption: The Absolute income, Relative income, Permanent income and Life-cycle income hypotheses, Kuznet's findings and Reconciliation of Marginal and Average Propensity to Consume, Random Walk hypothesis.                                                                                                                                                                                                                                               |
| <b>Saving and Investment</b>              | Definition of Investment, Real and Financial investment, Autonomous and Induced investment, Determinants of investment: The Present Value, IRR and benefit-cost ratio Criteria, Investment Demand and capital formation, Tobin's Q-Theory, Lags in Investment Demand, Volatility of Investment, Investment by the Public sector, Equilibrium in the goods market, Derivation of IS curve (graphic), Factors affecting the position and slope of the IS curve, The Saving-Investment gap in developing countries. |
| <b>The Demand for and Supply of Money</b> | The demand for money, The Quantity Theory of money, The Liquidity Preference/ Portfolio Balance Approach, the supply of money: M1, M2, M3 concepts, the money supply multiplier and the role of banking system, Instruments of credit control and the role of central bank, Equilibrium of the money market, Derivation of LM Curve (graphic), Factors affecting the position and slope of the LM curve.                                                                                                         |
| <b>Determination of National Income</b>   | The Keynesian model for a simple two-sector economy, inflationary and deflationary gaps, the concepts of multiplier, Interaction of the IS and LM functions and derivation of the                                                                                                                                                                                                                                                                                                                                |



|                                               |                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                               | aggregate demand function, The impact of changes in public expenditure, taxation and money supply on aggregate demand and the rate of interest.                                                                                                                                                                                                                                                                               |
| <b>Aggregate Supply and the Labour Market</b> | The aggregate production function, Demand for Labor in the short run, Labor market and its Equilibrium: The flexible and sticky wage models, the imperfect information model, the short-run aggregate supply function, The concepts of full-employment and underemployment, Supply shocks and stabilization, Interaction of aggregate demand and supply and price determination.                                              |
| <b>Inflation and Unemployment</b>             | Types, causes and measurement of inflation, the structural and monetary approaches to inflation, Global inflation and its impacts on developing economies, Transitory and structural unemployment, the problem of unemployment with special reference to Pakistan, the relationship between inflation and unemployment, Phillips curve in the short-run and long-run, Remedial measures to combat inflation and unemployment. |

#### Teaching Methodology:

- To deliver lectures on topics included in course outline
- To require each student to solve independent assignments on topics included in the course.

#### Evaluation Criteria:

| Evaluation Method   | Total Percentage |
|---------------------|------------------|
| Quizzes/Assignments | 25%              |
| Mid-Term Exam       | 35%              |
| Final-Term Exam     | 40%              |

#### Recommended Books:

- Shapiro, Edward – Macroeconomic Analysis- Latest Edition–Harcourt Brace Inc.
- Froyen, Richard–Macroeconomics: Theories and Policies–Latest Edition–Macmillan
- Mankiw, Gregory N.- Macroeconomics –Latest Edition, Worth Publishers, NY.
- Blanchard, Oliver – Macroeconomics –Latest Edition - Prentice Hall International.
- Dornbusch & Fisher – Macroeconomics- Latest Edition- McGraw Hill Inc.



**Course Code: ECON-203****Title: Research Methods- I****Credit Hours: 03****Prerequisite: Elementary Statistics****Course Objectives:**

The primary objective of this course is to develop research orientation among the students and to acquaint them with fundamentals of research methods. Specifically, research methods I aims at introducing the basic concept of research methodology, hallmarks of scientific research, hypothetico-deductive and inductive method of research, review of literature, bibliography manager software, components of theoretical framework, hypothesis formulation and elements of research design.

**Learning Outcomes:**

In the course the students should be able to:

- Define a research question
- Write a research proposal
- Perform literature search
- Review, summarize and synthesize paper found in search
- Present the information in a variety of formats to a variety of audiences.

**Course Contents:**

|                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Introduction to Research</b> | Meaning of research, objectives of research, research methods versus research methodology, types of research in three perspective. Application perspective: basic versus applied research. Objective perspective: qualitative versus quantitative research types. Mode of enquiry perspective: structured and unstructured approach. Classification of research, inductive versus deductive logic. Time in research: cross sectional versus longitudinal. |
| <b>Scientific Investigation</b> | The hallmarks of scientific research: purposiveness, rigor, testability, replicability, precision and confidence, objectivity, generalizability and parsimony. The seven-step process in hypothetico-deductive method: identification of broad problem area, defining problem statement, developing hypothesis, determine measures, data collection, data analysis and interpretation of data. Some obstacles to conducting research in economics.        |
| <b>Research Questions</b>       | How problem can be translated into problem statement? General and specific research questions, developing research                                                                                                                                                                                                                                                                                                                                        |

|                                                         |                                                                                                                                                                                                                                                                                                                                                                                                               |
|---------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                         | questions in different situations, role of research questions. Basic types of questions: exploratory and descriptive.                                                                                                                                                                                                                                                                                         |
| <b>The Critical Literature Review</b>                   | What is literature review? How to approach the literature review? Evaluating the literature and documenting the literature review. Familiarization with APA citation format for referencing relevant material. Ethical issues: plagiarism. Introduction to bibliography manager software.                                                                                                                     |
| <b>Theoretical Framework and Hypothesis Development</b> | Need of theoretical framework in deductive research, definitional concept of theoretical framework, four main types of variables: dependent, independent, moderating and mediating. How theory is generated? The components of a theoretical framework. Hypothesis development: directional and non-directional hypothesis.                                                                                   |
| <b>Elements of Research Design</b>                      | Research strategies: experiment, survey research, observation, case studies, grounded theory, action research and mixed methods. Extent of researcher interference with the study. Study setting. Units of analysis: individuals, dyads, groups, organizations and cultures. General overview of research methods, quantitative (multivariate) methods, qualitative (interpretive) methods and mixed methods. |

### Teaching Methodology:

- To deliver lectures on topics included in course outline
- To require each student to solve independent assignments on topics included in the course.
- To work on an independent research project.

### Evaluation Criteria:

| Evaluation Method   | Total Percentage |
|---------------------|------------------|
| Quizzes/Assignments | 25%              |
| Mid-Term Exam       | 35%              |
| Final-Term Exam     | 40%              |

### Recommended Books:

- Greenlaw, S., Doing Economics: A Guide to Understanding and Carrying Out Economics Research, South Western Cengage Learning.
- William, G., Zikmand, Business Research Methods, 7<sup>th</sup> Edition.

- Kothari, C.R, (2008). Research methodology: Methods and techniques. New Delhi, New Age International Publishers.

#### **Additional Readings:**

- Sekaran, U. (2006). Research methods for business: A skill building approach. Singapore, John Wiley and Sons.
- Teddlie, C., & Tashakkori, A. (2009). Foundations of mixed methods research: Integrating quantitative and qualitative approaches in the social and behavioral sciences. Sage Publication
- Singleton & Straits. Approaches to social research (5th edition.). Oxford University Press

#### **Course Code: ECON-204**

**Title: Islamic Economics**

**Credit Hours: 03**

**Prerequisite: Intermediate Microeconomics**

#### **Course Objectives:**

The main objective of this course is to acquaint students with the economic aspects of Islamic principles and their application to analyze contemporary economic problems. The course helps to build nexus between the Islamic principles and mainstream economic theories. The topics covered in this course range from microeconomic to macroeconomic concepts pertaining to fundamentals of Islamic thoughts.

#### **Learning Objectives:**

The students will learn:

- Difference between contemporary and Islamic methods of dealing with economic problems.

#### **Course Contents:**

|                                          |                                                                                                                                                      |
|------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Introduction and Theory</b>           | Economic Problem and Islamic View of Life, Sources of Islamic Economics, Nature & Need of Islamic Economics, Moral Values & Economic System of Islam |
| <b>Islam and Other Economics Systems</b> | Economic System of Islam, Capitalism, Socialism                                                                                                      |

|                                                                  |                                                                                                                                                                                                                                                              |
|------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Consumption and Production in Islam</b>                       | Islam & Consumption of Wealth, Theory of Consumer Behavior in an Islamic Society, Islamic Approach to Production Purpose, Musharka (Partnership), Modarbah, Behavior of Firm Under the influence of Islamic Spirit, Islam & Price Mechanism                  |
| <b>Islam and Distribution of Wealth</b>                          | Circulation of Wealth, Rent, Profit, Wages and Dignity of Labour, Measures to Stop Concentration of Wealth, Legal Measures (Zakat & Inheritance), Optional Measures (Sadqat & Alms, Charity of Surplus & Auqaf)                                              |
| <b>Prohibition of Interest &amp; Interest Free Banking</b>       | Prohibition of Interest, Interest Free Banking, Model of Interest Free Banking, Islamic Modes of Financing (Bai Salam, Bai Murabaha, Bai Muajjal, Ijara, etc.)                                                                                               |
| <b>Fiscal Policy</b>                                             | Islamic State & its Fiscal Policy, Instruments of Fiscal Policy in Islam, Bait-ul-Mal, Theory of Taxation in Islam, Public Expenditures in Islamic Economy, Zakat & Ushr Ordinance                                                                           |
| <b>Some Topics from current Perspective of Islamic Economics</b> | Economic implication of Zakat, Islamic Inheritance Law and its Economic Implications, Theory of Taxation in Islam (Ibn-e-Khaldun and Theory of Taxation), Takaful and Islamic insurance policies in Pakistan, Islamic Financial System: Theory and Practice. |

#### Teaching Methodology:

- To deliver lectures on topics included in course outline
- To require each student to solve independent assignments on topics included in the course.

#### Evaluation Criteria:

| Evaluation Method   | Total Percentage |
|---------------------|------------------|
| Quizzes/Assignments | 25%              |
| Mid-Term Exam       | 35%              |
| Final-Term Exam     | 40%              |

#### Textbooks and Supplementary Readings:

- Islamic Economics: Dar A.H & M. Akram Ilmi Kitab Khana, Lahore (latest ed.)
- Mirakhor, A., & Askari, H. (2017). Ideal Islamic economy: an introduction. Springer.
- Uusmani, M. T., & Taqī 'Uṣmānī, M. (2002). An introduction to Islamic finance (Vol. 20). Brill.

## **Course Code: ECON-301**

**Title: Advanced Microeconomics**

**Credit Hours: 03**

**Prerequisite: Intermediate Microeconomics**

### **Course Objectives:**

This course approaches microeconomic analysis at an advanced level, presenting some formal techniques used in economic research. The course covers the main topics of microeconomics from consumer and producer behavior, game theory and asymmetric information. The course is designed to make students comfortable with the tools and methods of microeconomic analysis and to teach how to structure and analyze problems on these areas from a quantitative perspective

### **Learning Outcomes:**

On completion of this course, students would be able to

- Explain how the consumers and producers will take decisions
- Understand the equilibrium states of consumer and producers in given economy states.
- Compare the modern cost theories with traditional cost theories.
- Analyze and study different market structures and compare the functioning of all the structures in the economy.

### **Course Contents:**

|                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                |
|---------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Review of Consumer Behavior</b>    | Budget constraints, preferences, choice, intertemporal choice, supply, equilibrium, perfectly competitive markets                                                                                                                                                                                                                                                                                                              |
| <b>Oligopoly</b>                      | Choosing a Strategy: Quantity vs Price Leadership model, Comparing Price Leadership and Quantity Leadership, Introduction to Simultaneous Quantity Setting, Cournot Equilibrium, Simultaneous Price Setting, Collusion.                                                                                                                                                                                                        |
| <b>Externalities and Public Goods</b> | Externalities, The Inefficiency of Competition with Externalities Supply-and-Demand Analysis, Cost-Benefit Analysis, Regulating Externalities Emissions Fees versus Standards under Uncertainty, Market Structure and Externalities Monopoly and Externalities Monopoly versus Competitive Welfare with Externalities, Taxing Externalities in Noncompetitive Markets, Public Goods, Types of Goods, Markets for Public Goods, |

|                                 |                                                                                                                                                                                                                                                                                               |
|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                 | Allocating Property Rights to Reduce Externalities, Open-Access Common Property, Overuse of Open-Access Common Property, Solving the Commons Problem, Free Riding, Reducing Free Riding, Valuing Public Goods.                                                                                |
| <b>Welfare</b>                  | Aggregation of Preferences, Social Welfare Functions, Welfare Maximization, Individualistic Social Welfare Functions, Fair Allocations, Envy and Equity                                                                                                                                       |
| <b>Asymmetric Information</b>   | Problems Due to Asymmetric Information, Signaling and Screening, Education as a Signaling Model, Adverse Selection, Adverse Selection Problem in Insurance Markets and Labour Markets, Hidden Actions, Moral Hazard in Insurance Markets and Product Markets, Employer-Employee Relationships |
| <b>Choice under Uncertainty</b> | Gambles and Contingent Commodities, Applications of Contingent Commodities, Insurance, Diversification, Risk Spreading                                                                                                                                                                        |
| <b>Game Theory</b>              | Introduction to games, Nash equilibrium, pure strategies, mixed strategies, dominated actions, bargaining, formation of beliefs, extensive games with perfect information, Stackelberg's model of duopoly,                                                                                    |

#### Teaching Methodology:

- To deliver lectures on topics included in course outline
- To require each student to solve independent assignments on topics included in the course.

#### Evaluation Criteria:

| Evaluation Method   | Total Percentage |
|---------------------|------------------|
| Quizzes/Assignments | 25%              |
| Mid-Term Exam       | 35%              |
| Final-Term Exam     | 40%              |

#### Recommended Books:

- Perloff, J. M. (2013). Microeconomics: Theory & Applications with Calculus, 3rd Edition. Pearson Supplementary Texts
- Nicholson, W. & Snyder, C. (2009). Intermediate Microeconomics and Its Application, 11th Edition. Cengage Learning.
- MaS-Colell, A. (1995). Microeconomic Theory, Oxford University Press: New York.
- Mansfield, E., & Yoke, G. (2010), Microeconomics, 11th Ed. Viva Books: New York.

- Jehle, G. A. and Philip J. R. (2000). Advanced Microeconomic Theory, Second edition, Addison Wesley.
- Mas, Colell, A. Whinston, M. D. Green, J. R. (1995). Microeconomic Theory, Oxford University Press.

## **Course Code: ECON-302**

**Title: Advanced Macroeconomics**

**Credit Hours: 03**

**Prerequisite: Intermediate Macroeconomics**

### **Course Objectives:**

The course is designed for senior level undergraduate students. The objective is to enhance their analytical skills and to clarify further their macroeconomic concepts with quantitative tools. They must have a strong theoretical foundation to understand the real problems of the economy with particular emphasis on inflation, unemployment, and instability, to which the developing countries are confronted at present.

### **Learning Outcomes:**

After the completion of this course, students would be able to:

- Develop the foundation of modern macroeconomic theory in the light of different schools of thought
- Analyze the working of the economy in the perspective of relevant policy issues
- Apply key concepts of modern macroeconomic in their own research

### **Course Contents:**

|                                              |                                                                                                                                                                                                                                                                                                                                                                                          |
|----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Aggregate Supply and the Labor Market</b> | The aggregate production function, Demand for Labor in the short run, Labor market and its Equilibrium: Rigidity versus flexibility of nominal wage rate, The flexible and sticky wage models, the imperfect information model, the short-run aggregate supply function, The concepts of full-employment and underemployment, Supply shocks and stabilization.                           |
| <b>The Supply Side Economics</b>             | Derivation of Aggregate Supply under the Classical and Keynesian assumptions, Interaction of aggregate demand and supply and price determination, Scarcity of raw material and energy, The supply side disturbances leading to stagflation, The problems of developing countries: shortage of supply vs. deficiency of demand, The scope for structural changes in developing economies. |
| <b>Inflation and Unemployment</b>            | The relationship between inflation and unemployment, Phillips curve in the short-run and long-run, The expectation-augmented Phillips Curve, The Perfect Foresight and adjustment, Remedial measures to combat inflation and unemployment. Types, causes and measurement of inflation, the structural and monetary approaches to inflation, Global                                       |



|                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                        | inflation and its impacts on developing economies, Transitory and structural unemployment, the problem of unemployment with special reference to Pakistan,                                                                                                                                                                                                                                                                               |
| <b>The Open Economy Macroeconomics</b> | The IS-LM-FB Model, The case of a small open economy with special reference to developing countries, The relative effectiveness of fiscal and monetary policies under the fixed and floating exchange rate regimes, The managed exchange rate systems, Determinants of Exchange rates, Devaluation and its impacts for developing countries, Problem of stabilization in global framework.                                               |
| <b>Macroeconomic Policy Debate</b>     | Policy Actions: Active or Passive, Ignorance, Expectations and the Lucas Critique, Conduct of Policy: Formula/Rule vs. Discretion, Distrust of Policy makers and Political Process, Macroeconomics Policy: the Fiscal, Monetary and Commercial policies, Policy formulation and implementation in a world of uncertainty, Seignorage and Inflation: Monetarists vs. Activists.                                                           |
| <b>Growth and Business Cycles</b>      | The accumulation of capital, Acceleration principle and the concept of steady state growth path, The steady state conditions, The basic Neoclassical (Solow) growth model, Golden rule for consumption and investment (capital accumulation), The role of technical progress in growth, Introduction to the theory of Real Business Cycles: Labor market, Technology shocks, Household Behavior, the persistence of output Fluctuations. |

### Teaching Methodology:

- To deliver lectures on topics included in course outline
- To require each student to solve independent assignments on topics included in the course.

### Evaluation Criteria:

| Evaluation Method   | Total Percentage |
|---------------------|------------------|
| Quizzes/Assignments | 25%              |
| Mid-Term Exam       | 35%              |
| Final-Term Exam     | 40%              |

### Recommended Books:

- Branson, William – Macroeconomics: Theory and Policy – Latest Edition, Harper and Row, New York.
- Froyen, R. Macroeconomics: Theories and Policies–Latest Edition. Prentice Hall



- Gupta, G. S. Macroeconomics: Theory and Applications- Latest Edition, Tata McGraw Hill Publishing Company, New Delhi
- Scarth, W. M. Macroeconomics: An Introduction to Advanced Methods-Latest Edition Dryden Press: Harcourt Brace & Company.
- Romer, D. Advanced Macroeconomics –Latest Edition. McGraw Hills, New York.
- Mankiw, Gregory- Macroeconomics- Latest Edition. Worth Publisher, New York.

## **Course Code: ECON-303**

**Title: Research Methods- II**

**Credit Hours: 03**

**Prerequisite: Research Methodology-I**

### **Course Objectives:**

Research methods II is divided into two main parts. Part I includes measurement of variables, operational definitions and scales, reliability and validity, data collection methods (primary, micro and macro) and sampling methods. Part II covers basic quantitative and qualitative data analysis using statistical software, elements of research proposal and steps in report writing.

### **Learning Objectives:**

In this course students will learn how to

- Define and clean data
- Work with secondary dataconduct research analysis.

### **Course Contents:**

| <b>Part I</b>                                         |                                                                                                                                                                                                                                                                                                    |
|-------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Measurement: Scaling, Reliability and Validity</b> | Four types of measurement scales and their properties. Rating scales: definition and its types. Ranking scales: definition and its types. Examples and exercises to practical scales. Construct validity: internal and external. Goodness of measures, reliability and stability measures.         |
| <b>Data Collection Methods</b>                        | Sources of data, experiments, survey design, field study, methods of collecting primary and secondary data, guidelines for questionnaire design, quantitative data collection, qualitative data collection, mixed data collection techniques, selection of appropriate method for data collection. |
| <b>Sampling</b>                                       | What is sampling process? Defining population, determining sampling frame, sampling design and appropriate sample size.                                                                                                                                                                            |

|                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                       | Discussion on simple random sampling, systematic sampling, clustered sampling, convenience sampling, quota sampling, judgement sampling and snowball sampling. How to select appropriate sampling technique? Issues of precision and confidence in determining a sample size.                                                                                                                                      |
| <b>Part II</b>                                        |                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Quantitative Data Analysis</b>                     | Coding and data entry in statistical software. Getting a feel for the data: frequencies, summary measures, bar charts, pie charts. Relationship between two nominal variables: chi-square test. Correlations and testing goodness of measures using statistical software. Basic regression analysis.                                                                                                               |
| <b>Quantitative Data Analysis: Hypothesis Testing</b> | Basic concepts and procedure for hypothesis testing, type I and type II errors, statistical power choosing appropriate statistical technique: univariate versus multivariate. Test of hypothesis about mean and difference between means, Test of hypothesis about proportions and difference between proportions, variance, equality of variance, and correlation coefficient. Limitations of test of hypothesis. |
| <b>Qualitative Data Analysis</b>                      | Three important steps in qualitative data analysis: data reduction, data display and drawing conclusions. Coding and qualitative data entry in statistical software. Reliability and validity in qualitative research. Compare and contrast content analysis, narrative analysis and analytic induction.                                                                                                           |
| <b>Research Proposal</b>                              | Meaning and need of research proposal, steps in writing a research proposal, layout of research proposal, types of research proposal and oral presentation.                                                                                                                                                                                                                                                        |
| <b>Steps in Report Writing</b>                        | Abstract, introduction, literature review, methods and data collection, findings and interpretation, conclusion, suggestions, future prospects, references.                                                                                                                                                                                                                                                        |

### Teaching Methodology:

- To deliver lectures on topics included in course outline
- To require each student to solve independent assignments on topics included in the course.
- To require each student to work on independent research project with data analysis.

### Evaluation Criteria:

| Evaluation Method   | Total Percentage |
|---------------------|------------------|
| Quizzes/Assignments | 25%              |
| Mid-Term Exam       | 35%              |
| Final-Term Exam     | 40%              |

### Recommended Books:

- Kothari, C. R (2008). Research methodology: Methods and techniques. New Delhi, New Age International Publishers.
- Saunders, M. (2005). Research methods for business studies. Singapore, Pearson Education
- Monette D. R., Sullivan, T.J., & Dejong, C.R. (2010). Applied social research: A tool for the human services. Belmont, CA: Linda Schreiber-Ganster.
- Sekaran, U. (2006). Research methods for business: A skill building approach. Singapore, John Wiley and sons.

### Course Code: ECON-304

**Title: History of Economic Thought**

**Credit Hours: 03**

**Prerequisite: Intermediate Microeconomics and Intermediate Macroeconomics**

### Course Objectives:

The main objective of this course is to introduce students with difference schools of thoughts in economics. This course surveys the main schools in the history of the development of economic thought, beginning with the Classical school and concludes with the recent work in field of economics.

### Learning Outcomes:

After the completion of this course, the students will be able to understand the following:

- Historical background of various schools of economic thought and social, political and economic environment within which a specific school emerged.
- The major tenets of the school, enabling a concise presentation of the essence of each school
- The beneficiaries of a specific school during its time period
- The utility and validity of the school in its time
- The tenets of the school which made the lasting contribution and helped in the emergence of modern economics
- The key debate and the socio-political context in which the main texts were written
- The ability to appreciate and analyze primary texts, including their contemporary relevance

## Course Contents:

|                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Introduction</b>                             | Importance of Economic History: Approaches to Study the Subject. Overview of Ancient Greek, Roman and Medieval Economic Thought. Overview of Ancient and medieval thoughts, Quensey's influential ideas, Feudalism, Mercantilism, Nature's Circular Flow, Process of modern theories and thoughts. Overview of Islamic thoughts: Ibne Khuldoon and thereafter. Overview of Salient Features of Muslim School of Thought.                                                                                                                                                                                                         |
| <b>Mercantilists and the Physiocrats</b>        | The Middle Ages and the Scholastics, Mercantilism – French and English, The Rise of Science and Scientific Method, The French Enlightenment and the Physiocrats, A doctrine of economic process, transition to liberalization: Wage theory to natural price, the task of government. Overtime, Changing Role of Government                                                                                                                                                                                                                                                                                                       |
| <b>Classical School</b>                         | The Contributions of Classical School: Adam Smith, Malthus (Life and Times, The Essay on Population, The Principles of Economics, Interpretative Issues), Ricardo (Life and Times – Utilitarians, Napoleon, Politics, Corn Laws, Ricardo's Contributions, Ricardians and the anti-Ricardo reaction, Interpreting Ricardo) and Karl Marx (Origins of Marx's Thought – Hegel, Utopian Socialists, and British Political Economy, Marx's Laws of Motion of Capitalism – Exposition and Appraisal, The Marxian Tradition).                                                                                                           |
| <b>The Marginal Revolution</b>                  | Hutchison's analysis of reaction to politics, theory, method, The Early Marginalists – dehomogenizing the emphasis on utility, Jevons – utilitarian theory, Menger – subjectivism and unintended consequences, Walras – general equilibrium and Pareto, The German Historical School and the <i>Methodenstreit</i> .                                                                                                                                                                                                                                                                                                             |
| <b>Alfred Marshall and the Cambridge School</b> | His life, his influence on Cambridge, his economics, The Principles of Economics, The Marginal Revolution Redux – the Mirowski Thesis                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Keynesian and Post Keynesian Schools</b>     | <p>The Keynesian and Post Keynesian Schools, The Great Depression: Economic Policies and Capitalist Instability between the World Wars. The 20th Century Paradigms. The Growth of International Economy. The rise and fall of Post-World War II. The American economic history and lessons.</p> <p>(i) The Emergence of Modern Economic Growth, Building Blocks, Industrial Revolution, Structuralist and Dependency School of thought; Sunkel, Amir Samer, Frank and others' contribution. Technological Change and Impacts. The Crisis of the 1970s and International Responses. Failure of Trickle-down effects. Rational</p> |

|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <p>Expectation Revolution. Basic Needs Approach and welfare. Emergence Famine and Poverty.</p> <p>(ii) Modern Economic Thought. Comparison of Post Keynesian and Monetarist Thoughts. Supply side economics. Globalization, Liberalization and New Weave of Economic Growth and Welfare. New Growth Theory. Convergence and Divergence Debate. Future of Economics, Quality and International Trade and emergence of markets, New Regionalism, New directions of research. Environment and sustainable growth.</p> |
|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### Teaching Methodology:

- To deliver lectures on topics included in course outline
- To require each student to solve independent assignments on topics included in the course.

### Evaluation Criteria:

| Evaluation Method   | Total Percentage |
|---------------------|------------------|
| Quizzes/Assignments | 25%              |
| Mid-Term Exam       | 35%              |
| Final-Term Exam     | 40%              |

### Recommended Books:

- Economic Thought Before Adam Smith (2009), Murray N. Rothbard
- Mark Blaug, Economic Theory in Retrospect, Cambridge University Press.
- John D. Henery. (1992). The Future of Economics, Blackwell Publishers.
- Ingrid Rima. (2009). Development of Economic Analysis, 7th edition, Routledge, Taylor & Francis Group, London.
- Henry W. Spiegel, The Growth of Economic Thought. 3rd ed. Durham: Duke University Press, 1991.
- Adam Smith, The Wealth of Nations. [1776] Chicago: University of Chicago Press, 1976.
- Robert Tucker, ed. The Marx-Engels Reader. 2<sup>nd</sup> ed. New York: Norton, 1978.
- Bruce Caldwell, Hayek's Challenge: An Intellectual Biography of F.A. Hayek. Chicago: University of Chicago Press, 2004.

**Course Code: ECON-305**

**Title: Econometrics-I**

**Credit Hours: 03**

**Prerequisite: Elementary Statistics, Probability and Probability Distributions**

### Course Objectives:

This is a foundation course for enabling the students of Economics to analytically formulate and statistically estimate the economic problems for verifying the empirical validity of theoretical models. The analytical skill is supported by the usage of different Econometrics Software.

### Lab work

The students are required to devote at least two hours per week to computer laboratory. They have to attend classes to learn various Econometrics software for practical application of Econometric models they have learnt so far.

### Learning Outcomes:

On completion of this course, students would be able to

- Have a complete knowledge of the basic concepts of the subjects thus enabling them to attain higher education in the field of econometrics
- This course will enable the students to apply tools of mathematics and statistics to to identify the relationships between economic variables
- This will help the students to estimate and evaluate some of the basic and advanced econometric models
- It will guide the students to use econometric models for forecasts and managerial decision making.
- It will also help the students in conducting the research work necessary for completing B.sc Honor programme

### Course Contents:

|                                      |                                                                                                                                                                                                                                                                                                                                                                                                                         |
|--------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Introduction</b>                  | Definition and scope of Econometrics, Economic theory as the basis for empirical analysis, Mathematical and Econometric models, Application of Statistical techniques to Economic data, Ingredients of Econometric modeling: Specification, Estimation, Evaluation and Forecasting. Data Types and Sources, Cross-sectional data, time-series data and pool data, Data at current and constant prices, Sources of data. |
| <b>The Simple Two-Variable Model</b> | Simple Regression function, population and regression function, linear regression function: linearity in variables and linearity in parameters, Simple Regression Analysis: Estimation, Method of Ordinary Least Squares (OLS), Estimation of regression equation using OLS, Standard error of estimates, Numerical properties of OLS estimators, Statistical properties of OLS estimators (BLUE),                      |



|                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|----------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                    | Classical Linear Regression Model (CLRM), Assumptions of CLRM, Estimation of the coefficient of determination, Interval estimation of regression coefficients, Classical Normal Linear Regression Model (CNLRM), Difference between CLRM and CNLRM, Properties of OLS estimates under CNLRM, Simple Linear Regression Analysis: Inference, Confidence interval approach for regression coefficients, Test of significance approach for regression coefficients, Analysis of variance (ANOVA), Test of the overall significance of the model, Simple Linear Regression Analysis: Extensions, Regression through the origin, Scaling and measurement of variables, Functional Forms of Regression Function, Log linear models, Cobb Douglas production function Log-Lin models, The constant growth model, Estimating the growth rate, Lin-Log models, Engel curve, Reciprocal models. |
| <b>The Multiple Linear Regression Model (MLRM)</b> | Multiple Regression Analysis, Difference between simple and multiple regression analysis, Interpretation of multiple regression function, Multiple Regression Analysis: Estimation, OLS estimation of multiple regression equation, Standard error of partial regression coefficients, Properties of OLS estimators, Coefficient of determination ( $R^2$ ), $R^2$ and Adjusted $R^2$ , Multiple Regression Analysis: Inference, Test of individual significance (t-test), Test of overall significance (F-test) Multiple Regression Function: Extensions, Testing the equality of parameters, Testing linear equality restriction, Testing for structural stability of regression models: The Chow test.                                                                                                                                                                            |
| <b>Categorical Variables in regression</b>         | Nature of dummy variables, ANOVA models with qualitative variables, Regression with dummy independent variables                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Deviation from the Classical Assumptions</b>    | Assumptions of the classical model and economic reality, Relaxation of the assumptions and estimation issues, Brief introduction to the nature of problems and alternatives.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Multicollinearity</b>                           | Linear relationship between any two explanatory variables, Nature and severity of the problem, OLS estimation of regression equation in the presence of perfect multicollinearity, Causes of multicollinearity, Distinction between perfect and partial Multicollinearity, Detection of the problem and remedial measures.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

### Teaching Methodology:

- To deliver lectures on topics included in course outline

- To require each student to solve independent assignments on topics included in the course including the lab work.

#### **Evaluation Criteria:**

| <b>Evaluation Method</b> | <b>Total Percentage</b> |
|--------------------------|-------------------------|
| Quizzes/Assignments      | 25%                     |
| Mid-Term Exam            | 35%                     |
| Final-Term Exam          | 40%                     |

#### **Recommended Books:**

- Gujrati, D. J. Basic Econometrics – 4th Edition (2003) McGraw-Hill Company.
- Maddala, G. S. (1988) Econometrics – McGraw-Hill Company.
- Dougherty, C. (2002). Introduction to Econometrics–2nd edition Oxford University Press.
- Pindyck & Rubinfeld (1992). Econometric Models & Economic Forecasts- 3rd Edition, McGraw-Hill Inc.
- Stock H. J. and M. W. Watson (2003). *Introduction to Econometrics*, India: Pearson Education.

#### **Course Code: ECON-306**

**Title: Development Economics-I**

**Credit Hours: 03**

**Prerequisite: Intermediate Macroeconomics and Intermediate Microeconomics**

#### **Course Objectives:**

This course examines the process of economic development, including the determinants of underdevelopment and poverty. We will focus on describing and understanding the many problems faced by developing countries and their citizens. Throughout the course, we will discuss policy implications of the topics we explore. We begin with an overview of economic development and models economists use to understand the process of economic development. Next, we consider the impediments to development and how they can potentially be addressed.

#### **Learning Outcomes:**

Upon completion of the course students should be able to:

- Explain the key concepts of Developments Economics and Identify its role in devising the economic policies of the country



- Identify various perspectives on current and controversial issues regarding development (dependence on foreign aid, pollution, trade-off between industrial and agricultural sector, trade openness). Identify their position and be able to support it
- Understand the economic theory that has shaped the disciplines of economic development
- Analyze the economic rationale underlying policy choices
- Critically reflect on the theories of development and develop an analytical view and coherent opinion about economic changes within low-income countries

#### Course Contents:

|                                                                |                                                                                                                                                                                                                                                                                                                                                                                                   |
|----------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>The Concept &amp; Nature of Development</b>                 | The Nature of Development Economics, Why Study Development Economics? Traditional Economic Measures, Amartya Sen's "Capability" Approach, development versus growth, development variables, Basic Indicators of Development: PPP (purchasing power parity), PQLI (physical quality life index), human development index (HDI) and Governance index (GI). Characteristics of developing countries. |
| <b>Contemporary Models of Development and Underdevelopment</b> | Underdevelopment as a Coordination Failure, Multiple Equilibria: A Diagrammatic Approach, The Big Push, Michael Kremer's O-Ring Theory of Economic Development, The Hausmann-Rodrik-Velasco Growth Diagnostics Framework.                                                                                                                                                                         |
| <b>Poverty and Inequality</b>                                  | Measuring Income Distribution & Poverty, Absolute Poverty, Social Welfare, Dualistic Development and Shifting Lorenz Curves: Some Stylized Typologies, Kuznets's Inverted-U Hypothesis, Economic Characteristics of High-Poverty Groups.                                                                                                                                                          |
| <b>Population Growth and Economic Development</b>              | The Basic Issue: Population Growth and the Quality of Life, Population Growth: Past, Present, and Future, The Demographic Transition, The Causes of High Fertility in Developing Countries: The Malthusian and Household Models, The Consequences of High Fertility: Some Conflicting Perspectives.                                                                                               |
| <b>Urbanization and Rural-Urban Migration</b>                  | Theories of Urban Growth, Agglomeration Economies, Urban Unemployment, Migration and Development, Toward an Economic Theory of Rural-Urban Migration, Comprehensive Migration & Employment Strategy                                                                                                                                                                                               |
| <b>Human Capital: Education and Health</b>                     | Education & Health in Development, International Trends, The Human Capital Approach, The Gender Gap: Discrimination in Education and Health, The Political Economy of Educational Supply and Demand: The Relationship between Employment Opportunities and Educational Demands, Social versus Private Benefits and Costs, Health Measurement and Distribution                                     |

**Teaching Methodology:**

- To deliver lectures on topics included in course outline
- To require each student to work on research project and present it.

**Evaluation Criteria:**

| Evaluation Method   | Total Percentage |
|---------------------|------------------|
| Quizzes/Assignments | 25%              |
| Mid-Term Exam       | 35%              |
| Final-Term Exam     | 40%              |

**Recommended Books:**

- Todaro & Smith (2014). Development Economics, 11th or 12th Edition Pearson
- Herrick B. and Kindleberger C., (latest eds.). Economic Development. McMillan, New York.
- Jones H. G., An Introduction to Modern Theories of Economic Growth, (Latest Edition), McGraw Hills

**Recommended Research Articles:**

- Cruz, M., Foster, J.E., Quillin, B. and Schellekens, P., 2015. Ending Extreme Poverty and Sharing Prosperity.
- Women and Contraceptives <http://www.cnn.com/2012/07/09/opinion/gates-contraception-families/index.html>
- Tacoli, C., McGranahan, G. and Satterthwaite, D., 2015. Urbanisation, rural-urban migration and urban poverty. Human Settlements Group, International Institute for Environment and Development.
- Economist "Healthcare in Poor Countries: For 80 cents More" August 15, 2002 <http://www.economist.com/node/1280587>

**Course Code: ECON-307****Title: Advanced Mathematical Economics****Credit Hours: 03****Prerequisite: Calculus -I & Calculus -II****Course Objectives:**

In this course, particular focus is on applying the mathematical tools on economic theory and models. The level of sophistication at which the material is to be taught is indicated by the Course Contents of the prescribed textbooks.

### Learning Outcomes:

- On completion of this course, students would be able to
- Apply mathematical techniques for better understanding of economic theory
- Relate mathematical methods most relevant to economics and fully explain each method
- Illustrate how the method is applied in economic analysis
- Use mathematical tools for understanding the relationship between Micro and Macroeconomic variable

### Course Contents:

|                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|--------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>First Order Linear Differential Equations</b>                                     | Meaning and Definition; Homogenous & non-Homogenous Cases. Solution of First Order Linear Differential Equation with Constant Coefficient & Constant Term and its Verification. Economic Applications: Dynamics of Market Price, Dynamics of Consumption using Differential Equations.                                                                                                                                                                                                                                                                                                                               |
| <b>Microeconomic Applications of First Order Linear Differential Equations:</b>      | Dynamics of Investment using Differential Equations. Dynamics of Value of Oil Well, Dynamics of Value of Gold Mine, Dynamics of Future Value of Investment, Dynamics of Fuel Consumption, Limited Growth, Dynamics of Ore Mining, Dynamics of Fish Stock.                                                                                                                                                                                                                                                                                                                                                            |
| <b>Macroeconomic Applications of First Order Linear Differential Equations:</b>      | National Income Determination using Differential Equations, Dynamics of Population, Dynamics of Government Spending, Dynamics of Unemployment.                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>First Order Differential Equation with Variable Coefficient and Variable Term</b> | Solution and its Verification of First Order Differential Equation with Variable Coefficient and Variable Term.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Non-linear Differential Equations of the First Order and First Degree.</b>        | Exact Differential Equation, its Solution and Verification. Production Function using Exact Differential Equations, Utility Function using Exact Differential Equations, Average Cost Function using Exact Differential Equations, Point Elasticity Formula using Exact Differential Equations. Separable Variables. Dynamics of Capital using Separable Variables Method. Bernoulli Equation. The Qualitative Graphic Approach. Concept of Phase Diagram, types of Time Paths and their Dynamic Stability. Economic Application: Solow Growth Model: Qualitative Analysis, Phase Diagram and Quantitative Analysis. |

|                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|-----------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Trigonometry and Complex Numbers</b>                   | Fundamentals of trigonometry, Trigonometric identities. Imaginary and Complex Numbers. Complex Roots. Circular Functions. Properties of Sine & Cosine functions. Euler Relations. Alternative Representation of Complex Numbers.                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Second Order Differential Equations</b>                | Solution and Verification of Second Order Linear Differential Equations with Constant Coefficient and Constant term-Distinct Real Roots, Repeated Real Roots and Complex Root Cases. Dynamic Stability of Equilibrium. Economic Applications: A Market Model with Price Expectations. The Interaction of Inflation and Unemployment. Phillips Curve: The Phillips Relation, Expectations Augmented Phillips Relation, Feedback from Inflation to Unemployment, Time Path of Expected Rate of Inflation, Time Path of Expected Rate of Inflation – Numerical, Time Path of Inflation – Numerical, Time Path of Unemployment – Numerical. |
| <b>Differential Equations with Variable Term</b>          | Differential Equation with Variable Term, Differential Equation with Variable Term: A Modification.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>First Order Difference Equations</b>                   | Solution and Verification of First Order Difference Equations. The Dynamic Stability of Equilibrium. Economic Applications – The Cobweb Model, A Market Model with Inventory. Harrod Model of Growth, Lagged Income Determination Model, Income Dynamics with Induced Investment, Capital Market, Money Market and Balanced Trade using Difference Equations. Investment Dynamics using Difference Equations. Fish Stock Analysis using Difference Equation, Difference Equations and Adaptive Expectations.                                                                                                                            |
| <b>Second Order Difference Equations</b>                  | Solution and Verification of Second-Order Linear Difference Equations with Constant Coefficients and Constant Term-Distinct Real Roots, Repeated Real Roots and Complex Roots cases. The Convergence of the Time Path. Economic Applications, Samuelson Multiplier-Acceleration Interaction Model. Inflation and Unemployment in Discrete Time. Second-Order Difference Equation with Variable Term, Second-Order Difference Equation with Variable Term ( $ct^n$ )                                                                                                                                                                     |
| <b>Higher Order Differential and Difference Equations</b> | Solution of Higher Order Differential Equations with Constant Coefficient and Constant Term. Convergence and the Routh Theorem. Higher Order Linear Difference Equations and their Solutions. Convergence and Schur Theorem.                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Linear Programming</b>                                 | Ingredients of linear Programming. Graphical approach, simplex method, economic application of linear programming. Concept of                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

|                                |                                                                                                                                                                                                                                                                                                                                                                                       |
|--------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                | primal & dual. Duality theorems. Solving of Primal via dual. Economic interpretation of a dual.                                                                                                                                                                                                                                                                                       |
| <b>Non-Linear Programming:</b> | The Nature of Non Linear Programming Non-Linearities in Economics. Kuhn Tucker Condition. Interpretation of Kuhn Tucker Condition. Kuhn Tucker Sufficiency Theorem: Concave Programming. Arrow Enthoven Sufficiency Theorem: Quasiconcave Programming. Economic Application-Utility Maximization, Least Cost Combination. Solving a Nonlinear Program via the Kuhn-Tucker Conditions. |

### Teaching Methodology:

- To deliver lectures on topics included in course outline
- To require each student to solve **independent** assignments on topics included in the course.

### Evaluation Criteria:

| Evaluation Method   | Total Percentage |
|---------------------|------------------|
| Quizzes/Assignments | 25%              |
| Mid-Term Exam       | 35%              |
| Final-Term Exam     | 40%              |

### Recommended Books:

- Chiang A.C. Fundamental Methods of Mathematical Economics McGraw Hill –Latest Edition.
- Weber E. Jean, Mathematical Analysis, Business and Economic Application (latest edition).
- Bradley, T. Essential mathematics for economics and business. Latest edition. John Wiley & Sons.
- Dowling, E. T. Theory and Problems of Introduction to Mathematical Economics. Latest Edition, McGraw-Hill, New York.
- Hoffman, L. D., & Bradley, G. L. Calculus for business, economics, and the social and life sciences. McGraw-Hill. Latest edition.
- Kolman, B., & Beck, R. E. (1995). Elementary linear programming with applications. Academic Press.
- Sydsæter, K. and Hammond, P. Essential Mathematics for Economic Analysis. Latest edition., Pearson Education Limited, England.

## Course Code: ECON-308

**Title: Monetary Economics**

**Credit Hours: 03**

**Prerequisite: Intermediate Macroeconomics**

### Course Objectives:

This course is intended to provide the students a thorough understanding of monetary theory, financial markets, importance of money and interest rates along with the conduct of monetary policy in closed and open economy. This course also introduces the alternative monetary policy stances. In addition, this includes the discussion of monetarists' and Keynesian views regarding inflation.

### Learning Outcomes:

At the end of the course, students will be able to:

Understand the nature and function of money

- Evaluate different policies for demand of money, determination of money supply, interest rates, inflation and output
- Understand the classical monetary theory and neutrality of money
- Understand the transmission mechanism of monetary policy and the monetary policy in closed and open economies in both theories and practice
- Use the monetary policy rule (simple and optimal)

### Course Contents:

|                                                           |                                                                                                                                                                                                                                                                                                                                       |
|-----------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Introduction and Development of Monetary Mechanism</b> | Evolution of Money and Payment System. Definition of Money, Function of Money & Measurement of Money, Demand for Money & Supply of Money. Definitions of Money, M1, M2, M3, etc. The Evolution of Monetary Thought, Quantity Theory of money, Keynesian Tradition, Monetarism and Modern Quantity Theory, Rational Expectation Theory |
| <b>Role of Money in the Economy</b>                       | Debate on neutrality and non-neutrality of Money. Classical Dichotomy and Keynesian Integration of Nominal and Real Sectors. Role of Money in the ISLM Framework. Money in Aggregate Demand & Aggregate Supply Analysis. Keynesian Vs. Monetarist views of Equilibrium Output, Employment and Prices                                  |
| <b>Theory of Demand for Money</b>                         | Quantity Theory of Money. Transaction Theories of Demand for Money. Portfolio Theories of Demand for Money. Baumol – Tobin Model of Cash Management. Friedman's Restatement of Quantity Theory of Money: Empirical Evidence on the Demand for Money                                                                                   |
| <b>The Money Supply Process</b>                           | Monetary Base, Keynesian Liquidity Preference Framework, Friedman's Modern Quantity Theory of Money. Multiple Deposit Creation: Introducing the Money Supply Process. Determinants                                                                                                                                                    |



|                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-----------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                     | of the Money Supply: Exogenous and Endogenous. Understanding Movements in the Monetary Base, Money Multiplier. Velocity of Money and its variability. Explaining Depositor and Bank Behaviour: The Complete Money Supply Model. Regressive Expectation Model                                                                                                                                                                                                                                                                                                                                                     |
| <b>The Conduct of Monetary Policy: Central Bank</b> | Structure and Functions of Central Bank. Independence of Central Bank and its role in Economic Growth. Tool of Central Bank for Money Control. Monetary Base. The Conduct of Monetary Policy: Goals and Targets. Tools of Monetary Policy: Three Major Tools of Monetary Policy. Required Reserve Ratio, Discount Window Loans and Open Market Operation. Advantages and Disadvantages and Impact Analysis of each Policy Tool. Central Bank as a lender of last Resort. Financial Intermediaries. Monetization of Public Debt. Ineffectiveness of Stabilization Policies. Targeting interest rate and inflation |
| <b>Money &amp; Interest Rates</b>                   | Determinants of interest rates. Kinds of Interest Rates. The Behavior of Interest Rates. Understanding & Measuring the Interest Rates. Real & Nominal Interest Rates, Theories of Interest Rate Determination. The Risk and Term Structure of Interest Rates. The Distinction Between Interest Rates and Returns. Maturity & Volatility of Bond Returns. Portfolio Choice: The Theory of Asset Demand                                                                                                                                                                                                            |
| <b>Money and Inflation</b>                          | What is Inflation & Causes of Inflation? Demand – pull Inflation, Cost – push Inflation, Stagflation & Hyperinflation. Inflation as a Monetary Phenomenon. The Philips Curve and Accelerating Inflation. Inflation in Pakistan: Sources and Managing Policies. Sources of Inflation in Pakistan: Spiracle Evidences. Policies to Combat Inflation in Pakistan                                                                                                                                                                                                                                                    |
| <b>Monetary Policy in International Framework</b>   | Money Demand and Empirical Evidences. Money in Growth Models. Inflationary Spiral. Current Issues in Monetary and Financial Sector Reforms. Conduct of Monetary Policy in Fixed and Flexible Exchange Rates. Perfect Mobility of Capital and Money Control, Effectiveness of Policy. Impact of Reserves on B.O.P and Exchange Rates. Targeting exchange rate. Managed Exchange Rate System and Monetary Policy. Targeting Inflation. Exchange Rate and Balance of Payment                                                                                                                                        |

### Teaching Methodology:

- To deliver lectures on topics included in course outline
- To require each student to solve independent assignments on topics included in the course.



**Evaluation Criteria:**

| <b>Evaluation Method</b> | <b>Total Percentage</b> |
|--------------------------|-------------------------|
| Quizzes/Assignments      | 25%                     |
| Mid-Term Exam            | 35%                     |
| Final-Term Exam          | 40%                     |

**Recommended Books:**

- Mishkin, Frederic S. (2001). the Economics of Money, Banking and Financial Markets. (Sixth edition). Addison Wesley, New York. Latest edition
- Bennett T. McCallum. (1989). Monetary Economics, Theory and Policy, McMillan. latest edition

**Additional Readings:**

- Fredric S. Mishkin, (1995). Financial Markets and Money, Harper & Row Publishers.
- Laider, David E.W. (1996). The Demand for Money: Theories, Evidence and Problems (Fourth edition), Harper & Row, Publishers, New York.
- Miller, R. L. and David VanHose, (2001). Money, Banking & Financial Markets. South Western, Singapore.
- Vanish K. (2000). Monetary Theory. Vikas Publishing House, Delhi.
- Patinkin Don, Money, Interest and Prices, Harper and Row Publishers, (Latest Edition)
- Handa J. (2000). Monetary Economics, London: Routledge

**Course Code: ECON-309****Course Title: Labor Economics****Credit Hours: 03****Prerequisite: Intermediate Microeconomics & Intermediate Macroeconomics****Course Objectives:**

This course provides an introduction to the broad field of labor economics. Students will become familiar with models of how firms make decisions about their need for workers, how workers make labor supply decisions, and how wages are determined. Other important issues that affect the labor market such as immigration, discrimination, and unions are also discussed. Throughout the course we will investigate the effects of relevant public policies (such as minimum wage, employment subsidies, and unemployment insurance) on the labor market.

**Learning Outcomes:**

Students will learn:

- the theoretical models explaining the dynamics of demand and supply of labour market.
- The issues of wage determination and how to explain the wage differentials.

#### Course Contents:

|                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Introduction</b>                      | What is Labor Economics? Importance of Labor Economics                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Labor Supply</b>                      | Basic Concepts and Definitions: unemployment, reasons of unemployment, labor force, labor participation, etc. with special reference to Pakistan, reservation wage, unemployment benefits, safety net.<br><br>Household labor supply model: Marginal Rate of Substitution, Constraints, solution to Worker's Optimization problem, Hours of Work Decision, Comparative Statics, Substitution Effect and Income Effect, determination of Reservation Wage, cost of working, Unemployment Benefits, Inter-temporal Labor Supply. |
| <b>Labor Demand</b>                      | The objective of the Firm, Labor Demand model: The Demand for Labor in the Short Run and Long Run, derived demand, Substitution and Scale Effect, Elasticity of Substitution, Elasticity of Labor Demand and its Determinants, Non-Wage Costs of Employment                                                                                                                                                                                                                                                                    |
| <b>Labor Market Equilibrium</b>          | Equilibrium in Competitive Labor Markets, Shifts in Demand and Supply, The Cobweb Model, Monopsonistic Labor Market, Minimum Wages in a Monopsonistic Labor Market, The Effects of Government Intervention on Equilibrium.                                                                                                                                                                                                                                                                                                     |
| <b>Education and Training</b>            | The Schooling Model, The Wage-Schooling Locus, The Stopping Rule, Signaling Model, Human Capital Model, On-the-job Training, Returns on Training with reference to Pakistan.                                                                                                                                                                                                                                                                                                                                                   |
| <b>The Distribution of Earnings</b>      | Wages and distribution of national income, GINI coefficient, wage-share and income-share, relation between income and savings rate                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Further Topics in Labor Economics</b> | Labor Market Discrimination, Employer Discrimination, Informal Labor Market, Employee Discrimination. Wage Inequality: A Simple Model of Labor Market, Reasons for wage inequality. Immigration (domestic and international)                                                                                                                                                                                                                                                                                                   |
| <b>Trade Unions</b>                      | Where and why Unions Exist? First Bargaining Structure, Leontieff's Monoploy union Model, Efficiency of Unions.                                                                                                                                                                                                                                                                                                                                                                                                                |

#### Teaching Methodology:

- To deliver lectures on topics included in course outline
- To require each student to solve independent assignments on topics included in the course.

**Evaluation Criteria:**

| Evaluation Method   | Total Percentage |
|---------------------|------------------|
| Quizzes/Assignments | 25%              |
| Mid-Term Exam       | 35%              |
| Final-Term Exam     | 40%              |

**Recommended Books:**

- Borjas, G.J. (2009). Labor economics. (Boston, Mass.; London: Irwin/McGraw-Hill) fifth edition [ISBN 9780070172708].
- Briggs, V.M., A.G. & Marshall, F.R. (1980). Labor Economics: Wages, Employment and Trade Union. Richard D. Irwing, Inc Home wood Illusions.
- Edgmand, M.R. (1979). Macroeconomic Theory and Policy, Prentice Hall.
- Ehrenberg, R.G., & Smith R.S. (1997). Modern Labor Economics: Theory and Public Policy. Addison-Wesley.
- Kaufman, B.E., & Hotchkiss, J.L. (2003). Labor Market Economics. Cengage Learning
- Lloyd Reynolds, Labor Economics and Labor Relations.
- Mc Connell, C-R., Bruce, S.L., & Macpherson, D.A. (2003). Contemporary Labor Economics. MCgraw-Hill.
- Smith Stephen (2003). Labour Economics, Routledge Publishers.

**Course Code: ECON-310****Title: Urban Economics****Credit Hours: 03****Course Objectives:**

The discipline of urban economics is defined by the intersection of geography and economics. Economics explores the choices people make when resources are limited. Households make choices to maximize their utility while firms maximize their profit. Geographers study how things are arranged across space. Urban economics puts economics and geography together, exploring the location choices of utility-maximizing households and profit-maximizing firms. Urban economics also explores the causes and consequences of urban problems such as congestion, pollution, and crime and evaluates alternative policy responses to these problems. Also, urban economics explores the efficiency and distributional effects of the policies of local governments.

**Learning Outcomes:**

The students should learn:

- how urban cities are formed?

- how do they manage resources to maximize efficiency?
- about the problems of urban cities, such as providing education and health facilities.

### Course Contents:

|                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                       |
|---------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Introduction</b>                                     | What is urban economics? What is a city? Why do cities exist? The five axioms of urban economics, Trading City, Trading Cities in Urban History, Factory City,                                                                                                                                                                                                                                                        |
| <b>Agglomeration Economies and Clustering</b>           | Sharing Intermediate Inputs, Self-Reinforcing Effects Cause Industry Clusters, Sharing a Labor Pool, Labor Matching, Knowledge Spillovers, Urbanization Economies, Utility and City Size, A System of Cities, Specialized and Diverse Cities,                                                                                                                                                                         |
| <b>Urban Growth</b>                                     | Where Do cities Develop? The Urban Labor Market, Public Policy and Equilibrium Employment, The Regional Context of Urban Growth, Consumer Cities and Central Place Theory.                                                                                                                                                                                                                                            |
| <b>Land Rent and Manufacturing Land</b>                 | Introduction to Land Rent, Bid-Rent Curves for the Manufacturing Sector, Bid-Rent Curves for the Information Sector, Office Bid-Rent Curves with Factor Substitution, Housing Prices, The Residential Bid-Rent Curve.                                                                                                                                                                                                 |
| <b>Spatial Distribution of Employment and Residence</b> | The Spatial Distribution of Population, The Rise of the Monocentric City, The Demise of the Monocentric City, Urban Sprawl, The Monocentric Model and Applications.                                                                                                                                                                                                                                                   |
| <b>Neighborhood and Land Use Policy</b>                 | Diversity versus Segregation, Sorting for Local Public Goods, Neighborhood Externalities, Neighborhood Choice, Racial Segregation, The Consequences of Segregation, Land-Use Zoning, Growth Control: Urban Growth Boundaries                                                                                                                                                                                          |
| <b>Urban Public Transit and Local Government</b>        | Autos and Highways, Congestion Externalities, The Congestion Tax, Practicalities of the Congestion Tax, Alternatives to a Congestion Tax, Commuting and Transit Ridership, The Cost of Travel and Modal Choice, The Efficient Volume of Ridership, Designing a Transit System, The Role of Local Government, Local Public Goods: Equilibrium versus Optimum, Natural Monopoly, Federalism and Metropolitan Government |
| <b>Education and Crimes</b>                             | Spending and Educational Achievement, The Education Production Function, School Inputs: The Importance of Teachers, Innovation: Charter Schools, Spending Inequalities and Public Policy, Education in Central Cities, Crime Facts, The Rational Criminal, The Equilibrium Quantity of Crime, Legal Opportunities and Education                                                                                       |

**Teaching Methodology:**

- To deliver lectures on topics included in course outline
- To require each student to solve independent assignments on topics included in the course.

**Evaluation Criteria:**

| Evaluation Method   | Total Percentage |
|---------------------|------------------|
| Quizzes/Assignments | 25%              |
| Mid-Term Exam       | 35%              |
| Final-Term Exam     | 40%              |

**Recommended Books:**

- O'Sullivan, Arthur (2019). *Urban Economics*. 9th Edition, NY, McGraw Hill Education
- John F. McDonald and Daniel P. McMillan (2011), *Urban Economics and Real Estate: Theory and Policy*, Malden, MA, Blackwell Publishing.
- Barry Bluestone, B., Stevenson, M. H. and Russell Williams (2008), *The Urban Experience: Economics, Society and Public Policy*, Oxford, U.K., Oxford University Press.
- Brendan O'Flaherty (2005), *City Economics*, Cambridge, MA, Harvard University Press.
- John McDonald (1997), *The Fundamentals of Urban Economics*, Upper Saddle River, New Jersey: Prentice-Hall.
- Relevant material in the context of Pakistan Urban conditions should be added by the course instructor.

**Course Code: ECON-311****Title: Comparative Economic System****Credit Hours: 03****Prerequisite: Principles of Microeconomics and Principles of Macroeconomics****Course Objectives:**

This course aims to make graduate students familiar with the major concepts, theories, and debates in comparative economic systems. The course is aimed to maintain the students' representations of the complexity of the economy of global market. Comparative economic systems are often viewed as a divergence of traditional economics types: free and planned economies with a new type of mixed systems. The course covers many issues to compare legal systems, corporate governance, financial market, energetic systems, human capital, inequality, social policies, etc. Additional attention was paid to informal institutions (corruption and organized crime), which could be "normal" institutions in economic sense.

**Learning Outcomes:**

Students by the end of this course, will learn:

- about historical debates of various economics system that existed.
- how the current system is a mixture of various systems that existed historically.

#### Course Contents:

|                                       |                                                                                                                                                                                                                                                                                                                                                                                                                              |
|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Evaluation of Economic Systems</b> | Classification of Economic Systems, Historical Development Criteria for Evaluation of an Economic System. Basic Differences between Economic System.                                                                                                                                                                                                                                                                         |
| <b>Capitalism</b>                     | Definition, Historical Development, Pre-Requisites of Capitalism, Competitive and Monopoly Capitalism. Economic Institutions of Capitalism. Distinction between private and Public Property and Production, Price System, Profit Maximization/Cost Minimization and Competition. Economic Decision Making under Capitalism. Evaluation of Capitalism in USA and Europe Capitalism, Market System and Environmental Problems. |
| <b>Communism and Socialism</b>        | Meaning and Nature, Communism and Power to the Centre. Concepts of Ownership and Production. Various types of Communism, Marxism, Socialism, Economic Planning and Decision Making. Achievements and Failures, Communism / Socialism in USSR, China and Other parts of the World. Modern Trend and its failure, Convergence of Capitalism and Socialism. Present Status of Socialism.                                        |
| <b>Economic System of Islam</b>       | Distinctive Features, Basic Economic Values of Islam. Economic Institutions and their Role, Property Rights, Production and Distribution System. The Concept of Welfare State. Comparative Analysis: Islamic System, Capitalism and Communism.                                                                                                                                                                               |
| <b>Economic System in Pakistan</b>    | Brief Economic History of Pakistan. The Mixed Economy, Nationalization and its Impacts. Operations under Mixed Economy, Privatization and Liberalization in the 1990's and 2000's. Islamisation in Pakistan: Process and Problem, Achievements and Experience. Islamic Economics as Global System of Equity and Welfare.                                                                                                     |

#### Teaching Methodology:

- To deliver lectures on topics included in course outline
- To require each student to solve independent assignments on topics included in the course.

#### Evaluation Criteria:

| Evaluation Method   | Total Percentage |
|---------------------|------------------|
| Quizzes/Assignments | 25%              |



|                 |     |
|-----------------|-----|
| Mid-Term Exam   | 35% |
| Final-Term Exam | 40% |

### Recommended Books:

- Carson, Richard L. (2002). Comparative economic systems. ME Sharpe.
- Bornstein, Morris. (2005). Comparative economic systems: Models and cases. McGraw-Hill/Irwin.
- Comparative Economic System by William, Louks and William G. Whitnery (latest edition).
- Martin C. Schmelzter and James W. Hordyke (1977). Comparative Economic Systems.
- Mifflin Company, Boston (1985). Comparative Economic Systems by Gregory, Stuart, Houghton.

### Course Code: ECON-312

**Title: Managerial Economics**

**Credit Hours: 03**

**Prerequisite: Intermediate Microeconomics, Calculus-II**

### Course Objectives:

The main objective of this course is to make the students familiar with economic theory and tools essential to the analysis and solution of those problems that have significant economic consequences both for the firms and society at large. After taking this course a student will be able to suggest solutions to such business problems as optimization of output mix, input combination and profit maximization

### Learning Outcomes:

At the end of the course, students will able to learn:

- Application and real world managerial practice
- Concepts will be applied in long run for efficiency of business

### Course Contents:

|                                                           |                                                                                                                                                                                                                                                                                                                  |
|-----------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Nature, Scope and Overview of Managerial Economics</b> | The Nature and Scope of Managerial Economics, Firms and Managerial Objectives, The Managerial decision making process, Economic Optimization, Firms and Profits, Marginal Analysis, Theory of the firm. Profit maximization, Cost Minimization. Economic optimization.                                           |
| <b>Market Forces: Demand and Supply Analysis,</b>         | Demand, Supply and Equilibrium, Demand Analysis, Elasticity, use of derivative in finding elasticity of demand & supply, use of various forms of elasticity in decision making, Measuring Economic Relationships, Statistical Relations and Regression Analysis, Simple linear regression model, Multiple linear |



|                                         |                                                                                                                                                                                                                                                                                                                                                                                       |
|-----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Estimation and Forecasting</b>       | regression model estimation. Forecasting, Methods of demand forecasting. Interpretation of estimated results                                                                                                                                                                                                                                                                          |
| <b>Production and Cost</b>              | Theory of Production and cost, Production functions. Economies of scale and Diseconomies of scale. Optimization of multivariate functions, maxima, minima, point of inflection and their use in economic decision making.                                                                                                                                                             |
| <b>Market Structure</b>                 | Pricing decision under Perfect Competition, Monopolies, Monopolistic Competition and Oligopoly. Pricing Strategies for Firms in the short run and long run. Mark up pricing. Price Discrimination, Monoposony and Multiplant Monopoly and measuring Firms' Performance, Making Decisions under Noncompetitive Conditions. Game Theory: Pricing Strategies for Firm with Market Power. |
| <b>Linear Programming</b>               | Linear Programming using Microsoft Excel solver: Ingredients of Linear Programming, Assumption etc. Application of Linear Programming. Profit Maximization problems using graphical approach. Cost Minimization problems using graphical approach. The concept of dual, duality theorem, solving primal via dual and their economic applications.                                     |
| <b>Capital Budgeting and Investment</b> | Project appraisal/ capital budgeting, Investment Criteria and Decisions. Cash flow estimation, Net present value (NPV) Internal Rate of Return (IRR)                                                                                                                                                                                                                                  |

#### Teaching Methodology:

- To deliver lectures on topics included in course outline
- To require each student to solve independent assignments on topics included in the course.

#### Evaluation Criteria:

| <b>Evaluation Method</b> | <b>Total Percentage</b> |
|--------------------------|-------------------------|
| Quizzes/Assignments      | 25%                     |
| Mid-Term Exam            | 35%                     |
| Final-Term Exam          | 40%                     |

#### Recommended Books:

- Baye, Michael, Managerial Economics and Business Strategy. Sixth Edition. Boston: McGraw-Hill Irwin, latest edition.
- Mark, Hirschey. (2003). Managerial Economics, latest edition, Thomson/South-Western College Publishing.

- Peterson, Craig, H., Lewis, W. Cris and Jain, Sudhir, K. (2009). Managerial Economics, latest Edition Pearson Education.
- Salvatore, D. (2001). Managerial Economics, McGraw Hill

## Course Code: ECON-313

**Title: Introduction to Political Economy**

**Credit Hours: 03**

### Course Objectives:

This course is designed to provide students with an introduction to the field of positive political economy. Positive political economy seeks to understand and explain policy outcomes and political behavior. The approach taken is an interdisciplinary one that utilizes tools and concepts from economics to study politics and political behavior. The course will presume that political actors are rational and goal oriented. Political outcomes are then explained by the interaction between these actors within the (institutional) constraints of their environment.

### Learning Outcomes:

Students should be able to:

- Comprehend what is mean by political economy.
- Understand the basic tools to understand politics and political behaviour.

### Course Contents:

|                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                              |
|------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>The Economic Way of Thinking: An Introduction</b>                   | What is political economy? For that matter, what is politics? What is economics? Why is it important to study the two fields together? What are the large debates that occupy the minds of political economists? The political economy of peanuts and Pintos, Contrasting approaches: rationality, culture and psychology, Beyond cash: Thinking in terms of costs, benefits, and resources, Who seeks rent? |
| <b>Rational Choice: The Economics of Human (&amp; Pirate) Behavior</b> | What can economists teach political scientists, if anything? Is it possible to explain human behavior and complex political phenomena based on a few simple assumptions about how people calculate costs and benefits? Why vote? Why not? A market for those with heart (two kidneys, lungs and a liver), Organizing an anarchic society: the economics of pirate democracy, Hazardous morals.               |
| <b>The Economics of Collective Action</b>                              | This section deals with one of the central themes in political economy: collective action. How do people come together to                                                                                                                                                                                                                                                                                    |

|                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|----------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                    | make group decisions regarding rules and regulations within society? By studying the creation of property rights in the anarchic West of the 1800s, we will get a better understanding of the solutions to collective action problems. Related issues will include the dilemma of the commons (a sub-species of collective action problems), principal-agent theory and the importance of transaction costs. Public goods, public bads, common pools, and exclusive clubs, Stampede: The dilemma of the commons, Can an invisible hand have a green thumb? |
| <b>Voting and Elections</b>                        | Downsian electoral Competition, median-voter equilibria, probabilistic voting, efficient and inefficient electoral competition, electoral accountability, Institutions and Their Impact, Bureaucracy, Delegation, and Principal Agent Theory                                                                                                                                                                                                                                                                                                               |
| <b>Topics in the Political Economy of Pakistan</b> | The economics of patronage, Seven pillars of good governance, The political economy of reforms, Pakistan - Dynamics of Elitist Model, Strategies for self-preservation, A military theocracy, Different regimes, Economic development, Ethnic conflict, The role of state, Emerging economic and political issues and their linkages, Regionalism, Poverty, Governance.                                                                                                                                                                                    |

### Teaching Methodology:

- To deliver lectures on topics included in course outline
- To require each student to solve independent assignments on topics included in the course.

### Evaluation Criteria:

| Evaluation Method   | Total Percentage |
|---------------------|------------------|
| Quizzes/Assignments | 25%              |
| Mid-Term Exam       | 35%              |
| Final-Term Exam     | 40%              |

### Recommended Books:

- Winter, H. 2005. Trade-Offs: An Introduction to Economic Reasoning and Social Issues. 2nd Edition. Chicago: University of Chicago Press. Winter provides a very accessible text that deals with policy issues from a cost-benefit perspective. There are no solutions, only trade-offs.
- Leeson, P. 2009. The Invisible Hook: The Hidden Economics of Pirates. Princeton: Princeton University Press. Aaargh! What could be more fun than reading about pirates?

- K. Shepsle and M. Bonchek, *Analyzing Politics: Rationality, Behavior, and Institutions* (W.W. Norton, 1997).
- W. Riker's *Liberalism Against Populism* (Waveland Press, 1988).

#### **Additional Reading:**

- Stiglitz, J. (2016). *The great divide*. Penguin
- Stiglitz, J. (2016). *Globalization and its discontents* (new ed.). Penguin
- Latest Research articles on political economy

### **Course Code: ECON-314**

**Title: Agricultural Economics**

**Credit Hours: 03**

**Prerequisite: Intermediate Microeconomics & Intermediate Macroeconomics**

#### **Course Objectives:**

This course focuses on the understanding of the basic concepts of economics and their application in agriculture. We focus first on individual consumer and producer decision making, followed by market equilibrium and economic conditions. Next we discuss government intervention in agriculture followed by macroeconomic policy.

#### **Learning Outcomes:**

Students should be able to:

- Comprehend the importance of agriculture economics.
- Understand the basic structure and characteristics of agriculture economics.
- Have a sense of the primary theoretical models explaining the economic conditions for agriculture.

#### **Course Contents:**

|                                                             |                                                                                                                                                                                                                                                                                                   |
|-------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Introduction and Importance of Agriculture</b>           | Agricultural Revolution, Importance of Agriculture, Food Safety, Agriculture vs. Industrial development Debate. Current State of Agricultural Development. Brief Overview of Sources of Growth. Introduction to issues in Agriculture etc.                                                        |
| <b>Structure and Characteristics of Agriculture in LDCs</b> | Traditional Agriculture and Productivity. Access to non-labor resources. The Framing Environment: Natural Hazards and Economic Uncertainties. The Food problem. Farm Size and Productivity. Farm Size in Pakistan and Productivity Commercial Farms. Impact of Agricultural Reforms on Farm Size. |

|                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|----------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Role of Agriculture in Economic Development</b>                         | A framework of Analysis. Product Contribution, Market Contribution, Factor's Contribution. Foreign Exchange Contribution. Agriculture vs. Industry Debate of Development.                                                                                                                                                                                                                                                                                                                                                                |
| <b>Theory of Rent and the Concept of 'Surplus'</b>                         | Introduction. Economic Rent. The Theory of Rent. Rent and Quasi-Rent the Ricardian 'Corn Rent. The Rental Market. Agricultural Surplus. Characteristics of Landownership in Underdeveloped Agriculture and Pakistan. The Theory of Share Tenancy. Some Extensions of the Share Tenancy Model. Tenant's Status in Pakistan. Agricultural productivity in Pakistan and use of inputs. Green revolution, its origin, and impact on agriculture in Pakistan.                                                                                 |
| <b>Agriculture in Dualistic Development Models</b>                         | Introduction. The Lewis Model. The Ranis-Fei (FR) Model. The Jorgenson Model. Kelley, Williamson, Cheetham Model, Criticism: Dual Economy Models. Agricultural Adequacy; Mellor's Model.                                                                                                                                                                                                                                                                                                                                                 |
| <b>Resource Use Efficiency and Technical Change in Peasant Agriculture</b> | Efficiency of Resource Utilization. Technological Change in Agriculture. Generation of new Agricultural Technology. Factor-biased Technological Change and its Distributional Consequences. Agricultural Technical Change and Agricultural Employment: Empirical Evidences. Agricultural Resources and Technical Change in LDCs: with Special Reference to Pakistan.                                                                                                                                                                     |
| <b>Supply Response</b>                                                     | Introduction. The Cobweb Model: An Illustration. Price-Supply Response in Backward Agriculture. A simple Supply Response Model. Supply Response in the Underdeveloped Agricultural labour Market, Specifically in Pakistan. The Concept of 'Marketed Surplus': Some Methods of Estimation. Some Criticisms of Krishna's Method and the Alternative Approach of Behrman Perennial Crops and Marketed Surplus in Pakistan. Green Revolution in Pakistan and Responses. Present Status of Green Revolution in Pakistan: Empirical Analysis. |
| <b>Population and Food Supplies</b>                                        | The Classical Model. Contra-Malthusian Model. Ecological Disequilibrium. Synthesis of Population and Food Supply Theories. Malnutrition in Developing Countries like Pakistan and Remedial Manners. Choice of Planning Strategy for Agriculture Development, Agricultural Project Planning.                                                                                                                                                                                                                                              |

#### **Teaching Methodology:**

- To deliver lectures on topics included in course outline
- To require each student to solve independent assignments on topics included in the course.

#### **Evaluation Criteria:**

| Evaluation Method   | Total Percentage |
|---------------------|------------------|
| Quizzes/Assignments | 25%              |
| Mid-Term Exam       | 35%              |
| Final-Term Exam     | 40%              |

### Recommended Books:

- Johnston, Bruce, F & Fellir Kily. Agricultural and Structural Transformation, Economic Strategies in late Developing Countries.
- Khan Mahmood-ul-Hassan. (1975). The Economics of Green Revolution in Pakistan, New York.
- Shafi M. (1995). Pricing of Farm Produce in Pakistan, Objectives, Practices, and Experiences. Print Associates International, Islamabad.
- Yujiro Hayami and Vernon W. Ruttan, Agricultural Development, John Hopkin University Press, London. (Latest edition).
- Zaidi, S. Akbar (2005). Issues in Pakistan's Economy, Oxford University Press.
- Pakistan Economic Survey (various issues)

### Course Code: ECON-315

**Title: Human Resource Development**

**Credit Hours: 03**

**Prerequisite: Intermediate Microeconomics & Intermediate Macroeconomics**

### Course Objectives:

This course will help the students understand meaning and need of human resource development. It will also make them familiar with various types of human resource activities. They will also learn organizational process of human resource development.

### Learning Objectives:

The students should be able to:

- Learn about the basics of human resource and its role in an organization.
- Understand the primary responsibilities of human resource department.
- Learn about primary theoretical models explaining human resource development.

### Course Contents:

|                                                       |                                                                                                                                                                                                                                                                               |
|-------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Meaning and Need of Human Resource Development</b> | The progression towards a field of Human Resources Development. The early training programs for semiskilled and unskilled workers. The human relations movement. The establishment of training profession. Emergence of human resources development. The relationship between |
|-------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|



|                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                | human resources management and HRD training. Secondary HRM functions. Human resource development functions. Training and development, organizational development and career development. Roles and competence of an HRD professional. The HRD executive/manager. Other HRD roles and outputs for HRD professionals. Challenges to organizations and to HRD professionals.                                                                                                                                                                                                                                  |
| <b>Formal and informal organizations involved in human resource development</b>                | Definition and purpose of needs assessment. Components of person analysis. Performance appraisal in the person analysis process. Recruitment, and selection and performance management. Management promotion and performance criteria. Rewards and compensations. Training, development and employee wellbeing. Work design basing upon employee involvement. Organization development theory and concepts. Model of planned change. The role of labor unions. Types of intervention (techno structural, sociotechnical systems and organizational change). Educational, training and social institutions. |
| <b>Meaning and inter-relationship of organization structure and human resource development</b> | Introduction to organizational culture. Labor market changes and discrimination. Impact of recent immigration patterns. Adapting to demographic changes. Affirmative action programs. Cross-cultural education and training programs. Human resources development programs for culturally diverse employees. Socialization and orientation. Career development.                                                                                                                                                                                                                                            |
| <b>Meaning and Types of Human Resource Development Activities</b>                              | Defining of HRD activities. Designing HRD programs. Defining objectives of HRD intervention. Scheduling and HRD programs. Training delivery methods. On the job training (OJT) methods. Job instruction training and job rotation. Class room training approach. Discussing methods and methods of Audio/Visual media. Experiential methods. Computer based training. Arranging the physical environment. Designing E-Learning material.                                                                                                                                                                   |
| <b>Organizational Processes involved in human resource development</b>                         | Extent of management development activities. Approach to understand the job of managing authority. Importance of needs assessment in determining managerial competence. Executive education programs. Company designed courses. Corporate universities. On job experiences and designing effective management for development programs. Up gradation of educational technical and socio-political atmosphere.                                                                                                                                                                                              |



|                            |                                                                                                                                                                                                                                                                                                                                     |
|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Motivation</b>          | Multiple Intelligence Model. Two factor model of intelligence. Primary Mental abilities. Nature and Nurture Controversy. Emotional Intelligence Framework. Hygiene. Motivator factors and achievement need.                                                                                                                         |
| <b>Power and Authority</b> | Ten Myths and truths of corporate creativity. Development needs associated with the power and authority. Use of power and authority for all. The HRD executive manager and other HRD roles and outputs for HRD professionals.                                                                                                       |
| <b>Leadership</b>          | Qualities of leadership. Extent of management development activities. Describing the manager's job. Command over the hearts of all the stake holders. Management training and experiences. Company designed courses, corporate universities and on the job experiences.                                                             |
| <b>Communication</b>       | Communications for the purpose of involvement and participation concept of "soft" HRM model. Communication with renewed interests and policy initiatives supported by the government. The central concept of HRD, the individual employee. Oriented approach. Difference between the employ participation and employed involvement. |
| <b>Conflict</b>            | The concept of going on strike and lock ups. Utilization of employ participation and involvement. Damages of conflicts. Important role of the management and government.                                                                                                                                                            |
| <b>Decision Making</b>     | Investment criterion and concept of cost drivers. Consideration of operating cost curves and planning cost curves. Consideration of internal and external market conditions. Availability of the resources and the profit margin.                                                                                                   |

### Teaching Methodology:

- To deliver lectures on topics included in course outline
- To require each student to solve independent assignments on topics included in the course.

### Evaluation Criteria:

| <b>Evaluation Method</b> | <b>Total Percentage</b> |
|--------------------------|-------------------------|
| Quizzes/Assignments      | 25%                     |
| Mid-Term Exam            | 35%                     |
| Final-Term Exam          | 40%                     |

### Recommended Books:

- Reynolds, L. G. (1974). Labor Economics and labor relations (6th ed.). New Jersey: Prentice Hall.
- Ehrenber, R. G., & Smith, R. S. (1997). Modern labor economics: theory and public policy (6th ed.). New York: Addison Wesley.
- Marshall, F. R., King, A. G., & Briggs, V. M. (1980). Labor economics: wages employment and trade unionism. USA: Richard D. Irwin.
- Peter M. Blau and W. Richard Scott. (1962). Formal Organizations: A Comparative Approach, Stanford University Press.
- Charles Perrow. (1971). Organizational Analysis: A Sociological View, Tavistock Publications Ltd.
- John W. Newstrom and Keith Davis. (2002). Organizational Behaviour: Human Behavior at Work, 11th edition, New York: McGraw-Hill-Irwin.
- Randy Hodson and Teresa A. Sullivan. (2012). The Social Organization of Work, 5th edition, Cengage Learning.
- Rolf P. Lynton and Udai Pareek, (2011). Training for Development, 3rd edition, New Delhi: Kumaria Press.
- Heinz Weihrich and Harold Koontz. (2004). Management: A Global Perspective, 11th edition New York: McGraw-Hill, Inc.

### Course Code: ECON-316

**Title: Money, Banking and Financial Institutions**

**Credit Hours: 03**

### Course Objectives:

To provide students with a solid awareness about the fundamentals of banking and financial institutions.

### Learning Outcomes:

Students will be:

- Introduced to financial markets and how it intersects with economic systems.
- Learning about the monetary policy and international finance.

### Course Content:

|                                                     |                                                                                                                                                                                                                                                                                              |
|-----------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Financial Markets and Financial institutions</b> | Banking and financial markets? an overview of financial system, debt and equity markets, exchange and over-the counter markets, Financial Structure, debt and equity contracts, understanding interest rates, the behavior of interest rates, the risk and term structure of interest rates. |
|-----------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|                                                    |                                                                                                                                                                                                                                        |
|----------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                    |                                                                                                                                                                                                                                        |
| <b>An economic analysis of financial structure</b> | Basic facts about financial structure, financial development and economic growth, factors causing financial crises, dynamics of financial crises in emerging market economies. Economic analysis of financial regulations.             |
| <b>International finance and monetary policy</b>   | PICIC, IDBP, ZTBL, HBFC, NIT, SMEDA, World Bank, IMF (Brief intro and contribution to the economy), Foreign exchange market, long run and short run exchange rate behavior, comparing expected returns on domestic and foreign assets. |

### Teaching Methodology:

- To deliver lectures on topics included in course outline
- To require each student to solve independent assignments on topics included in the course.

### Evaluation Criteria:

| Evaluation Method   | Total Percentage |
|---------------------|------------------|
| Quizzes/Assignments | 25%              |
| Mid-Term Exam       | 35%              |
| Final-Term Exam     | 40%              |

### Recommended Books:

- The Economics of Money, Banking and Financial Institutions by FREDERICS S. MISHKIN (Latest Edition)
- Miller, R. L., & VanHoose, D. D. (1993). *Modern money and banking*. McGraw-Hill College.
- Luckett, D. G. (1980). *Money and banking*. McGraw-Hill Companies.

### Course Code: ECON-317

**Title: Economic Principles, Strategies and Techniques of Marketing**

**Credit Hours: 03**

### Course Objectives:

The course will introduce students to economic principles and their correlation with different marketing strategies. The course aims primarily at developing the student's ability to analyze marketing activity in industrial and organizational markets. The structure of the course is to build upon basic marketing principles, and reveal the interfaces of marketing philosophies and their linkage with socio-economic factors with

their impact on inter-firm business functions, especially channels, sales, supply chains /value chain. In this course the learners will examine the points of retreat with consumer markets, be more managerial rather than descriptive in approach, and emphasize operational aspects.

### **Learning Outcomes:**

Students by the end of the course should learn about:

- Key marketing concepts and how it interrelated with supply chain management.
- International product management and marketing techniques.

### **Course Contents:**

|                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                          |
|------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Economics</b>                                                       | Understanding fundamental economic principles and concepts that impact business operations Introduction to Contemporary Business Practices, Macroeconomics in a Global Context and World Economy.                                                                                                                                                                                                        |
| <b>Introduction to Marketing</b>                                       | Key Marketing Concepts, Scope of Marketing, Evolution of Marketing, Importance of Marketing, Marketing as an Exchange Process, Marketing as a Demand Management Tool. Marketing Cost and Price Relationship.                                                                                                                                                                                             |
| <b>Supply Chain / Value Chain Management</b>                           | Primary and Secondary activities in Supply Chain / Value Chain. Identify and Maximizing linkages within the Various Activities. Relationships & E-commerce, Managing Logistics and Supply Chain.                                                                                                                                                                                                         |
| <b>Strategic Marketing Planning Process</b>                            | The SMP Process. The BCG Matrix, GE Nine Cell Model and Porter's Generic Strategies Models of SMP.                                                                                                                                                                                                                                                                                                       |
| <b>International Marketing: The Target Markets &amp; Marketing Mix</b> | Identify Target Markets. Forecasting Market Demand, The Target Market Strategies, Key Elements of Marketing Mix, Local Vs. International Marketing, International Marketing Strategy and Planning, Organizational Structure of International Markets, Exporting, Contracting, International Corporations and their Marketing Techniques. The Promotional Program, the Key Components of Promotional Mix. |
| <b>Product Management</b>                                              | The Consumer and Industrial Products. Product Planning and Development, Product Adoption and Diffusion, Product Life Cycle, Product Positioning and Differentiation, Product Mix Strategies.                                                                                                                                                                                                             |
| <b>Marketing Cost, Finance and Cost Management</b>                     | Analysis of Cost, Marketing Cost for Products, Reasons for Higher Cost and Methods to Minimize. Abuses and Advantages of Marketing Cost Debate over Legitimacy Solution, Need for Marketing Finance. Kinds of Marketing. Financing. Significance of                                                                                                                                                      |

|                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-----------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                           | Prices, Methods of Price Determination. Key Pricing Strategies. Price wars.                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Distribution Management and Market Intermediaries</b>  | Designing Distribution Networks. Channels Conflict and Channel Power. Physical Distribution. The Wholesale Market, Retail Marketing.                                                                                                                                                                                                                                                                                                                     |
| <b>Consumer's and Buyer's Behaviours</b>                  | Consumer Motive to Marketing, Models of Consumer, Buyer Behaviour, Influences on Buyer's Behaviour, Industrial Buyer Behaviour. Decision making behaviour.                                                                                                                                                                                                                                                                                               |
| <b>Managing Marketing in Various Forms of Competition</b> | Industrial Analysis, Industry and Market Concepts of Competition, Strategies of Leaders, Challengers, Followers and Nichers. Global Marketing, Industrial Marketing, Services Marketing, Agricultural Marketing.                                                                                                                                                                                                                                         |
| <b>Managing Total Marketing Effort</b>                    | Key Organizations for Marketing, Market Control, Marketing Audit.                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Segmentation and Targeting</b>                         | Significance, Advantages of Market Segmentation, Segmentation Variables, Segmenting Industrial Markets, Effectives of Segmentation, Market Targeting, Strategic Opinion.                                                                                                                                                                                                                                                                                 |
| <b>Pricing Strategies and Sales</b>                       | Theories of Pricing and Value. Supply and Demand Model, Mark-up Pricing, Dumping, Economic Choice, Pricing and Market Orientation, Market Share, Cost based Pricing, Customer based Pricing Methods, Competitor based Pricing war. Short-run and Long-run Pricing, Predicting Future Conditions Forecasting Sales and difference Pricing Policies, Data Analysis and Sales Forecasting, Survey and Price Analysis, Profit and other Motives for Pricing. |

### Teaching Methodology:

- To deliver lectures on topics included in course outline
- To require each student to solve independent assignments on topics included in the course.

### Evaluation Criteria:

| <b>Evaluation Method</b> | <b>Total Percentage</b> |
|--------------------------|-------------------------|
| Quizzes/Assignments      | 25%                     |
| Mid-Term Exam            | 35%                     |
| Final-Term Exam          | 40%                     |

### Recommended Books:

- Jerome McCarthy. Basic Marketing 8th Edition, Prentice Hall.
- Boyd H.W. Marketing Management: Strategic Approach with a Global Orientation. 3rd Edition, latest edition.
- Evans Burman. Marketing. 4th Edition, McMillan Publishing Co., latest edition.
- R.D. Tousley & Eugene Clark. Principles of Marketing, latest edition.
- Etzel, Walker & Stanton. The Fundamentals of Marketing. 11th Edition, McGraw Hill (or latest ed.
- Kardes, Cronley & Cline. (2015). Consumer Behavior. South Western College, (ISBN: 978-1-133-58767-5); older editions are also acceptable.

## **Course Code: ECON-318**

**Title: Population Economics**

**Credit Hours: 03**

### **Course Objectives:**

This course introduces the students to the subject of demography. The course aims to educate students regarding basic demographic theories, study of demographic processes and population characteristics, demographic trends, basic methods of demographic analysis and the important demographic transitions. In the latter part of the course the focus will be on the relationship between population and issues such as urbanization, family change, population aging and health.

### **Learning Outcomes:**

Students should be able to learn about:

- Demography and the role it plays in the economic growth.

### **Course Contents:**

|                     |                                                                                                                                                                                                                         |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Introduction</b> | Understanding Demography, Importance of Demography, Sources of Demographic Data, Processes of Demography                                                                                                                |
| <b>Fertility</b>    | Understanding the Concept; Intercourse, Conception & Gestation Variables, Measuring Fertility, Period Measures of Fertility, Cohort Measures of Fertility, Fertility Trends, Fertility & Population Growth in Pakistan. |
| <b>Mortality</b>    | Components of Mortality; Lifespan, Longevity, Causes of Death, Measuring Mortality, Differentials in Mortality, Estimates of Mortality in Pakistan.                                                                     |
| <b>Migration</b>    | Definition & Measurement, Causes of Migration, Internal Vs International Migration, Individual & Social Consequences of                                                                                                 |

|                                                          |                                                                                                                                                                                                                                         |
|----------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                          | Migration, Migration Trends in Pakistan: Causes & Consequences.                                                                                                                                                                         |
| <b>Theories of Population</b>                            | Mercantilist Perspective, Malthusian Theory of Population, The Marxist Perspective, John Stuart Mill, Lujo Brentano's Theory, Emile Durkheim's Perspective, The Theory of Demographic Transition, Theory of Relative Income.            |
| <b>Population Composition: Age &amp; Sex Composition</b> | Measuring the Age Composition; Measuring the Sex Composition; Impact of Population Processes on the Age Composition; Impact of Age Composition on Population Processes; the Dynamics of Age/Sex Compositions; Age-Stratification Theory |
| <b>Population Growth &amp; Ageing</b>                    | Demographic Composition of the Elderly: Dynamic Scenarios, Biological & Social Aspects of Aging, Implications for the Society                                                                                                           |
| <b>Urbanization</b>                                      | Demographic Components of Urbanization, Internal Rural-to-Urban Migration, Impact of Population Processes, and Urbanization in Pakistan: Trends, Causes & Consequences.                                                                 |
| <b>Population Policy</b>                                 | Need of Population Policy, Influencing Population Processes, Case Studies of Pakistan.                                                                                                                                                  |

#### Teaching Methodology:

- To deliver lectures on topics included in course outline
- To require each student to solve independent assignments on topics included in the course.

#### Evaluation Criteria:

| Evaluation Method   | Total Percentage |
|---------------------|------------------|
| Quizzes/Assignments | 25%              |
| Mid-Term Exam       | 35%              |
| Final-Term Exam     | 40%              |

#### Recommended Books:

- Weeks J. R. (1989). Population: An Introduction to Concepts and Issues, Wadsworth Publishing Company, USA.
- Poston D. L. & L. F. Bouvier (2010). Population and Society: An Introduction to Demography, Cambridge University Press, UK.
- John Piggott & Alan Woodland. (2016). Handbook of the Economics of Population.



## Course Code: ECON-401

**Title: Public Sector Economics**

**Credit Hours: 03**

**Prerequisite: Intermediate Microeconomics and Intermediate Macroeconomics**

### Course Objectives:

The main objective of this course is to develop an understanding of the impact of government policies on resource allocation and equitable distribution of income. Public sector economics helps to analyze and identify the effect of public policy on various macroeconomic indicators. This course will help the students to enhance their understanding of various public policies and their effects on the economy.

### Learning Outcomes:

The market mechanism that leads to efficient resource use

- Principle of pareto efficiency and pareto improvement
- Causes and solutions of market failures and government failures
- Centralized and decentralized decision making to form structure of fiscal federalism
- The criteria that should be applied when one is judging the merit of various budget policies
- Structure of taxes, types of taxes and concept of buoyancy of tax system
- The social, political and historical forces which have shaped the present fiscal and monetary institution's tax and expenditure measures and have determined the formulation of contemporary fiscal and monetary policy

### Course Contents:

|                              |                                                                                                                                                                                                                                    |
|------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Introduction</b>          | Introduction to public sector economics, concept of public budget, Objectives of budget, Types of budget deficit, surplus and balanced budget, Distinction between public and private expenditure, Main sources of public revenue. |
| <b>Public Expenditures</b>   | Causes of public expenditure growth, Major theories of public expenditure, Wagner's Law of public expenditure, Colin Clark (Critical Limit) Hypothesis.                                                                            |
| <b>Market Failure</b>        | Externalities, Types of externalities- negative externality, socially optimal output, Positive externality, internalizing externalities, Coase theorem, Implications for income distribution.                                      |
| <b>Public Goods</b>          | Public goods excludable, public goods non excludable, Rivalrous and non rivalrous in consumption, Excludability and non-excludability, Concept of free rider, Asymmetric information, Adverse selection and moral hazard.          |
| <b>Cost Benefit Analysis</b> | Private and social costs, Types of cost and benefits, Criteria for cost benefit analysis, Project evaluation, Concepts of NPV (Net                                                                                                 |

|                                                 |                                                                                                                                                                                                                                                                                                                                                                                                        |
|-------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                 | Present Value), IRR (Internal Rate of Return) and their economic applications.                                                                                                                                                                                                                                                                                                                         |
| <b>Introduction to Taxation</b>                 | Desirable characteristics of a tax system, choosing among different tax system, Types of taxes- direct vs indirect taxes Tax incidence in competitive markets and without perfect competition, Equivalent Taxes, Factors affecting tax incidence.                                                                                                                                                      |
| <b>Taxation and Economic Efficiency</b>         | Theories of taxation- The cost theory, the benefit theory, the ability to pay theory- Subjective approach, Objective approach, Progressive, Proportional and Regressive taxes, Degressive tax, specific and advalorem tax, single and multiple taxes, Impact of tax vs incidence of a tax. Specific tax incidence, Differential tax incidence, balanced budget incidence. Concept of optimal taxation. |
| <b>Deficit Financing and Public Enterprises</b> | Consequences of Government deficits, improving the budgetary process, Managerial problems of public enterprises in Pakistan, Theories of pricing in public sector enterprises- no profit no loss theory, Managerial cost of production theory, Average cost of production theory, Theory of making profit.                                                                                             |
| <b>Policies</b>                                 | Social Insurance and Security, Income Distribution and Welfare Reforms State and Local Public Economics: Overview of Fiscal Policy in Pakistan.                                                                                                                                                                                                                                                        |

#### Teaching Methodology:

- To deliver lectures on topics included in course outline
- To require each student to solve independent assignments on topics included in the course.

#### Evaluation Criteria:

| Evaluation Method   | Total Percentage |
|---------------------|------------------|
| Quizzes/Assignments | 25%              |
| Mid-Term Exam       | 35%              |
| Final-Term Exam     | 40%              |

#### Recommended Books:

- Stiglitz E. Economics of The public sector, W.W. Norton & Co. (Latest Edition)
- Ted Gayer and Harvey S. Rosen. (2010). Public Finance- 10th edition.
- Ronald C. Fisher, (2006). State and local public finance-3rd edition.
- Govt of Pakistan, Economic Survey, yearly publication, Ministry of Finance, Economic Advisory Wing, Islamabad
- Reports of State Bank of Pakistan
- Reports of World Bank

**Course Code: ECON-402****Title: Growth Theories and Models****Credit Hours: 03****Prerequisite: Advanced Macroeconomics, Development Economics-II****Course Objectives:**

This course aims at developing understanding of economic growth models and theories at graduate level. The goal of this course is to comprehend the foundations and principles to make countries successful in economic terms. It focuses on both the historical experience of countries that are currently rich and are in the process of catching –up.

**Learning Outcomes:**

On completion students would have:

- An enhanced knowledge of development related economic topics
- An improved awareness of the causes, consequences and possible remedies for serious third world problem
- An analytical appreciation of policy choices employed by developing country governments
- Exposure to development issues and improved ability to articulate the problems in light of economic theory and empirical evidence

**Course Contents:**

|                                       |                                                                                                                                                                                                                                                              |
|---------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Introduction and Overview</b>      | This section includes discussions on the facts about economic growth, differences in the level of income among countries, differences in the rate of income growth, growth before and after 19th century, the production function and the lessons from data. |
| <b>Factors of Production</b>          |                                                                                                                                                                                                                                                              |
| <b>Physical Capital</b>               | The nature of capital, capital's role in production, the Solow growth model, the relationship between investment and saving, and exogenous versus endogenous factors.                                                                                        |
| <b>Population and Economic Growth</b> | Population and output in the long-run, the Malthusian model, population growth in the Solow model, demographic transitions, future population trends, and the economic consequences of demographic change.                                                   |
| <b>Human Capital</b>                  | Human capital in the form of health, human capital in the form of education, modeling the interaction of health and incomes, changes in the level of education, education and variation in income across countries.                                          |
| <b>Productivity and Technology</b>    | Measuring productivity, differences in productivity and growth rates of productivity among countries, the role of Technology in growth, the nature, transfer and determinants of technology, research and development, patents, modeling the relationship    |

|                                                 |                                                                                                                                                                                                                                                                                                                 |
|-------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                 | between Technology creation and growth, barriers to technology transfers, the cutting edge technology, the industrial revolution, the technology production function, differential technological progress, efficiency, differences in efficiencies, types of inefficiencies, misallocation of factors.          |
| <b>Growth in the Open Economy</b>               | Autarky versus openness, Globalization, the effect of openness on economic growth, effects of geographical barriers, openness and factor accumulation, growth with capital mobility, openness and productivity, trade as a form of technology, openness and technological progress, and opposition to openness. |
| <b>Fundamentals behind Economic Growth</b>      |                                                                                                                                                                                                                                                                                                                 |
| <b>Government</b>                               | Government's role in economy, governance and rule of law, taxation, efficiency and the size of government, planning and policies, corruption and Kleptocracy, democracy and economic growth, causation from government quality to income.                                                                       |
| <b>Income Inequality</b>                        | Facts about income equality, the Kuznets Hypothesis, effects of income inequality on the accumulation of physical and human capital, and economic growth.                                                                                                                                                       |
| <b>Culture</b>                                  | The effect of culture on economic growth, openness to new ideas, hard work, saving behaviour, trust, social capital, cultural change, determinants of cooperation.                                                                                                                                              |
| <b>Geography, Climate and Natural Resources</b> | Location, trade and growth, geographic concentration and spillovers, climate and productivity, climate and disease, natural resource endowment and economic growth, the curse of natural resources, incorporating natural resources into the analysis of economic growth.                                       |

### Teaching Methodology:

- To deliver lectures on topics included in course outline
- To require each student to solve independent assignments on topics included in the course.
- To read publish paper on the topics included in the course and present on one of the topics in class.

### Evaluation Criteria:

| <b>Evaluation Method</b> | <b>Total Percentage</b> |
|--------------------------|-------------------------|
| Quizzes/Assignments      | 25%                     |
| Mid-Term Exam            | 35%                     |
| Final-Term Exam          | 40%                     |

**Recommended Books:**

- David N. Weil, Economic Growth, (2012). Third Edition. Pearson Addison-Wesley.
- Charles I. Jones and Dietrich Vollrath. (2013). Introduction to Economic Growth, Third edition. W. W. Norton & Company.
- Elhanan Helpman, The Mystery of Economic Growth, (2010). Belknap, Press of Harvard University Press.
- William R. Easterly. (2008). The Elusive Quest for Growth: Economists' Adventures and Misadventures in the Tropics. The MIT Press

**Course Code: ECON-403****Title: Applied Economics****Credit Hours: 03****Prerequisite: Advanced Microeconomics, Advanced Macroeconomics, Mathematical Economics-I and Econometrics-II****Course Objectives:**

This course is designed for senior students having gone through theory of Macroeconomics, Microeconomics, Statistics, Mathematics and Econometrics. The important areas will be emphasized in this course are reading articles, data analysis/estimation and writing ability. The students will also be taught how to build economic model and estimate them. The software programs such as EVIEWS, SPSS and STATA are recommended for this course. This course is enormously demanding and students will have to make efforts and hard work. Consistency and seriousness will be of paramount importance for performing well in this course. At the end of the course, the students are expected to independently write a research paper on their own topic of interest.

**Learning Outcomes:**

Students will be learning about:

- the significant sources of secondary data
- cleaning and using the data for an empirical analysis
- various methods of creating variables, especially indexes.

**Course Contents:**

|                                          |                                                                                                                                                                                                                         |
|------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Review of Basic Research concepts</b> | Defining a problem, identifying a research question, conducting literature review, building a theoretical and econometric model, defining dependent, explanatory and control variables and collection of secondary data |
| <b>Introduction to Survey Analysis</b>   | Structure and methodology of surveys, Data management of survey data, cleaning the data, Important surveys in Pakistan: Pakistan Social and Living Standards Measurement (PSLM),                                        |

|                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|---------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                             | Household Integrated Economic Survey (HIES), Multiple Indicator Cluster Survey (MICS), Labor Force Survey (LFS), Demographic and Health Survey (DHS)                                                                                                                                                                                                                                                                                    |
| <b>Time series Database</b>                 | Structure of databases, variables availability, definitions of variables, Structure of the following databases: World Development Indicators (WDI), International Financial Statistic's (IFS), Global Financial Development (GFD), Data access from State bank of Pakistan and Pakistan Bureau of Statics                                                                                                                               |
| <b>Techniques for handling missing data</b> | Types of Missing Data: Missing completely at random, Missing at Random, Missing not at random; Techniques for handling cross-section/survey missing data, Techniques for handling timeseries missing data, Imputation and multiple imputation                                                                                                                                                                                           |
| <b>Methods of Index Construction</b>        | Principal Component Analysis, Exploratory factor analysis, Human Development Index method. Publication and Resources, Abstract, Introduction, Literature review, Methods and data collection, Findings and interpretation, Conclusion and suggestions, References, Introduction to journal databases like Google scholar, science direct etc., JEL Classification, Categories of Journals by HEC: W, X Y, Z, Plagiarism and similarity. |

### Teaching Methodology:

- To deliver lectures on topics included in course outline
- To require each student to write an empirical research paper independently.

### Evaluation Criteria:

| Evaluation Method   | Total Percentage |
|---------------------|------------------|
| Quizzes/Assignments | 25%              |
| Mid-Term Exam       | 35%              |
| Final-Term Exam     | 40%              |

### Recommended Books:

- Kothari, C. R (2008). Research methodology: Methods and techniques. New Delhi, New Age International Publishers.
- Cameron, A. Colin, and Pravin K. Trivedi. Microeconometrics Using Stata, Second Edition, College Station, TX: Stata Press, 2010.
- Saunders, M. (2005). Research methods for business studies. Singapore, Pearson Education
- Monette D. R., Sullivan, T.J., & Dejong, C.R. (2010). Applied social research: A tool for the human services. Belmont, CA: Linda Schreiber-Ganster.



**Course Code: ECON-404****Course Title: International Trade****Credit Hours: 3****Pre-requisite(s): Intermediate Microeconomics and Intermediate Macroeconomics****Course Objectives:**

The main objective of the course is to offer a detailed introduction to theory and practice of international trade and of trade-related policies. It focuses on analysing the gains from trade, the changing patterns of trade, the income distributional consequences of liberalizing foreign trade, and the reasons for and consequences of trade policies. The course relies predominantly on a standard collection international trade models to understand the motivations behind modern trade policies.

**Learning Outcomes:**

By the end of the course, students should be able to

- Analyze causes and effects of exchange rate variations and apply that knowledge to the analysis of current issues in international finance
- Understand the policies that government may use to address internal and external imbalance
- Demonstrate the ability to think critically and comment on current debates on international policy

**Course Contents:**

|                                     |                                                                                                                                                                                                                                                              |
|-------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Introduction</b>                 | Why We Study International Trade? Basis of Trade, Gains from Trade. Absolute Advantage: Adam Smith, Trade Based on Comparative Advantage: David Ricardo, Comparative Advantage and Opportunity Costs and Free Trade Equilibrium. Disagreement on Free Trade. |
| <b>Modeling International Trade</b> | The Production Frontier with Increasing Costs, Community Indifference Curves, The Marginal Rate of Substitution, Equilibrium in Isolation, The Basis for and Gains from Trade with Increasing Costs, Trade based on Differences in Tastes.                   |
| <b>Equilibrium in Trade models</b>  | The Equilibrium-Relative Commodity Price with Trade-Partial Equilibrium Analysis, Offer Curves, General Equilibrium Analysis and Relationship Between General and Partial Equilibrium Analysis and Terms of Trade                                            |



|                                   |                                                                                                                                                                                                                                                                                                                                                                        |
|-----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Heckscher-Ohlin Theory</b>     | Assumption of the Theory, Factor Intensity, Factor Abundance and The Shape of Production Frontier, Factor Endowments and the Heckscher-Ohlin Theory. Factor price equalization. Factor Intensity Reversal                                                                                                                                                              |
| <b>International Trade Policy</b> | Trade as engine of growth, the terms of growth and economic development, export instability, import substitution.<br>Tariff and small country, effect of Tariff, price and demand for imports, welfare impacts, impacts of tariff on government revenue. TOT of production and welfare. Impacts of tariff on domestic and world welfare, tariff and the large country. |
| <b>Tariffs and Restrictions</b>   | Stolper-Samuelson theorem, The Optimum Tariff and retaliation, tariff and distribution of income. Import Quotas, Other Non-Tariff, Growth of factors of production, Rybczynski, growth and trade, Immiserizing Growth                                                                                                                                                  |
| <b>Further Topics in Trade</b>    | New trade theories and economies of scale, trade based on product differentiation, measuring intra- industry trade, technology gap and product cycle model, cost of transportation and environment standard                                                                                                                                                            |

#### Teaching Methodology:

- To deliver lectures on topics included in course outline
- To require each student to solve independent assignments on topics included in the course.

#### Evaluation Criteria:

| Evaluation Method   | Total Percentage |
|---------------------|------------------|
| Quizzes/Assignments | 25%              |
| Mid-Term Exam       | 35%              |
| Final-Term Exam     | 40%              |

#### Recommended Books:

- International Economics: Theory and Policy by Paul R. Krugman and Maurice Obstfeld, Latest edition.
- International Economics by Donomick Salvatore, Latest edition.
- International Trade (3rd Edition) by Robert C. Feenstra and Alan M. Taylor, Worth Publishers.

## Course Code: ECON-405

**Title: Econometrics-II**

**Credit Hours: 03**

### Course Objectives:

This course follows up the data analysis and data estimation techniques included in Econometrics I. The major objective of this course is to enable the student for competing in a job market where positive analysis is increasingly becoming subject to highly intensive and extensive analytical formulations, largely owing to the unprecedented and remarkable development in information technology. this course is designed for senior undergraduates more inclined towards quantitative studies. The objective is to enable the students dive deep into complex problems of the real world economic life. The students have to learn certain computer packages like, E-views and Stata besides Excel.

### Lab work

The students are required to devote at least two hours per week to computer laboratory. They have to attend classes to learn various Econometrics software for practical application of Econometric models they have learnt so far.

### Learning Outcomes:

On successful completion of this course, students will be able to:

- Research with econometrics
- Explain econometrics concepts and results intuitively
- Derive econometric results mathematically

### Course Contents:

|                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Heteroskedasticity</b>  | Meaning of Heteroskedasticity, The nature of the problem with reference to economic theory, Cross-section data and the problem of non-constant variances, Consequences for OLS estimators, Detection of the problem and remedial measures in brief, Introduction to the Generalized Least-Squares model (GLS).                                                                                                                                             |
| <b>Model Specification</b> | Model selection criteria, Types of specification errors, Consequences of model, specification errors, Tests of specification errors, Errors of measurement, Model selection criteria, Endogeneity: where X is not fixed in repeated sampling, Nature of Endogeneity, OLS estimation in presence of Endogeneity, Detection of Endogeneity, BLUE estimator in the presence of Endogeneity, Consequences of Endogeneity in OLS estimation, Remedial measures. |

|                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|--------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Simultaneous Equation Models &amp; Estimation Methods</b> | Simultaneous equation models, Nature of simultaneous equations, Examples of simultaneous equation models from economic theory, Inconsistency of OLS estimators, Identification problem, Notations and definitions, Unidentified, exactly identified and over identified, Rules for identification, Simultaneous equation approaches to estimation, Method of indirect least squares (ILS), Method of two stage least squares (2SLS), Instrumental Variable approach to 2SLS. |
| <b>Time Series Econometrics</b>                              | Concept of Stationarity, Tests of Stationarity, Unit Root test, Transforming Non-stationary Time Series, ARMA and ARIMA Models, Comparison of forecast based on ARIMA and regression models, Co-integration and Error Correction Mechanism (ECM),                                                                                                                                                                                                                            |
| <b>Autocorrelation</b>                                       | Autocorrelation and its causes, Time-series data and emergence of the problem with reference to economic theory, Serial Correlation, The AR (1) process, Consequences of Autocorrelation for OLS estimators, Detection of the problem and remedial measures                                                                                                                                                                                                                  |

#### Teaching Methodology:

- To deliver lectures on topics included in course outline
- To require each student to solve independent assignments on topics included in the course and lab work done during the course.

#### Evaluation Criteria:

| Evaluation Method   | Total Percentage |
|---------------------|------------------|
| Quizzes/Assignments | 25%              |
| Mid-Term Exam       | 35%              |
| Final-Term Exam     | 40%              |

#### Recommended Books:

- Stock H. J. and Watson M. W. (2003), Introduction to Econometrics, India: Pearson Education. Latest edition
- Gujarati, D. Porter, D. (2009). Basic Econometrics, McGraw-Hill Company. 5th edition
- Jeffrey M. Wooldridge J. M., (2001). Econometric Analysis of Cross Section and Panel Data", The MIT Press, Latest edition.
- Johnston, J. & John D. (1997). Econometric Methods. The McGraw Hill Companies, Inc, Singapore. Latest edition/
- Greene W. H (latest edition), Econometrics Analysis, Pearson Education, Inc.

**Course Code: ECON-406****Title: Development Economics-II****Credit Hours: 03****Prerequisite: Development Economics-I****Course Objectives:**

This is the second course in the development economics stream. This course will primarily deal with the major sectors in and the macroeconomics of development. A predominant portion of the course covers the essential theoretical and historical elements of development and international economics. Some specific issues currently debated in the development economics field will also be discussed.

**Learning Outcomes:****Course Contents:**

|                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Agriculture and Development</b>     | Agricultural Growth: Past Progress and Current Challenges, The Structure of Agrarian Systems in the Developing World, The Important Role of Women, The Transition from Peasant Subsistence to Specialized Commercial Farming, The Economics of Sharecropping and Interlocking Factor Markets, The Transition to Mixed or Diversified Farming, From Divergence to Specialization: Modern Commercial Farming.                      |
| <b>Industry and Development</b>        |                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>The Environment and Development</b> | Environment and Development: The Basic Issues, Rural Development and the Environment: A Tale of Two Villages, Global Warming and Climate Change: Scope, Mitigation, and Adaptation, Economic Models of Environment Issues, Urban Development and the Environment, Policy Options in Developing and Developed Countries.                                                                                                          |
| <b>Trade Policies</b>                  | Five Basic Questions about Trade and Development, Demand Elasticities and Export Earnings Instability, The Terms of Trade and the Prebisch-Singer Hypothesis, The Traditional Theory of International Trade (Comparative Advantage, Relative Factor Endowments and International Specialization: The Neoclassical Model, Trade Theory and Development: The Traditional Arguments), Traditional Trade Strategies for Development: |

|                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|--------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                    | Export Promotion versus Import Substitution, The Industrialization Strategy Approach to Export Policy.                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Balance of Payments, Debt &amp; Macroeconomic Stabilization</b> | The Balance of Payments Account, The Issue of Payments Deficits, Accumulation of Debt and Emergence of the Debt Crisis, Attempts at Alleviation: Macroeconomic Instability, Classic IMF Stabilization Policies, and Their Critics, “Odious Debt” and Its Prevention, Resolution of 1980s–1990s Debt Crises and Continued Vulnerabilities, The Global Financial Crisis and the Developing Countries.                                                                                                                    |
| <b>Foreign Finance, Investment, and Aid</b>                        | The International Flow of Financial Resources, Private Foreign Direct Investment and the Multinational, Private Foreign Investment: Some Pros and Cons for Development, Private Portfolio Investment: Benefits and Risks, The Role and Growth of Remittances, Foreign Aid: The Development Assistance Debate, Conceptual and Measurement Problems, Amounts and Allocations: Public Aid, Why Donors Give Aid, Why Recipient Countries Accept Aid, The Role of Nongovernmental Organizations in Aid, The Effects of Aid. |

### Teaching Methodology:

- To deliver lectures on topics included in course outline
- To require each student to solve independent assignments on topics included in the course.

### Evaluation Criteria:

| <b>Evaluation Method</b> | <b>Total Percentage</b> |
|--------------------------|-------------------------|
| Quizzes/Assignments      | 25%                     |
| Mid-Term Exam            | 35%                     |
| Final-Term Exam          | 40%                     |

### Recommended Books:

- Todaro, Michael P. and Stephen C. Smith, Economic Development, 12th Edition, Pearson 2015.
- Perkins Dwight H., Steven Radelet, and David L. Lindauer, Economics of Development, 7th Edition, New York, W. W. Norton, 2006.
- Hirshman A. O., (1960). Strategy of Economic Development, Yale University, Press.
- Jones H. G., An Introduction to Modern Theories of Economic Growth, (Latest Edition), McGraw Hills.
- Debraj Ray, Development Economics.

- David N. Weil, Development Economics, (Latest Edition).

Research Articles (Teach at least 2 research articles from given list or of instructor's choice)

- Kniivilä, M., 2007. Industrial development and economic growth: Implications for poverty reduction and income inequality. *Industrial development for the 21st century: Sustainable development perspectives*, 1(3), pp.295-333.  
<http://132.248.45.5/cegaDEMEX/DOCS/nudesalnddevsxxi.pdf#page=303>
- 3 Classical Trade Theories <https://www.economicdiscussion.net/international-trade/3-classical-trade-theories-discussed/4239>
- Overseas Development Institute (1995). "Poor Country Debt: A Never Ending Story", *Briefing Paper No.1*. <https://cdn.odi.org/media/documents/2650.pdf>
- Foreign Aid: Good or Bad?  
<https://www.imf.org/en/News/Articles/2015/09/28/04/53/soint090910a>

## Course Code: ECON-407

**Title: Major Issues in Pakistan Economy**

**Credit Hours: 03**

**Prerequisite: Intermediate Microeconomics and Intermediate Macroeconomics**

### Course Objectives:

This course is designed for the students to understand composition of Pakistan Economy and its management during different government regimes. They will also learn issues related to agriculture, manufacturing, and services sectors. They will also be familiarized with various issues like unemployment, inflation, poverty, and income distribution.

### Learning Outcomes:

On the completion of this course, students would be

- Well aware of the strengths and weaknesses of different sectors of Pakistan's economy
- Able to draw comparative analysis of Pakistan's Economy with the regional economics and evaluate where we stand

### Course Contents:

|                                     |                                                                                                                                                                                                                                                                    |
|-------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Overview of Pakistan Economy</b> | Development Experience, Approaches, Policies, and Outcomes. Identification of Issues: The era of 1950s, 1960s, 1970s, 1980s, 1990s and 2000s. Structural Change and Sources of Growth. Emergence of economic issues overtime with limited development in Pakistan. |
|-------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                       |
|------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Development Planning and Resource Mobilization</b>            | Major Targets of Develop Plans and Emerging Issues; Neglects and Successes: Mobilization of Domestic Resources; Shortages, Deficits and Role of Foreign Aid. Land Reforms and Its Impacts, the Role of Green Revolution and its Impacts: Present Status.                                                                                                                                                              |
| <b>Agriculture and Industrial Development: Emerging Issues</b>   | Agricultural Development Policies and Priorities, Agricultural Adequacy, Agricultural Price Policy and Income Tax. Sectoral Terms of Trade. Industrial Development Policies and Strategies. Development of Large and Small Industries. Value Added: Manufacturing Goods vs. Primary Goods Production. Agriculture vs. Industry: Development Debate.                                                                   |
| <b>Sectoral Development, Employment Pattern and Unemployment</b> | Sectoral Priorities and Development Issues. Human Resource Development and Emerging Issues: Population Growth, Labour Force Participation Rate and Employment Pattern, Unemployment and Underemployment, Good Governance, Social Action Plan and its Impact. Role of Institution in Development. Social Sectors development vs. High Return Sectors: Growth trade off.                                                |
| <b>International Debt and Dependency</b>                         | Concepts of Foreign Aid and Debt. Borrowing vs. Domestic Reserve Mobilization (failure). Size of Foreign Debt, Debt Saving and its Impacts. Strategies to combat with High Debt: Saving Policy, Foreign Trade Promotion, Cutting non-development Expenditures, Rescheduling and its Impacts. Debt Management in Pakistan and its Impacts. Debt Modeling and Future Implications.                                      |
| <b>Poverty and Income Distribution</b>                           | Pattern of Income Distribution: Rural and Urban. How to Combat Poverty: Growth Strategy, Basic Needs, Labour Intensive Investment: Education / Training etc. and Social Action Plan (SAP), its Role and Critical Review, Evasion of Policies / Strategies to Combat Poverty and Improving Income Distribution: Critical Evaluation. Neglect of Human Resource Development. Child Labour. Factors Productivity Issues. |
| <b>Inflation, Foreign Trade Deficit and Emerging Issues</b>      | Sources of Inflation in Pakistan. Policies to Combat Inflation and their Impacts. Trade Performance, Instability and its Impacts. Policies to Combat Deficit and Trade Instability. WTO and its Impacts. Reforms and Further Needs. Expected Impacts of WTO and Challenges. Terms of Trade Issues, Market Access and Health Related Rates.                                                                            |

### Teaching Methodology:

- To deliver lectures on topics included in course outline



- To require each student to solve independent assignments on topics included in the course.

#### **Evaluation Criteria:**

| <b>Evaluation Method</b> | <b>Total Percentage</b> |
|--------------------------|-------------------------|
| Quizzes/Assignments      | 25%                     |
| Mid-Term Exam            | 35%                     |
| Final-Term Exam          | 40%                     |

#### **Recommended Books:**

- S. Akbar Zaidi, (2011). Issues in Pakistan's Economy, 2nd edition. Oxford University Press.
- Shahrukh Rafi Khan. (2004). Fifty Years of Pakistan's Economy: Traditional Topics and Contemporary Concerns, Karachi: Oxford University Press.
- Mahboob ul Haq, (1989). A Profile of Poverty in Pakistan, Oxford University Press for the Centre for Human Development (MHCHD).
- Mahboob ul Haq. (1999). Human Development in South Asia: The Crisis of Governance, Karachi: Oxford University Press for the Human Development Centre.
- World Bank, World Development Reports.

#### **Course Code: ECON-408**

**Title: Institutional Economics**

**Credit Hours: 03**

**Prerequisite: Intermediate Microeconomics, Intermediate Macroeconomics and Development Economics-I**

#### **Course Objectives:**

Almost all economic activities are moderated by formal or informal—norms, mores and conventions—institutions. At the same time, research emanating out of international financial institutions (IFIs) like the World Bank also highlight the crucial importance of sound institutions for economic growth, especially for emerging economies. Yet, there is no consensus on how to approach fundamental questions about institutions: what are institutions; from where do they come; and how do they change? In this course, we try to analyze these questions as well as look at some applied challenges like designing appropriate—efficient and fair—institutions for the global Commons.

#### **Learning Outcomes:**

By the completion of the course, students should be able to:

- understand what institutions are

- how they influence the economic decision making
- primary theoretical models explaining the characteristics of institutions

### Course Contents:

|                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|----------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>An Introduction to Institutional Economics:</b> | The concept of institution: attempts at definition. Institutions and organizations. Functions of social institutions. Interaction situations and the types of norms: prisoners' dilemma-type situation; co-ordination situation; inequality situation. Enforcement characteristics. Institutional structure of a society. Formal and informal institutions. Sanctions for disobeying norms (self-enforcing sanctions, guilt, shame, informational sanctions, bilateral costly sanctions, multilateral costly sanctions). Conditions of norms' effectiveness. Interaction of formal and informal institutions. The limits of imitations of institutions from best-performing countries. The problems of their enforceability. The New Institutional Economics and modern institutionalism. Old institutional economics. |
| <b>Property rights</b>                             | Property rights in different legal traditions (common law and civil law traditions). The property rights approach: some basic concepts. Specification of property rights, the bundle of rights, partitioning of property rights, attenuation of property rights. Assigning of property rights: the internalization of externalities. The Coase Theorem. Critic of Coase (dynamic effects of alternative legal rules, wealth effect, distributional effects, strategic behavior and the problem of holding-out, endowment effect, sociological critic, unrealistic assumption about zero transaction costs). Communal property. Optimal group size. Private property. Moral and economic aspects of private property.                                                                                                   |
| <b>Contracts</b>                                   | The definition of a contract. Legal and economic approach to contracts. Freedom of contract. Bounded rationality and contractual incompleteness. Asymmetric information (hidden characteristics, hidden information/ hidden action, hidden intentions) and opportunistic behavior. Adverse selection and the closing of markets. Signalling, screening and self-selection. Asset plasticity and moral hazard. Principal-agent problem and agency costs. A simple principle-agent experiment in the classroom. Controlling and preventing moral hazard (controlling the agent, incentive contracts, bonding, do-it-yourself method).                                                                                                                                                                                    |
| <b>The new institutional theory of the firm</b>    | Neoclassical theory of the firm. Explanations of the firm in the new institutional theory (F. Knight, R. Coase, A. Alchian and H. Demsetz, O. Williamson, O. Hart). The market and the firm. Comparative analyses of the alternative coordination forms.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

|                                                   |                                                                                                                                                                                                                                 |
|---------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                   | Internal market and influence costs. The boundaries of the firm. Ownership structure of the firm. Separation of ownership and control in the open corporation.                                                                  |
| <b>The new institutional theory of the state:</b> | Social mechanisms for constraining open access. Contractual theories of the state (Locke, Rousseau), Hobbes predatory theory of the state. North's model of the state. The regulatory role of the state in the Russian economy. |

### Teaching Methodology:

- To deliver lectures on topics included in course outline
- To require each student to solve independent assignments on topics included in the course.

### Evaluation Criteria:

| Evaluation Method   | Total Percentage |
|---------------------|------------------|
| Quizzes/Assignments | 25%              |
| Mid-Term Exam       | 35%              |
| Final-Term Exam     | 40%              |

### Recommended Books:

- Elinor Ostrom. Understanding Institutional Diversity. latest Edition, Princeton University Press.
- Marc Tool. Institutional Economics Theory, Method, Policy. Latest Edition.
- North D. (1990) Institutions, Institutional Change and Economic Performance. Cambridge University Press.
- Tullock G. Rent-seeking. (1987) In: The New Palgrave: A Dictionary of Economics. - L., Macmillan.

### Recommended Research Articles:

- Wilber, C. K. and Harrison, R. S. 1978. "The Methodological Basis of Institutional Economics: Pattern Model, Storytelling, and Holism." Journal of Economic Issues 12 (March): 61 - 89.
- Rutherford, Malcolm 2001. "Institutional Economics: Then and Now." The Journal of Economic Perspectives 15.3 173-194
- Dugger, William. 1990 "The New Institutionalism: New But Not Institutionalism" Journal of Economic Issues, 24.2: 423-431
- North, Douglass 1991. "Institutions" The Journal of Economic Perspectives, 5.1: 97- 112.
- North, Douglass and Barry Weingast: The Evolution of Modern Institutions of Growth. In AEN: 129-164
- Afzal, A. (2017). "What does the Glorious Revolution really tell us about Economic Institutions?" IJPEE.

## Course Code: ECON-409

**Title: Health Economics**

**Credit Hours: 03**

### Course Objectives:

This is a course on the economics of health and health care, with an emphasis on the Pakistan's health care system. The course examines the demand for health and health care, the nature of health care as a commodity and the behaviour and organization of health care providers, health care financing and insurance, health care funding and payment of providers, innovation in health care (pharmaceuticals) and methods for evaluating health care programs and interventions, and UHC Particular attention will be given to current public policy issues for which economic analysis has potential to contribute constructively.

### Learning Outcomes:

The students should be able to:

- Understand the concept of health care systems
- Understand the supply and demand mechanisms of health systems
- Role of innovations in today's health system

### Course Contents:

|                                          |                                                                                                                                                                                                                                                                                                                                                                                    |
|------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Introduction</b>                      | What is health economics?, Normative issues in health economics                                                                                                                                                                                                                                                                                                                    |
| <b>Demand for Health Care</b>            | The health capital model and the demand for health (includes demand for bads such as smoking or junk food, and addictions as seen by economists), Demand for health care as a derived demand. Price and income elasticities of demand for health care (the Rand Health Insurance Experiment).                                                                                      |
| <b>Supply of health care</b>             | Health care professionals and institutions, their objectives and the environment in which they work (economic perspectives). Physician agency. What can we learn from geographic variations in health care utilization? What do we know about supplier-induced demand?                                                                                                             |
| <b>Third-party payers in health care</b> | The demand for health insurance and welfare consequences of moral hazard, Competition on the health insurance market and adverse selection as a market failure. Reclassification risk (also known as premium risk) in competitive markets as another potential source of market failure, Paying health care providers (funding of health care) and insurers as statistical buyers. |

|                                                   |                                                                                                                                                                                                                                                  |
|---------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Innovation and cost-control in health care</b> | The supply of R&D in health (special emphasis on pharmaceuticals), Evaluating innovation (methods of evaluation and technology assessment in health care), How much innovation do we want? The Value of Life approach and equity considerations. |
|---------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### Teaching Methodology:

- To deliver lectures on topics included in course outline
- To require each student to solve independent assignments on topics included in the course.

### Evaluation Criteria:

| Evaluation Method   | Total Percentage |
|---------------------|------------------|
| Quizzes/Assignments | 25%              |
| Mid-Term Exam       | 35%              |
| Final-Term Exam     | 40%              |

### Recommended Books:

- Jeremiah Hurley, 2010, Health Economics,
- Sherman Folland, Allen Goodman and Miron Stano, The Economics of Health and Health Care, several editions.
- Hurley's textbook Health Economics
- Feldstein, Paul J. 2011. Health Care Economics, 7th Edition. Cengage Learning.
- Henderson, James W. 2014. Health Economics and Policy, 6th Edition. Cengage Learning.
- Feldstein, Paul J. 2011. Health Policy Issues: An Economic Perspective, 5th Edition.
- Bodenheimer, Thomas, and Kevin Grumbach. 2012. Understanding Health Policy: A Clinical Approach, 6th Edition. McGraw Hill Lange.

### Course Code: ECON-410

**Title: Project Appraisal and Investment Analysis**

**Credit Hours: 03**

**Prerequisite: Public Sector Economics**

### Course Objectives:

This course aims to cover the working of public sector, that is, how the projects are evaluated, how do governments decide which project is worth spending on. In addition, how do governments arrange finances and conduct investment analysis.

### Learning Outcomes:

At the end of this course, students will be able to

- Analyze different phases of project cycles
- Critically evaluate various aspects of agriculture projects
- Evaluate different types of projects on the basis of discounted measures

#### Course Contents:

|                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|----------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Introduction</b>                                                        | Meaning of Project Appraisal, usefulness and significance, Aspect of project appraisal with a special focus on economic versus financial analysis, Contours and Dimensions of a project and its essential ingredients, Project Vs. Plans. Project Cycle.                                                                                                                                                                                                                                                 |
| <b>Identify Project Benefits and Costs</b>                                 | Objectives of Costs and Benefits, The Incremental / Net Benefits and “With” and “Without” Comparisons. Categories of Costs and Benefits.                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Pricing Project Costs and Benefits</b>                                  | Prices reflecting values, Market price and financial analysis, finding market prices and related problems, Change in relative prices and inflation: Shadow Prices and economic analysis, Removal of market price distortions in traded and non-traded goods. Premium on foreign exchange through use of Shadow exchange rate and conversion factor approach                                                                                                                                              |
| <b>Comparing Project Costs and Benefits (Measuring of a Project Worth)</b> | Undiscounted measures of project worth, Discounted measures of NPW, IRR, BCR, Net benefit – Investment Ratio, Comparative assessment of discounted measures. Social vs. Economic benefits and selection of projects                                                                                                                                                                                                                                                                                      |
| <b>Applying Discounted Measures of Project Worth</b>                       | Sensitivity analysis, switching value, Choice among mutually exclusive alternatives, Domestic resource cost of foreign exchange earning / saving.                                                                                                                                                                                                                                                                                                                                                        |
| <b>Cost Effectiveness Analysis</b>                                         | Areas and elements of analysis, Methods of analysis: Constant Effect Method, Constant Cost Method, Measuring of cost effectiveness, Present Worth, Annual Worth, Cross over discount rate, Limitation of analysis.                                                                                                                                                                                                                                                                                       |
| <b>Financial (Investment) Analysis</b>                                     | Theoretical and empirical examination and saving and investment, Concept of capitalization types of securities, non-banking financial institution. Development financing, Asset Structure, Stock Prices, Money Market, Portfolio Theory and Investment Analysis. Financing the industrial sector; Large scale and small scale, Interest rate. Bond market, Real and financial investment. Financial intermediaries; Speculation and Inter – relationship of financial and real variables in the economy. |

#### Teaching Methodology:

- To deliver lectures on topics included in course outline
- To require each student to solve independent assignments on topics included in the course.

#### **Evaluation Criteria:**

| <b>Evaluation Method</b> | <b>Total Percentage</b> |
|--------------------------|-------------------------|
| Quizzes/Assignments      | 25%                     |
| Mid-Term Exam            | 35%                     |
| Final-Term Exam          | 40%                     |

#### **Recommended Books:**

- Hughes, A., & D.J. Storey. (1994). Finance and The Small Firm” (Ed) Routledge, London.
- Bruno Solmik. (1996). International Investment, Addison Wesley.
- Burno Solnik. (1996). International Investments” 3rd Ed. Addison, Wesley publishing Co.
- Donald, E., Fischer, Arnold, J. (1997). Security Analysis and Portfolio Management. Prentice Hall, Delhi.

#### **Course Code: ECON-411**

**Title: Environmental and Natural Resource Economics**

**Credit Hours: 03**

**Prerequisite: Intermediate Microeconomics, Intermediate Macroeconomics, Development Economics-I**

#### **Course Objectives:**

According to Global Climate Risk Index (2019), Pakistan was the eighth most-affected country by extreme weather events for the period 1998-2017. At the same time, Pakistan’s environment and natural resources are facing a lot of pressure from the country’s significant population growth. Linkages between climate change, the ecology, natural resources and economic growth necessitate a careful analysis of how humans utilize natural resources—drawn from the ecological system—for economic production so that the rate of climate change can be mitigated, while also ensuring the conservation of the ecology as well as natural resources. This course will focus on using an economic perspective to carefully analyze the mentioned linkages. Some topics that will be covered include: climate change, resource and pollution management models, sustainable development and economic growth. Students will also be required to develop the expertise of conducting benefit/cost analysis using various methods, including Net Present Value (NPV).



### Learning Outcomes:

By the completion of the course, students should be able to:

- what are the market failures where government have to intervene and play their roles.
- what are various methods of cost-benefit analysis and how government makes decisions regarding projects up take.

### Course Contents:

|                                                                |                                                                                                                                                                                                          |
|----------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Introduction</b>                                            | What is environmental economics; review of microeconomics and welfare economics.                                                                                                                         |
| <b>Limitation of Market</b>                                    | Excludable and rival goods, common pool resources and public goods                                                                                                                                       |
| <b>The Theory of Externalities</b>                             | Pareto optimality and market failure in the presence of externalities; property rights and the Coase theorem.                                                                                            |
| <b>Valuing the Environment: Methods</b>                        | Introduction: theoretical aspects, Economic valuation of the environment: Methodologies and Applications                                                                                                 |
| <b>Cost-Benefit Analysis and Other Decision Making Metrics</b> | Framework of BCA and its applications, Divergence of Social and Private Discount Rates, Cost-Effectiveness Analysis, Impact Analysis                                                                     |
| <b>The Design and Implementation of Environmental Policy</b>   | Overview; Pigouvian taxes and effluent fees; tradable permits; choice between taxes and quotas under uncertainty; selection of environmental policy instruments; implementation of environmental policy. |
| <b>International Environmental Problems</b>                    | Trans-boundary environmental problems; global pollution; economic significance of biodiversity; economics of climate change; trade and environment, dispute resolutions.                                 |
| <b>Natural Resource Management</b>                             | Renewable resources, Non-renewable resources                                                                                                                                                             |
| <b>Measuring the Benefits of Environmental Improvements</b>    | Non-Market values and measurement methods; risk assessment and perception. Incentive based solutions                                                                                                     |
| <b>Contemporary Issues in Environmental Economics with</b>     | Energy crisis, Water footprints, Forests, Fisheries, Economics of Pollution                                                                                                                              |

|                                |                                                                                                                         |
|--------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| <b>Reference to Pakistan</b>   |                                                                                                                         |
| <b>Sustainable Development</b> | Concepts; Sustainability of Development, The Growth-Development Relationship measurement (conventional and alternative) |

### Teaching Methodology:

- To deliver lectures on topics included in course outline
- To require each student to solve independent assignments on topics included in the course.

### Evaluation Criteria:

| <b>Evaluation Method</b> | <b>Total Percentage</b> |
|--------------------------|-------------------------|
| Quizzes/Assignments      | 25%                     |
| Mid-Term Exam            | 35%                     |
| Final-Term Exam          | 40%                     |

### Recommended Books:

- Tietenberg, Tom and Lynne Lewis. Environmental and Natural Resource Economics, (11th Edition). Publisher: Pearson Addison Wesley, 2018.
- David A. Anderson. (2014). Environmental Economics and Natural Resource Management”, Routledge, 4th Edition.
- Robert N. Stavins. (2005) Economics of the Environment: Selected Readings, W. W. Norton, 5th edition.
- Ian Wills. (2007). Economics and Environment; A signaling and incentive approach, 2nd edition.
- Thomas Sterner. (2003). Policy Instruments for Environmental and Natural Resource Management, Resource for the future.

### Course Code: ECON-412

**Title: Economic Dynamics**

**Credit Hours: 03**

**Prerequisite: Advanced Microeconomics, Advanced Macroeconomics and Calculas-II**

### Course Objectives:

This is an introduction to the analysis of difference and differential equations, including the solution of the simplest linear dynamic systems, the linearization of non-linear systems and

the use of phase diagrams. It aims at providing necessary techniques for graduate students to analyze economic dynamics.

### Learning Outcomes:

By the completion of the course, students should be able to:

- Learn the macroeconomic demand supply dynamics
- Keynesian model and how it explains the changes in the various macroeconomic variables in the short run and long run and they interact with each other.

### Course Contents:

|                                        |                                                                                                                                                                                                                                                                |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Introduction</b>                    | Definitions and concepts, Dynamic models, Deterministic dynamical models, Difference equations, Attractors and repellers, Nonlinear dynamical systems, Continuous models Continuous differential equations.                                                    |
| <b>Demand and Supply Dynamics</b>      | The linear cobweb model, Different expectations, The Goodwin model of expectations, Nonlinear cobwebs, Ceilings and floors, Cobwebs in interrelated markets                                                                                                    |
| <b>Simple Keynesian Dynamics</b>       | The Keynesian-cross diagram, The dynamic multiplier, A dynamic model with taxes, The multiplier-accelerator model, Introduction of net exports                                                                                                                 |
| <b>IS-LM dynamics</b>                  | The static model, Instantaneous money market adjustment, a continuous IS-LM model A discrete version of the IS-LM model, Interest rate ceiling                                                                                                                 |
| <b>Inflation-unemployment dynamics</b> | The Phillips curve, a simple macroeconomic model of inflation, the dynamics of the simple model, Dynamic model with positive inflation, a change in the money supply A change in the natural level of unemployment.                                            |
| <b>Dynamics of the firm</b>            | Introduction, Monopoly and advertising, Advertising model: discrete version, Diffusion models, Static theory of oligopoly, Discrete dynamic model with output adjusting Instantaneously, Dynamic oligopoly with three firms, Partial-adjustment duopoly model. |
| <b>Fiscal Dynamics</b>                 | Budgetary concepts, Budget dynamics with no inflation and no monetary financing, Budget dynamics with inflation and money plus bond financing, Budget criteria of the Maastricht Treaty.                                                                       |

**Teaching Methodology:**

- To deliver lectures on topics included in course outline
- To require each student to solve independent assignments on topics included in the course.

**Evaluation Criteria:**

| Evaluation Method   | Total Percentage |
|---------------------|------------------|
| Quizzes/Assignments | 25%              |
| Mid-Term Exam       | 35%              |
| Final-Term Exam     | 40%              |

**Recommended Books:**

- Shone. R. (2001). An introduction to economic dynamics. Cambridge University Press.
- Shone. R. (2002). Economic Dynamics: Phase Diagrams and their applications

**Course Code: ECON-413****Course Title: International Finance****Credit Hours: 3****Pre-requisite(s): Microeconomics, Macroeconomics, International Trade****Course Objectives:**

This course deals with the analysis of three important and related macroeconomics issues in open economies: exchange rates, capital flows and financial crises. The objectives of the course are two-fold: to introduce and critically evaluate the main relevant economic theories, models and empirical works in these three key areas of International Finance; and to apply these analytical tools to build an understanding of relevant economic developments and policy issues in the global markets. The basic tenets of international finance are presented through a core textbook, assigned readings and online lectures.

**Lecture Outcomes:**

By the completion of the course, students should be able to:

- Understand the basic concepts of international finance, such as interest rates, exchange rates, balance of payments.
- They should get a sense of how trade has an impact on the financial outcomes of the country.

## Course Contents:

|                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|---------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Introduction and Basic Terminology</b>   | Introduction of International Finance; International Financial Markets: The basis of the foreign exchange market; Organization of the market; Exchange rate systems (Foreign Exchange Rates (FER), Real, Nominal, and Effective Rates. Fixed and Floating exchange rates, Determinants of Foreign Exchange Rate)                                                                                                                                                                                                                                      |
| <b>Interest and Exchange Rates</b>          | Dynamic Theory of Interest and Exchange Rate, The Theory of Spot Exchange Rate Determination, Capital Flow and Speculations, Forward Exchange Market, Interest Arbitrage, Interest Parity Theory, Speculators, Hedgers and Traders, Forward Exchange Rate Policy                                                                                                                                                                                                                                                                                      |
| <b>Balance of Payment</b>                   | <p>Purchasing Power Parity (PPP): Non-Trade Goods, Balance of Payment Effect. Test of Exchange Rate Instability. Impacts of FER. Rate Instability. BOP: Income Absorption Approach to BOP Elasticities of Exports and Imports and Impact on BOP. The Optimum Quantity of Reserves, Optimum Interest Rate. The Marshall-Larner Conditions.</p> <p>Disequilibria in balance of payments of corrective measures; Price Adjustment Mechanism; The partial equilibrium model; Price elasticities of demand for imports and exports; The J-Curve Effect</p> |
| <b>International Financial Institutions</b> | Introduction of International Institutions, Role and Major Activities of IMF. IMF Credit Facilities. Loan Mechanism, Value of SDRs. Euro-Dollar, Dollar Exchange and Impacts. Role of IMF in BOP Support and Stabilization. WTO: GATT and Emergency of WTO. Liberalization and BOP; Special Status of Developing Countries                                                                                                                                                                                                                            |
| <b>Risk and Return</b>                      | Facts on International Capital Flow: ASEAN Crisis and Role of International Institution and Criticism. Motive for Capital Movement: Return and Risk Diversification. Why disagreement on International Movement of Labor? Welfare Impacts of Labor Movements. Factors of Production and Price Equalization Issues                                                                                                                                                                                                                                     |
| <b>Further Topics</b>                       | Major Trade Rules and Their Expected Impacts on Trade, BOP Deficit. WTO and Fair Trade / Free Trade. Free Trade will it be? Issue of "Are Free Trade Gains more than Restricted Trade"? Emerging Regional Trading, Blocks, Special Preference and their                                                                                                                                                                                                                                                                                               |

|  |                                                                                                              |
|--|--------------------------------------------------------------------------------------------------------------|
|  | Impacts. Issues relating to the Contributions of International Financial Institutions and Their Evaluations. |
|--|--------------------------------------------------------------------------------------------------------------|

### Teaching Methodology:

- To deliver lectures on topics included in course outline
- To require each student to solve independent assignments on topics included in the course.

### Evaluation Criteria:

| Evaluation Method   | Total Percentage |
|---------------------|------------------|
| Quizzes/Assignments | 25%              |
| Mid-Term Exam       | 35%              |
| Final-Term Exam     | 40%              |

### Recommended Books

- International Finance: Theory and Policy by Steve Suranovic, Latest Edition
- International Trade Theory and Policy by Chacholiades, Miltiades, Latest Edition
- International Economics by Donomick Salvatore, Latest edition.
- International Economics: Theory and Policy by Paul R. Krugman and Maurice Obstfeld, Latest edition.
- World Trade and Payments. An Introduction by Richard E. Caves, Jeffrey A. Frankel, Ronald W. Jones. Latest edition.

### Course Code: ECON-414

**Title: Development Policy**

**Credit Hours: 03**

**Prerequisite: Advanced Microeconomics, Advanced Macroeconomics and Development Economics-I**

### Course Objectives:

The aim of this course is to introduce students to some of the impediments faced by development policy makers and how they can potentially be addressed. The course begins by examining the question of why some countries are rich and others poor which would provide the basis of addressing the central policy challenges. The course further identifies the main development challenges confronting policy makers in the contemporary world with emphasis on poverty and inequality, human capital development, social protection, environment and climate urbanization and rural-urban

migration and finance and fiscal development.

### Learning Outcomes:

By the completion of the course, students should be able to:

- Students should be able to connect the development economic topics that they studies with policy making and how those policies are evaluated.
- Understand the hiccups and problems faced in policy making and its implementations in various sectors.

### Course Contents

|                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|---------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Institutions, Poverty and Inequality</b> | <ul style="list-style-type: none"> <li>• Douglass C. North. The role of Institutions in Economic Development, Discussion Paper Series No. 2003.2, October 2003</li> <li>• Daron Accemonglu and James Robinson, Why Nations Fail: The Origins of Power, Prosperity and Poverty, Profile Books London, 2012, Chapter 3, 13 and 15</li> <li>• Banerjee, Abhijit. "Big Answers for Big Questions: The Presumption of Growth Policy." Paper for the Brookings conference on "What Works in Development? Thinking Big and Thinking Small." (2008).</li> </ul>                                                                                                                                                                                                                                       |
| <b>Human Capital: Education and Health</b>  | <ul style="list-style-type: none"> <li>• Case, Anne "The primacy of Education" Chapter 18 in <i>Understanding Poverty</i>, A. Banerjee, R. Benabou, D. Mookherjee (eds.), pp. 269-284, Oxford University Press (2005)</li> <li>• Andrabi, T, Das, J and Asim A. Khwaja "Education Policy in Pakistan: A Framework for Reform" (2010)</li> <li>• Connecting the Dots, Strategy Note 2016-2021, HIV, Health and Development UNDP</li> <li>• Nishtar, Sania 2010. "Health Systems in Pakistan" chapter 2 in Choked Pipes: Reforming Pakistan's Mixed Health Systems. Karachi, Oxford University Press,</li> <li>• Nishtar, Sania 2010. "Reform within the Healthcare System" chapter 11 in Choked Pipes: Reforming Pakistan's Mixed Health Systems. Karachi, Oxford University Press,</li> </ul> |
| <b>Social Protection</b>                    | <ul style="list-style-type: none"> <li>• Gazdar, H. (2014). 'Political Economy of Reform: social protection reform in Pakistan' in Development and Welfare Policy in South Asia (ed.) Gabriele Koehler and Deepta Chopra</li> <li>• Khatak, S. G. (2013). 'Social Protection: Extending Exclusion or Ending Exclusion' in Anita M. Weiss and Saba Gul Khattak (ed.). Development Challenges Confronting Pakistan. Boulder, Kumarian Press.</li> <li>• Graduation Strategies for Safety Net Beneficiaries", R.G. Hermosillo and Asad Sayeed. 2010</li> </ul>                                                                                                                                                                                                                                   |
| <b>The Environment</b>                      | <ul style="list-style-type: none"> <li>• Fifth Assessment Report, Climate Change 2014: Impacts, Adaptation, and Vulnerability, Summary for Policy Makers, Working Group II</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |



|                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>and Climate : Basic issues and Policy options in developing countries</b> | <p>Contribution to the of the Intergovernmental Panel on Climate Change (IPCC), 2014</p> <ul style="list-style-type: none"> <li>• Ghazanfar, M. (2009). Environmental Case of Sindh. Lahore Journal of Policy Studies 3(1).</li> <li>• Akmal Hussain, Pakistan's Water Crisis, The Express Tribune, Karachi, 16 August 2011: <a href="http://tribune.com.pk/story/231905/pakistans-water-crisis/">http://tribune.com.pk/story/231905/pakistans-water-crisis/</a></li> <li>• Ahmed, Mansoor, Water – Pakistan's most critical challenge, The Dawn News, 26 April 2016 : <a href="http://www.dawn.com/news/1254171">http://www.dawn.com/news/1254171</a></li> <li>• Connecting the Drops, An Indus Basin Roadmap for Cross-Border Water Research, Data Sharing, and Policy Coordination, Indus Basin Working Group, Stimson Centre, 2013</li> </ul> |
| <b>Urbanisation and Rural – Urban Migration</b>                              | <ul style="list-style-type: none"> <li>• Kundu, Amitabh, Urbanisation and Migration: An Analysis of Trend, Pattern and Policies in Asia, Human Development Research Paper 2009/16</li> <li>• Tacoli, C, MacGranahan, G, Satterthwaite, D, World Migration Report 2015, Urbanisation, Rural – Urban Migration and Urban Poverty, Human Settlements Group, International Institute for Environment and Development, International Organisation for Migration</li> </ul>                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Finance and Fiscal Policy for Development</b>                             | <ul style="list-style-type: none"> <li>• Abed, George T., Liam Ebrill, Sanjeev Gupta, Benedict Clements, Ronald M Morran, Anthony Pellecchio, Jerald Schiff, and Marijn Verhoeven, 1998, <i>Fiscal Reforms in Low-Income Countries</i>:</li> <li>• <i>Experience Under IMF-Supported Programs</i>, IMF Occasional Paper No. 160 (Washington: International Monetary Fund)</li> <li>• <i>Independent Evaluation Office</i>, 2003, "Evaluation Report: Fiscal Adjustment in IMF-Supported Programs" (Washington: International Monetary Fund); <a href="http://www.imf.org/External/NP/ieo/2003/fis/index.htm">http://www.imf.org/External/NP/ieo/2003/fis/index.htm</a></li> </ul>                                                                                                                                                                 |

### Teaching Methodology:

- To deliver lectures on topics included in course outline
- To require each student to write a research report and present it in class.

### Evaluation Criteria:

| <b>Evaluation Method</b> | <b>Total Percentage</b> |
|--------------------------|-------------------------|
| Quizzes/Assignments      | 25%                     |
| Mid-Term Exam            | 35%                     |
| Final-Term Exam          | 40%                     |

**Course Code: ECON-415****Title: Experimental and Behavioral Economics****Credit Hours: 03****Prerequisite: Intermediate Microeconomics****Course Objectives:**

Given the recent growth of interest in behavioral considerations, experiments are increasingly used in economics to study human behavior. Behavioral economists study models of human decision making and the interaction among such decision makers in games and markets. As behavioral economists aspire to construct "realistic" models, many of the models are inspired by Psychological and experimental studies. Students will also learn how to design experiments and interpret their results and will also overview some of the most important existing results.

**Learning Outcomes:**

By the completion of the course, students should be able to:

- Understand the concept of experimental economics and its pros and cons over other empirical analysis.
- Learn how behavioural economics is different from mainstream economics and the way.

**Course Contents:**

|                                              |                                                                                                                                                                                                                                                                                                                                                                                                       |
|----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Why Experiments?</b>                      | Brief history of experimental economics, experimental methods: advantages and disadvantages, some procedural and design considerations.                                                                                                                                                                                                                                                               |
| <b>Game Theory</b>                           | Extensive Form Games, solution by backwards induction, Strategies Normal Form Games, Dominance, Nash Equilibrium, Dominance Solvable Games.                                                                                                                                                                                                                                                           |
| <b>Individual Preferences: Time and Risk</b> | <p>Lottery choice experiments, financial incentives and control for wealth, preference elicitations: problems and application, preference reversals, inducing risk preferences, information processing: Bayes Rule and Biases.</p> <ul style="list-style-type: none"><li>• Holt, C. and S. Laury (2002), "Risk Aversion and Incentive Effects," American Economic Review, 92(5), 1644-1655.</li></ul> |
| <b>Dictator and Ultimatum Games</b>          | Ultimatum Game and Dictator game: basic results, Methodological variables, demographic variables culture. What is                                                                                                                                                                                                                                                                                     |

|                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                      | altruism and Fairness? Dictator Game and Altruism. Testing theories of altruism via Dictator and Prisoners' Dilemma game experiments; internal versus external validity.                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Trust Games</b>                                   | Trust game: basic results, pure and impure altruism, trustworthiness vs. altruism and fairness, a complex omnibus game, multistage trust game, gift exchange in experimental labor market, inequality aversion theories, fairness equilibrium, extensive-form fairness equilibrium.                                                                                                                                                                                                                                                                                                                                                    |
| <b>Coordination and public goods contribution</b>    | Public Good Game: theory and Practices, Free Riding and Voluntary Contribution Dilemma, Factors to Alleviate Free Riding, matching games, asymmetric players (Battle of Sexes: outside communication, evolution of meaning, external assignment), market entry games.                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Bargaining and Auctions</b>                       | Unstructured bargaining without side payments, bargaining over an externality, structured bargaining, auctions with fixed supply, first-price auctions with fixed supply.                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Gender differences and Experimental Economics</b> | <ul style="list-style-type: none"> <li>• Dolan, Kathleen, and Kira Sanbonmatsu. "Candidate gender and experimental political science." <i>Cambridge handbook of experimental political science</i> 289 (2011).</li> <li>• Bellemare, Charles, Patrick Lepage, and Bruce Shearer. "Peer pressure, incentives, and gender: An experimental analysis of motivation in the workplace." <i>Labour Economics</i> 17.1 (2010): 276-283.</li> <li>• Chattopadhyay, Raghavendra, and Esther Duflo. "Women as Policy Makers: Evidence from a Randomized Policy Experiment in India." <i>Econometrica</i> 72, no. 5 (2004): 1409-1443.</li> </ul> |

### Teaching Methodology:

- To deliver lectures on topics included in course outline
- To require each groups of students to do a project designing their own research question and relevant experiment.

### Evaluation Criteria:

| Evaluation Method   | Total Percentage |
|---------------------|------------------|
| Quizzes/Assignments | 25%              |
| Mid-Term Exam       | 35%              |
| Final-Term Exam     | 40%              |

### Recommended Books:

- Camerer, Colin F. Behavioral game theory: Experiments in strategic interaction. Princeton university press, 2011.
- Davis, Douglas D., and Charles A. Holt. Experimental economics. Princeton university press, 1993.
- Durlauf and Blume. Behavioral and Experimental Economics. Latest Edition.
- Falk and Gächter. Experimental Labor market. Latest Edition.
- Gerber, Alan and Donald Green. (2012). Field Experiments. W. W. Norton & Company

### Course Code: ECON-416

**Title: Gender Analysis in Economics**

**Credit Hours: 03**

**Prerequisite: Intermediate Microeconomics and Intermediate Macroeconomics**

### Course Objectives:

This course explores the gender dimensions of economic life in developing countries and how these can be examined in economic data analysis and policy research. It draws on the rich body of research of gender-aware economics and includes: household economics, paid and unpaid work, labor markets, entrepreneurship, migration, agriculture, trade and gender-responsive budgeting. The treatment of these gender-related topics include a feminist economics perspective and focus on its application in developing countries and in the process of economic development.

### Learning Outcomes:

By the completion of the course, students should be able:

- understand the concept of gender and what it means in general.
- how the gender dimensions and lens change the outlook of all the economics outcomes.
- how to measure the unpaid care work and what methodologies are used to value them.

### Course Contents

|                                             |                                                                                                                                                                                                                                                                           |
|---------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>What is Gender in Economic Analysis?</b> | Subject and methods of gender economics. Gender statistics. System of gender inequality factors, Institutionalization of gender economic studies. Methods of gender economics. Inter-disciplinary approach in gender studies. Macro- and micro- level of studies, General |
|---------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                 | principles of and requirements for data collection in the framework of an integrated system of gender state statistics, the development factor, discrimination and behavioural factors, Gender order, Gender gap in access to resources and control over economic resources. Gender risk groups.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Economic Growth and Gender Equality</b>      | Impact of economic growth on gender (in-)equality. Impact of gender equality on economic growth and socio-economic development. Core principles of feminist economics and its relevance for macroeconomics, Understanding the economy as a gendered structure (concepts and data), Analyzing multiple linkages between macro-economic policies and gender inequalities, Concept of the “Feminisation of Poverty”. Basic causes of the feminisation of poverty. Problems of measuring the feminisation of poverty. Macro-level and micro-level measurement of the feminisation of poverty. Life cycle and gender differences in incomes. Women’s contribution to GDP. Estimation of women’s unpaid work (GDP satellite accounts).                                                   |
| <b>Gender Inequality in The Labor Market</b>    | Economics of the labor market, Inequality in the labour market: entry into and exit from the labour market, careers, wages, skills training. Inequality and discrimination. Types of segregation. Measurement of segregation. Models of gender inequality and gender discrimination in the labour market. Factors in wage disparities, role of culture, norms and identity.                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>The Gender Factor In Household Economics</b> | Household economics: theoretical concepts and gender critique. Gender critique of the neo-Marxist approach. The gender contract, negotiation models within families. Household time budgets: structure, and problem of measurement. Distribution of time: work in the labour market, work in the household, leisure time, Unpaid work and care work, Distribution of incomes (expenditures) in the household: inter-generational and gender aspect. Model of investment in human capital. Function of production of human capital. Investments in human capital at household level and the sex factor. Accumulation of human capital over a life cycle. Household factors (reproductive, productive, structural, income-based, power-related) and employment of household members. |

|                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                            | Marital and age-dependent models of female and male employment.                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Measuring and Valuing Unpaid Domestic and Care Work</b> | What is unpaid domestic and care work? What is unpaid domestic and care work? Time Use Surveys, Descriptive statistics related to unpaid and care work, Policies targeting unpaid domestic work and care work.                                                                                                                                                                                                                                                             |
| <b>Gender Policy and Gender inequality indicators</b>      | Incorporation of gender policy elements into socio-demographic policies in various countries. Main gender issues in socio-economic development in developed countries, developing countries and CIS countries. Regional particularities of gender policy. Gender issues in the Millennium Development Goals. Indicators of gender differences in socio-economic development. Incorporation of the gender factor into the Human Development Index, Gender Equality Indices. |
| <b>Gender Budgets</b>                                      | The government budget, Definition of gender responsive budgeting (GRB), Tools for integrating a gender perspective into different levels of the budget, Gender analysis of different components of the budget (Public Expenditure and Taxation), Methodology of gender budget preparation. Approaches and principles of gender budgeting. Experiences of different countries in gender budgeting.                                                                          |

#### Teaching Methodology:

- To deliver lectures on topics included in course outline
- To require each student to solve independent assignments on topics included in the course.

#### Evaluation Criteria:

| Evaluation Method   | Total Percentage |
|---------------------|------------------|
| Quizzes/Assignments | 25%              |
| Mid-Term Exam       | 35%              |
| Final-Term Exam     | 40%              |

#### Recommended Books:

- Beneria, L2003. Gender, Development and Globalisation. Economics As If All People Mattered. New York: Routledge
- Jacobsen, J 2007. The Economics of Gender. Cambridge:Wiley Blackwell.
- Rai, S. and G. Waylen (eds). 2014. New Frontiers in Feminist Political Economy. London. Routledge.

#### **Recommended Research Articles:**

- Kabeer, N. (1994) Reversed realities: gender hierarchies in development thought, London: Verso
- Kabeer, N. (2012) Women's economic empowerment and inclusive growth: labor markets and enterprise development
- Macdonald, M. (1995) Feminist Economics: From Theory to Research, The Canadian Journal of Economics, 28(1), 159-176.
- Chant, S. (2007) Gender Generation and Poverty: exploring the 'feminization of poverty in Africa, Asia and Latin America.

#### **Course Code: ECON-417**

**Title: Poverty and Income Distribution**

**Credit Hours: 03**

**Prerequisite: Development Economics-I**

#### **Course Objectives:**

The aim of this course is to introduce students to the subject of poverty and inequality. The emphasis will be on problems in the modern contemporary economy, but the topic will also be approached from an historical perspective, taking a longer term view of the distribution of income in human societies. In particular, we will examine why some individuals and groups have less access to resources and income than others. Students will learn the fundamental concepts, theories and methodologies that are used by economists to measure and understand questions related to the distribution of income. By the end of the course, students should expect to not only have a greater knowledge of the forces determining the extent of poverty and the distribution of income, but to have also gained some understanding of the methods used by economists and policy-makers to deal with them.

#### **Learning Outcomes:**

By the completion of the course, students should be able to:

- Learn traditional and contemporary measures of poverty and income inequality.
- Learn about Pakistan's income inequality profile.

#### **Course Contents:**



|                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|--------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Traditional Measures of Poverty</b>           | What Is Poverty and Why Measure It? The Concepts of Well-Being and Poverty, Measuring Poverty, steps in Measuring Poverty, monetary poverty, capability approach, Poverty Lines, Relative Poverty, Absolute Poverty, Issues in Choosing an Absolute Poverty Line, Measures of Poverty, Headcount Index, Poverty Gap Index, Squared Poverty Gap (Poverty Severity) Index,                                                                                                                                                                                                           |
| <b>New Dimensions of Poverty</b>                 | Multidimensional Poverty Index, Sen Index, The Sen-Shorrocks-Thon Index, FGT class of measures, multidimensional poverty index, Poverty Indexes: Checking for Robustness, Sampling Error, Measurement Error, Equivalence Scales, Choice of Poverty Line and Poverty Measure                                                                                                                                                                                                                                                                                                        |
| <b>Measuring Income Inequality</b>               | Definition of Inequality, Measures of Inequality, Decile Dispersion Ratio, Gini Coefficient of Inequality, Generalized Entropy Measures, Atkinson's Inequality Measures, Inequality Comparisons, Measuring Pro-Poor Growth, Decomposition of Income Inequality, Poverty Profiles, Additive Poverty Measures, Determinants of Poverty, Household and Individual-Level Characteristics, Analyzing the Determinants of Poverty: Regression Techniques, Poverty Reduction Policies, The Analysis of Poverty over Time, Vulnerability to Poverty, Quantifying Vulnerability to Poverty. |
| <b>Poverty and Income Inequality in Pakistan</b> | Poverty and inequality in Pakistan- How does Pakistan compare? Historical Trends- the longer term picture                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

### Teaching Methodology:

- To deliver lectures on topics included in course outline
- To require each student to solve independent assignments on topics included in the course.

### Evaluation Criteria:

| <b>Evaluation Method</b> | <b>Total Percentage</b> |
|--------------------------|-------------------------|
| Quizzes/Assignments      | 25%                     |
| Mid-Term Exam            | 35%                     |
| Final-Term Exam          | 40%                     |

### Recommended Books:

- Sen, A. (2002). On Economic Inequality. New Delhi: Oxford University Press.
- Shorrocks, A., & Hoveen, R.V.D. (Eds.) (2005). Growth, Inequality, and Poverty: Prospects for economic development. New York: Oxford University Press.

- Jonathan Haughton and Shahidur Khandker (2009). Handbook on Poverty and Inequality. (World Bank Training Series). ISBN-10: 0821376136.
- Atkinson, A. B., B. Cantillon, E. Marlier and B. Nolan (2002). Social Indicators. The EU and Social Inclusion. Oxford: Oxford University Press.
- Cowell, F. (1992). Measuring Inequality, 2nd Edn. Hemel Hempstead: Harvester Wheatsheaf.
- Fields, G. S. (2001). Distribution and Development. Cambridge, MA: MIT Press.
- Foster, J. E., J. Greer and E. Thorbecke (1984). A class of decomposable poverty measures. Econometrica, 52(3), 761-66.
- Kakwani, N. (2003). Issues in Setting Absolute Poverty Lines. Poverty and Social Development Paper 3. Asian Development Bank. .
- Ravallion, M. (1994). Poverty Comparisons, Chur: Harwood Academic Publishers.
- Sachs, Jeffrey. (2005). End of Poverty: Economic Possibilities for our Time (New York, Penquin).

## **Course Code: ECON-418**

**Title: Energy Economics**

**Credit Hours: 03**

**Prerequisite: Intermediate Macroeconomics**

### **Course Objectives:**

The main focus of the course is to study economic relationships in energy production, consumption, demand and supply, pricing and conservation, energy policy and the development of new and renewable energy sources. The aim of the course is to provide students with the opportunity to study and develop a broader understanding of the economics of energy. There is mounting evidence that the current global energy system has been growing far beyond what is globally sustainable and already poses a serious, and potentially irreversible, threat to global environmental quality and stability in future decades.

### **Learning Outcomes:**

By the completion of the course, students should be able to learn:

- How energy production, demand and supply side are dealt with
- What are various energy producing sources and what are the benefits and problems of each?
- The overview of industries involved in energy production.

### **Course Contents:**

|                                          |                                                                                                                                                                                    |
|------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Overview and Fundamental Concepts</b> | Energy in the Economy; Global Energy Problems and Issues, Fundamental Concepts: Energy, Power; Measuring energy: units and conversion factors, Laws of Thermodynamics, 1st Law and |
|------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|                                                                                     |                                                                                                                                                                                                                                                                                                                                                              |
|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                     | Entropy law, Input and output energy, energy efficiency, Energy conversions (technologies), Global Energy system sustainability issues, Energy markets and energy prices; competitive market valuation and social valuation, Market Failures: Market power (monopoly, oligopoly, and cartels), Externalities, Public goods and common property resources.    |
| <b>Oil Resources and Economic Issues</b>                                            | Discounted cash flow analysis, reserve and resources, resource substation, forecasting prices and speculation, Natural resource (oil) demand and supply, OPEC                                                                                                                                                                                                |
| <b>Overview of Energy Economics and Global Energy Sustainability</b>                | Primary energy supply, secondary energy and energy end use demand/consumption, Energy, economic activity, and growth: energy intensity, global energy use forecasts, Energy conversion, energy transportation and/or transmission, and clean energy use, Conditions for a sustainable global energy system, climate change and clean energy                  |
| <b>Clean Energy Supply from Non-Conventional, Alternative and Renewable sources</b> | Climate change and primary and secondary energy use, Energy Conservation policies, demand side management, Non-conventional and renewable alternative energy: Solar, Biomass, Wind, Geothermal, Tidal & other, Hydrogen energy, fuel cells                                                                                                                   |
| <b>Energy Supply from Non-Renewable Fossil Fuel Resources</b>                       | Fossil fuel resources and reserves model, Cartel models of global energy markets, Overview of Oil, Natural gas, and Coal industries, Backstop technologies and Non-renewable Resource Pricing (Hotelling model), Clean fossil fuel use; carbon sequestration                                                                                                 |
| <b>Energy Conversion and Supply by Electric Utility Industries</b>                  | Electric Power demand and load duration vs. Electric energy consumption, time-of-use, Electricity production technologies and electricity production cost trade-offs, Thermal, hydro, nuclear and renewables production and transmission of electricity, Electricity pricing issues, regulation and deregulation issues, Nuclear energy issues and prospects |
| <b>Sustainable Energy Policy</b>                                                    | Jaccard's Sustainable Energy System in 2100, Energy Policy of Pakistan, Sustainable energy options and criteria for comparison, Sustainable Energy Policy Alternatives and Climate Change, International Policy Initiatives                                                                                                                                  |
| <b>China-Pakistan Economic Corridor (CPEC)</b>                                      | What is CPEC? Central Role and Key Areas, Energy sector projects, Project Financing, Benefits to Pakistan, Benefits to China and other Countries, Challenges                                                                                                                                                                                                 |

### Teaching Methodology:

- To deliver lectures on topics included in course outline

- To require each student to solve independent assignments on topics included in the course.

#### **Evaluation Criteria:**

| <b>Evaluation Method</b> | <b>Total Percentage</b> |
|--------------------------|-------------------------|
| Quizzes/Assignments      | 25%                     |
| Mid-Term Exam            | 35%                     |
| Final-Term Exam          | 40%                     |

#### **Recommended Books:**

- Tom Tietenberg and Lynne Lewis. (2012). Environmental & Natural Resource Economics Pearson Education, Inc., 9th Edition.
- Mark Jaccard. (2005). Sustainable Fossil Fuels: The Unusual Suspects in the Quest for Clean and Enduring Energy, Cambridge University Press.
- Subhes C. Bhattacharyya. (2011). Energy Economics Concepts, Issues, Markets and Governance Springer-Verlag London Limited.
- J. M. Griffin, and H. B. Steele (1985). Energy Economics and Policy, Academic Press

#### **Course Code: ECON-419**

**Title: Islamic Banking and Finance**

**Credit Hours: 03**

**Prerequisite: Principles of Microeconomics and Principles of Macroeconomics**

#### **Course Objectives:**

Islamic banking course provides an opportunity to explore, investigate and become professionally competent. According to an estimate upto 20% of the world population utilizes or actively seeks Islamic banking services. With no previous financial experience of knowledge of banking required, students will be introduced to the concept of Islamic banking and the way in which it differs from traditional banking. It covers the primary and unique characteristics of Islamic finance, along with how Islamic law applies to and affects the financial dealings. It is based on the principles of socio-economic justice.

#### **Learning Outcomes:**

By the completion of the course, students should be able to learn:

- Various tools used in Islamic Economics to provide banking services keeping in view the Islamic rulings.
- Financial products offered under Islamic Banking

#### **Course Contents:**

|                                                 |                                                                                                                                                                                                                                                                                                                                                 |
|-------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Introduction of Islamic Banking</b>          | Provision of a comprehensive introduction to Islamic banking. Discovering the financial instruments offered by Islamic banks. Awareness of the theory behind Islamic banking practices                                                                                                                                                          |
| <b>Basics of Islamic Banking and Finance</b>    | Exploring the philosophy of Islamic banking and financing alongwith the series of key alternative financing principles. Knowledge about Mudarabah, Musharkah, Murabaha-Mu'ajjal and Salam                                                                                                                                                       |
| <b>Islamic Banking and Conventional Banking</b> | Highlighting the key differences between Islamic banking and conventional banking. Islamic banking model and its comprehensive definition and overview of how the model work in a contemporary business setting. Exploration of the difference between Islamic banking and others and magnitude of difference from conventional western banking |
| <b>Islamic Law of Contract</b>                  | Contract laws under the rules of Islamic banking and representation of balancing situation. Explanation of the key elements of Islamic contracts, the various types of contracts relevant to Islamic banking and their framework                                                                                                                |
| <b>Islamic Financial Products</b>               | Deeper insight into specific Islamic financial products, touching upon Islamic financial institutions, investment banking, and introduction to Islamic principles in relation to stock and sukuk (Islamic bonds for the generation of returns to investors)                                                                                     |
| <b>Islamic Business Ethics</b>                  | Analyzing the Islamic business ethics, incorporating essential teachings on gentleness, honesty, legitimate earnings and justice                                                                                                                                                                                                                |
| <b>Globalization of Islamic Banking</b>         | Islamic banking and finance on a global basis and representation of the contemporary international Islamic financial institutions including the Islamic Financial Services Board and the International Islamic Rating Agency                                                                                                                    |

### Teaching Methodology:

- To deliver lectures on topics included in course outline
- To require each student to solve independent assignments on topics included in the course.

### Evaluation Criteria:

| <b>Evaluation Method</b> | <b>Total Percentage</b> |
|--------------------------|-------------------------|
| Quizzes/Assignments      | 25%                     |
| Mid-Term Exam            | 35%                     |
| Final-Term Exam          | 40%                     |

### Recommended Books:

- Ahmad, K. (1976). Islami Nazriya Hayat. Karachi: Institute of policy studies.
- Chapra, M. U. (1992). Islam and the economic challenge. Lahore: The Islamic foundation.
- Dar, A. H., & Akram, M. M. (2007). Islamic Economics. Lahore: Ilmi Kitab Khana.
- Mannan, M. A. (2006). Islamic Economics : theory and practice. Lahore: Sh. Muhammad Ashraf.
- Mulathawi, M. H. (1995). Muashiyat Islam. Karachi: Manghu Peer Sharif.
- Naqvi, S. N. H.(2003). Perspectives on Morality and Human Well-Being: A Contribution to Islamic Economics. UK: The Islamic Foundation.
- Naqshbandi, Q. M. A. Islamic Economics. Lahore: Iqra Books

### Course Code: ECON-420

**Title: Impact Evaluation Methods**

**Credit Hours: 03**

**Prerequisite: Intermediate Microeconomics, Intermediate Macroeconomics, Econometric I & II**

### Course Objectives:

The course will present an overview of major techniques for impact evaluation and an introduction to the mixed methods approach. In addition, it will be drawing from the good international practices and literature in the field. A dominant feature of empirical research in economics in the past two decades has been the “credibility revolution” in empirical research that has prioritized the generation of credible causal estimates of the impact of policies and programs (Angrist and Pischke, 2010).

### Learning Outcomes:

By the completion of the course, students should be able to learn:

- The evidence-based policy and why evaluation of the development projects really important.
- What is inferences and the conditions required to determine causality.
- The main evaluation methods used to examine the impact of policies and the data sets required for each.

### Course Contents:

|                                          |                                                                                                                                                                                                                                                                                      |
|------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Introduction to Impact Evaluation</b> | What is evidence-based policy making, what is impact evaluation? Impact evaluation for policy decisions, deciding whether to evaluate, cost effectiveness analysis, prospective versus retrospective analysis, efficacy and effectiveness studies, determining evaluation questions, |
|------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|



|                                                                      |                                                                                                                                                                                                                                                                                                                                      |
|----------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                      | types of evaluation questions, theories of change, the result chain, hypothesis for the evaluation, selecting performance indicator.                                                                                                                                                                                                 |
| <b>Evaluation questions, Causal Inference and Counterfactuals</b>    | Causal inference, estimating counterfactual, two counterfeit estimates of the counterfactual                                                                                                                                                                                                                                         |
| <b>Experimental Designs: Randomized Control Trials</b>               | Randomized assignment of the treatment, two variations in the randomized assignment, estimating impact under randomized offering, read a research article that uses randomized control trial and understand its design, the selection problem, how random assignments help in selection problem, regression analysis of experiments. |
| <b>Quasi - Experimental Designs: Instrumental Variable Technique</b> | Instrumental Variables and causality, Two-Stage Least Squares, Grouped Data and 2SLS, Asymptotic 2SLS Inference, Over-identification, Local Average Treatment Effects, IV in Randomized Trials, Fixed Effects, Random Effects                                                                                                        |
| <b>Quasi - Experimental Designs: Matching Estimation</b>             | When is matching estimation used? What are its assumptions? How does it work? Propensity score matching and its various techniques, limitation of matching methods, read papers relevant to propensity score matching and understand their application.                                                                              |
| <b>Quasi - Experimental Designs: Difference-in-Difference (DD)</b>   | How is DD helpful? How does it work? Limitation of DD method, read papers relevant of difference-in-difference and understand their application.                                                                                                                                                                                     |
| <b>Quasi - Experimental Designs: Regression Discontinuity (RD)</b>   | How is RD helpful? How does it work? Limitation of RD method, read papers relevant to regression discontinuity and understand their application.                                                                                                                                                                                     |
| <b>Operationalizing Impact Evaluation</b>                            | Choosing Evaluation Method, determine if evaluation ethical? How to set up the evaluation? How to time and budget the evaluation? What kinds of Data is required? Power Calculations Sampling Strategy? Developing and testing Questionnaires, Conducting Fieldworks, Processing and Validating Data.                                |

### Teaching Methodology:

- To deliver lectures on topics included in course outline
- To require each student to solve independent assignments on topics included in the course.



**Evaluation Criteria:**

| Evaluation Method   | Total Percentage |
|---------------------|------------------|
| Quizzes/Assignments | 25%              |
| Mid-Term Exam       | 35%              |
| Final-Term Exam     | 40%              |

**Recommended Books:**

- P. Gertler, S. Martinez, P. Premand, L. Rawlings, and C. Vermeersch, *Impact Evaluation in Practice*, The World Bank, 2016.
- J. Angrist and J. Pischke, *Mostly Harmless Econometrics: An Empiricist's Companion*, Princeton University Press, 2009.

**Recommended Research Articles:**

- Dehejia, R. H., & Wahba, S. (2002). Propensity score-matching methods for nonexperimental causal studies.
- Finklestein, et al. (2012) The Oregon Health Insurance Experiment: Evidence from the First Year.
- Madestam et al. (2013). Do political protests matter? Evidence from the Tea Party Movement.
- Crepon, et al. (2013). Do labor market policies have displacement effects? Evidence from clustered randomized experiment.
- Clampet- Lundquirt, et al. (2006) Moving At-Risk Youth Out of High-Risk Neighborhoods: Why do Girls Fare Better than Boys?
- Card, D. (1990) The Impact of the Mariel Boatlift on the Miami Labor Market.
- Cohodes, S. & Goodman, J. (2014). Merit Aid, College Quality, and College Completion.

**Course Code: ECON-421****Title: World Economic History****Credit Hours: 03****Prerequisite: Intermediate Microeconomics and Intermediate Macroeconomics****Course Objectives:**

This course surveys the history of colonial expansion from the discovery of the Americas up to the colonization of the Middle East following the end of the First World War. The course is intended to give students a chronological account of the rise of imperialism and the colonial expansion into Latin America, Asia and Africa. While emphasis is placed on social, political and economic developments within the European nations which prompted colonial expansion, the primary focus is on how the actual colonies were acquired and the

subsequent changes that took place. The course attempts to provide an essential background to understanding the historical processes by which modern developed nations became developed, and how the underdeveloped “Third World” originated as part of the same historical process.

### **Learning Objectives:**

By the completion of the course, students should be able to:

- Learn about the economic history and how the economic systems evolved over time.

### **Course Contents:**

|                                                                                             |                                                                                                                                                                                                                              |
|---------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Introduction</b>                                                                         | What is Economic History? What is World Economic History? Why study World Economic History                                                                                                                                   |
| <b>The world economy during the medieval period: The beginnings of the Great Divergence</b> | The first Global Economy? The Sung Miracle, the Golden Age of Islam and Medieval Europe, The Plague and the beginnings of the Great Divergence.                                                                              |
| <b>The Discovery of the Americas to the Rise of Industrial Capital, 1492 – 1770</b>         | Introduction: Dependency Theories, The Discovery of America, The Origins of the Slave Overland Trade Routes and the Middle East, Origins of Colonial Expansion in Asia, Overview of Africa’s Economy from ancient to modern. |
| <b>Industrial Capitalism to Monopoly Capitalism, 1771 – 1870</b>                            | The First Industrial Revolution, The Latin American Revolutions, The End of the Slave Trade, Expansions into the Middle East, The Conquest of India, Big Push industrialization.                                             |
| <b>Monopoly Capitalism to The First World War, 1870 – 1917</b>                              | The Second Industrial Revolution, The Scramble for Africa, The Opium Wars and the Onslaught on East Asia, Russian Political and Economic Developments, The Japanese Response                                                 |
| <b>Nationalist Thoughts, 1917 - onwards</b>                                                 | Nationalist Thought and Indigenous Responses to the West, Origins of Nationalist Thought I, Origins of Nationalist Thought II, The Russian Revolution., Rise of East Asia, (Re)-Globalization                                |

### **Teaching Methodology:**

- To deliver lectures on topics included in course outline
- To require each student to solve independent assignments on topics included in the course.

### **Evaluation Criteria:**

| Evaluation Method   | Total Percentage |
|---------------------|------------------|
| Quizzes/Assignments | 25%              |
| Mid-Term Exam       | 35%              |
| Final-Term Exam     | 40%              |

### Recommended Books:

- L. S. Stavrianos (1981). Global Rift: The Third World Comes of Age, (New York: William Morrow and Company.
- Allen, Robert C. (2001). Global Economic History: A Very Short Introduction (Oxford, OUP.
- Harriss, John, Janet Hunter, and Colin Lewis. (2009). The new institutional economics and third world development. Routledge.

### Additional Readings:

- Kuhn, Frithjof. (1987). Causes of underdevelopment and concepts for development. An introduction to development theories." The Journal of Institute of Development Studies, NWFP.
- Arrighi, Giovanni. (1998). Globalization and the Rise of East Asia Lessons from the Past, Prospects for the Future. International Sociology 13(1), 59-77.
- Findlay, Ronald, and Kevin H. O'Rourke. (2007). Power and plenty: trade, war, and the world economy in the second millennium. Princeton, NJ: Princeton University Press.
- Stiglitz, Joseph E. (2002). Globalization and its DisCourse Contents. Vol. 500. New York: WW Norton.
- Stiglitz, Joseph E. (2003). Globalization, technology, and Asian development. Asian Development Review, 20(2) 1-18

## Course Code: ECON-422

**Title: General Equilibrium and Welfare Economics**

**Credit Hours: 03**

**Prerequisite: Intermediate Microeconomics**

### Course Objectives:

It is a fundamental microeconomics course which studies the interdependency between individual decision making and market clearing price formation. From a methodological viewpoint, the general equilibrium approach views the economy as a closed and interrelated system in which the equilibrium values of all variables are simultaneously determined. Welfare economics deals with the interaction between "positive economics" (i.e. statements of what will happen under different policies) and "normative economics" (statements about what policies should be followed). Welfare economics considers how

value judgments involving interpersonal comparisons can be incorporated into the basic framework of positive economics in order to reach normative conclusions.

### Learning Outcomes:

By the completion of the course, students should be able to:

- Learn the concept of general equilibrium of the economy
- Understand the welfare theorems and optimal allocations

### Course Contents:

|                                                                                     |                                                                                                                                                                                                                                                            |
|-------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Concept and History of General Equilibrium Theory</b>                            | Introduction to General Equilibrium Theory, Partial and General Equilibrium: Development of the field, History of General Equilibrium Theory, Problems with Partial Equilibrium Analysis.                                                                  |
| <b>An elementary General Equilibrium Model: The Robinson Crusoe Economy</b>         | Centralized Allocation, Decentralized Allocation, Pareto Efficiency of the Competitive Equilibrium Allocation: First Fundamental Theorem of Welfare Economics                                                                                              |
| <b>The Edgeworth Box</b>                                                            | Geometry of the Edgeworth Box, Pareto Efficiency and its constituents, Consumption Efficiency, Production Efficiency, Product Mix Efficiency, Calculating an Efficient Allocation, Competitive Market solution in the Edgeworth Box, Walrasian Equilibrium |
| <b>Theorems</b>                                                                     | Envelope theorem, Kakutani fixed point theorem                                                                                                                                                                                                             |
| <b>Integrating production and multiple consumption decisions: A 2 X 2 X 2 model</b> | A 2 X 2 X 2 model, Proof of existence of general equilibrium, Technical Efficiency and Pareto Efficiency, Prices and Decentralization, Efficient Market Hypothesis                                                                                         |
| <b>Welfare</b>                                                                      | First and second theorems for welfare economics, Social Justice and the Social Optimum, Markets as the solution of efficient resource allocation problem                                                                                                   |

### Teaching Methodology:

- To deliver lectures on topics included in course outline
- To require each student to solve independent assignments on topics included in the course.

### Evaluation Criteria:

| <b>Evaluation Method</b> | <b>Total Percentage</b> |
|--------------------------|-------------------------|
| Quizzes/Assignments      | 25%                     |
| Mid-Term Exam            | 35%                     |
| Final-Term Exam          | 40%                     |

### **Recommended Books:**

- L. S. Stavrianos (1981). Global Rift: The Third World Comes of Age, (New York: William Morrow and Company. Starr, Ross M. (20110). General Equilibrium Theory: An Introduction, 2nd Edition, Cambridge University Press
- Walter Nicholson. (2005). Microeconomic Theory: Basic Principles and Extensions, 9th edition. Thomson South-Western.
- James M. Henderson and Richard E. Quandt. (1971). Microeconomic Theory: A Mathematical Approach, 2nd edition. McGraw Hill Kogakusha Ltd., Tokyo.
- Eugene Silberberg. (1990). The Structure of Economics: A Mathematical Analysis, 2nd edition, New York: McGraw Hill Publishing Co.
- P. R. G. Layard and A. A. Walters. (1978). Microeconomic Theory, McGraw Hill Book Company.
- Jeffrey M. Perloff. (2014). Microeconomics, 7th edition, Prentice Hall
- Allen, Robert C. (2001). Global Economic History: A Very Short Introduction (Oxford, OUP.
- Harriss, John, Janet Hunter, and Colin Lewis. (2009). The new institutional economics and third world development. Routledge.

### **Course Code: ECON-423**

**Title: Regional Economics**

**Credit Hours: 03**

**Prerequisite: Calculus-II and Development Economics**

### **Course Objectives:**

Regional Economics is an introductory course, which will provide the students with the basic concepts related to regional economic growth and analysis. The course will also touch on the concept of sustainability and prevalent approaches to policy design in this context.

The objective of this course is to apply the methods of microeconomic analysis to gain an understanding of both why this is the case, and of the forces that shape land development and resource allocation in urbanized areas. A general objective is to improve your ability to apply microeconomic analysis. The more specific objective is to build a working understanding of the economics of urban areas: (1) economic explanations of why cities exist and where they develop and why they grow; (2) how and why urban land develops as it does; and (3) the roles of local governments in influencing the allocation of resources in urban areas. Income determination will also be an important area of study, looking at how economic growth is distributed, and the role of growth in improving household well-being.

### **Learning Outcomes:**

By the completion of the course, students should be able to:

- Learn about the regional dynamics of various countries who are located geographically together
- What are the economic benefits and costs that these countries need to be mindful of.

**Course Contents:**

|                                                               |                                                                                                                                                                                                                                                                                 |
|---------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Introduction</b>                                           | Economics and space, Location and Physical-metric space, Regional Growth local development, Theories of convergence and divergence                                                                                                                                              |
| <b>Agglomeration and Location</b>                             | Agglomeration economies and transportation costs, Localization Economies and transportation cost, Market size and Transportation cost, Economies of scale and transportation cost, spatial demand, market equilibrium and firm location, Interdependency in location choices.   |
| <b>Accessibility and location</b>                             | Accessibility and transportation cost, The location of agriculture activities: The Von Thunen model, The urban location of firms: The Alonso model, The urban location of household, Recent developments: General equilibrium model                                             |
| <b>Productive Structure and development</b>                   | The different interpretations of regional growth and development, the different conceptions of space, the theories of the stages of development, Stages of development and disparities, Industrial structure and regional growth: shift share analysis, The centrality approach |
| <b>Demand</b>                                                 | Demand and Regional Growth, interregional relations, the exporter region: the export-base model, critical assessment of the model, input-output analysis, the importer region: The Harrod-Domar model, Balance of payment and local growth                                      |
| <b>Factor Endowment</b>                                       | Factor endowment and regional growth, regional growth and factor mobility, factor mobility specialization and well-being, absolute vs comparative advantages in regional growth                                                                                                 |
| <b>Territorial Competitiveness and Exogenous Development</b>  | The components of territorial Competitiveness, the growth-pole theory, the role of multinational companies in local development, the spatial diffusion of innovation, infrastructure and regional development, new communication technologies and regional development          |
| <b>Territorial Competitiveness and Endogenous Development</b> | The endogenous sources of competitiveness: agglomeration economies, space and static efficiency, space and dynamic efficiency, the urban structure and regional development                                                                                                     |



|                                                                        |                                                                                                                                                                                                                                                                |
|------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Territorial Competitiveness and Cumulative Demand/Supply Growth</b> | Increasing returns, Competitiveness and cumulative growth, equilibrium in condition of non-linearity, increasing returns external to the firm: the circular and cumulative causation model, increasing return internal to the firm: the new economic geography |
| <b>Territorial Competitiveness and Endogenous Growth</b>               | Endogenous growth and increasing returns, the endogenous sources of growth: the knowledge stock and learning                                                                                                                                                   |

### Teaching Methodology:

- To deliver lectures on topics included in course outline
- To require each student to solve independent assignments on topics included in the course.

### Evaluation Criteria:

| <b>Evaluation Method</b> | <b>Total Percentage</b> |
|--------------------------|-------------------------|
| Quizzes/Assignments      | 25%                     |
| Mid-Term Exam            | 35%                     |
| Final-Term Exam          | 40%                     |

### Recommended Books:

- Capello, R. (2007). Regional Economics. New York: Routledge.
- Hoover, E. M., & Giarratani, F. (1999). An introduction to Regional Economics (3rd ed.). West Virginia: Knopf.
- Richardson, H.W. (1972). Input-Output & Regional Economics. London: Weidenfeld and Nicolson.
- Temple, M. (1994). Regional Economics. Great Britain: St. Martin's Press.
- Armstrong, Harvey, (2000). Regional Economics and Policy, Black wall Publishing Company.
- Choi, Rok-Byung, (2003). High Technology Development in Regional
- Davis H., Craig. (1990). Regional Economics, Impact Analysis and Project Evaluation, Amazon.com.
- Hijkamp (Editor). Handbook of Regional and Urban Economics, North Holland.
- Lee Roger (ed). (1997). Geographics of Economics, John Wiley and Sons. 106
- Mcker David L., (1970). Regional Economics: Theory and Practice, The Free Press, New York.
- Philip McCann, (2001). Urban and Regional Economics, Oxford University Press





## **Compulsory Courses (Non-Economics)**

**Code: ISE-111**

**Title: Islamiyat**

**Credit Hours: 02**

### **Course Objectives:**

This course is aimed:

- To provide basic information about Islamic Studies
- To enhance understanding of the students regarding Islamic Civilization
- To improve Students skill to perform prayers and other worships
- To enhance the skill of the students for understanding the issues related to faith and religious life

### **Course Contents**

#### **Introduction to Quran Studies**

- Basic Concepts of Quran
- History of Quran
- Uloom-ul –Quran

#### **Study of Selected Text of Holly Quran I**

- Verses of Surah Al-Baqra related to Faith (Verse No. 284-286)
- Verses of Surah Al-Hujrat related to Adab Al-Nabi (Verse No. 1-18)
- Verses of Surah Al-Mominoon related to Characteristics of faith (Verse No. 1-11)
- Verses of Surah al-Furqan related to Social Ethics (Verse No. 63-77)
- Verses of Surah Al-anam related to Ahkam(Verse No.152-154)

#### **Study of Selected Text of Holy Quran II**

- Verses of Surah Al-Ahzab related to Adab al-Nabi (Verse No. 6, 21, 40, 56, 57, 58.)
- Verses of Surah Al-Hashar (18, 19, 20) related to thinking, Day of Judgment
- Verses of Surah Al-Saf related to Tafakar,Tadabar (Verse No. 1,14)

#### **Seerat of Holy Prophet (S.A.W) I**

- Life of Muhammad Bin Abdullah (before Prophet hood)
- Life of Holy Prophet (S.A.W) in Makkah
- Important Lessons derived from the life of Holy Prophet in Makkah

#### **Seerat of Holy Prophet (S.A.W) II**

- Life of Holy Prophet (S.A.W) in Madina
- Important Events of Life of Holy Prophet in Madina
- Important Lessons derived from the life of Holy Prophet in Madina

#### **Introduction to Sunnah**

- Basic concepts of Hadith

- History of Hadith
- Kinds of Hadith
- Uloom-ul-Hadith
- Sunnah & Hadith
- Legal Position of Sunnah

### **Selected Study from Text of Hadith Introduction to Islamic Law & Jurisprudence**

- Basic Concepts of Islamic Law & Jurisprudence
- History & Importance of Islamic Law & Jurisprudence
- Sources of Islamic Law & Jurisprudence
- Nature of Differences in Islamic Law
- Islam and Sectarianism

### **Islamic Culture & Civilization**

- Basic concepts of Islamic Culture & Civilization
- Historical development of Islamic Culture & Civilization
- Characteristics of Islamic Culture & Civilization
- Islamic Culture & Civilization and Contemporary Issues

### **Islam & Science**

- Basic concepts of Islam & Science
- Contributions of Muslims in the Development of Science
- Quran & Science

### **Islamic Economic System**

- Basic Concepts of Islamic Economic System
- Means of Distribution of wealth in Islamic Economics
- Islamic concept of Riba
- Islamic Ways of Trade & Commerce

### **Political System of Islam**

- Basic concepts of Islamic Political System
- Islamic concept of Sovereignty
- Basic Institutions of Govt. in Islam

### **Islamic History**

- Period of Khlafat-e-Rashida
- Period of Umayyads
- Period of Abbasids

### **Social System of Islam**

- Basic concepts of Social System of Islam
- Elements of Family
- Ethical Values of Islam

### **Recommended Books:**

- Hameed ullah Muhammad. Emergence of Islam. IRI, Islamabad
- Hameed ullah Muhammad. Muslim Conduct of State.
- Hameed ullah Muhammad. Introduction to Islam.
- Hussain Hamid Hassan. An Introduction to the Study of Islamic Law. Leaf Publication Islamabad, Pakistan.
- Ahmad Hasan. (1993). Principles of Islamic Jurisprudence. Islamic Research Institute, International Islamic University, Islamabad.
- Mir Waliullah. (1982). Muslim Jurisprudence and the Quranic Law of Crimes. Islamic Book Service.
- H.S. Bhatia. (1989). Studies in Islamic Law, Religion and Society. Deep & Deep Publications New Delhi.
- Dr. Muhammad Zia-ul-Haq. (2001). Introduction to Al Sharia Al Islamia. Allama Iqbal Open University, Islamabad

### **Code: ISE-111**

**Title: Ethics (For Non-Muslims)**

**Credit Hours: 02**

### **Course Objectives:**

This course is designed for Non-Muslims and it introduces them various concepts in the daily life.

### **Course Contents**

Definition of Ethics, Different Concepts of Ethics (Ancient and Contemporary), Types of Ethics, Good Ethics, Bad Ethics

### **Importance of Ethics in Human Life**

Individual Life, Family Life, Social Life, Importance of Ethics in Economic Life, Importance of Ethics in Politics

### **Ethical Teachings and Values in Different Religions**

- Hinduism
- Buddhism
- Zoroastrianism
- Christianity
- Judaism
- Sikhism
- Islam

### **Ethical Values of the above Religions**

- Truthfulness
- Trustworthiness

- Service to Humanity
- Tolerance, Endurance
- Respect for Others
- Cooperation, Mutual Help, Selflessness
- Justice (Social Justice, Economic Justice)
- Equality

### **Concept of Virtue and Evil in Different Religions**

### **Concept of “Flah” in Different Religions**

### **Attitude Towards Other Religions**

### **Recommended Books:**

- John S. Mackenzie, A Manual of Ethics, 4th edition. Hinds, Hayden & Eldredge, Inc.
- Harold H. Titus. (1976). Ethics for Today, 5th edition Van Nostrand Reinhold Co.
- B. A. Dar. (1970). Quranic Ethics, Orientalia Art Ltd.
- Muhammad Hamidullah, Introduction to Islam, Darul Ishaat Pakistan.
- Syed Ameer Ali, The Spirit of Islam

**Code: PST-111**

**Title: Pakistan Studies**

**Credit Hours: 02**

### **Course Objectives:**

Develop vision of historical perspective, government, politics, contemporary Pakistan, ideological background of Pakistan. Study the process of governance, national development, issues arising in the modern age and posing challenges to Pakistan.

### **Course Contents**

#### **Historical Perspective**

- Ideological rationale with special reference to Sir Syed Ahmed Khan, Allama Muhammad Iqbal and Quaid-i-Azam Muhammad Ali Jinnah.
- Factors leading to Muslim separatism
- People and Land
  - Indus Civilization
  - Muslim advent
  - Location and geo-physical features.

#### **Government and Politics in Pakistan**

Political and constitutional phases:

- 1947-58
- 1958-71

- c. 1971-77
- d. 1977-88
- e. 1988-99
- f. 1999 onward

### **Contemporary Pakistan**

- a. Economic institutions and issues
- b. Society and social structure
- c. Ethnicity
- d. Foreign policy of Pakistan and challenges
- e. Futuristic outlook of Pakistan

### **Recommended Books:**

- Burki, Shahid Javed. (1980). State & Society in Pakistan, The Macmillan Press Ltd.
- Akbar, S. Zaidi. (2000). Issue in Pakistan's Economy. Karachi: Oxford University Press.
- S.M. Burke and Lawrence Ziring (1993). Pakistan's Foreign policy: An Historical analysis. Karachi: Oxford University Press.
- Mehmood, Safdar. (1994). Pakistan Political Roots & Development. Lahore.
- Wilcox, Wayne. (1972). The Emergence of Bangladesh., Washington: American Enterprise, Institute of Public Policy Research.
- Mehmood, Safdar. Pakistan Kayyun Toota, Lahore: Idara-e-Saqafat-e- Islamia, Club Road
- Amin, Tahir. Ethno - National Movement in Pakistan, Islamabad: Institute of Policy Studies, Islamabad.
- Ziring, Lawrence. (1980). Enigma of Political Development. Kent England: WmDawson & sons Ltd.
- Zahid, Ansar. (1980). History & Culture of Sindh. Karachi: Royal Book Company

## **Course Code: STAT-101A**

**Title: Elementary Statistics**

**Credit Hours: 03**

### **Course Objectives:**

The aim of this course is to familiarize and give students a thorough understanding of statistics particularly topics related to process of statistical solution to a problem, descriptive statistics including classification, organization and graphical presentation of data, measures of central tendency, dispersion and a survey of basic probability concepts.

### **Course Contents**

#### **Introduction**

Descriptive and inferential statistics, basic terminologies including variable and constant, population and sample, parameter and statistic. Sequence of statistical solution to a problem. Type of measurement scales: nominal, ordinal, interval and ratio. Types of data: univariate, bivariate and multivariate data, primary and secondary data, quantitative data and qualitative data, time series, cross-sectional and pooled data. The concept of significant digits and rounding off numbers; errors: biased and unbiased.

### **Presentation of Data**

Classification and tabulation, the frequency distribution, the cumulative frequency distribution, the relative frequency distribution, The percentage frequency distribution; Graphic and diagrammatic representation: simple and multiple bar charts, pie chart, histograms and frequency curves.

### **Measure of Central Tendency**

Types of averages for grouped and ungrouped data including arithmetic mean, weighted mean, geometric mean, harmonic mean, median, quartiles, deciles, percentiles, mode. The summary measures and box plot. Empirical relation between mean, median and mode. Advantages and limitations of mean, median and mode.

### **Measures of Dispersion, Skewness and Kurtosis**

Absolute and relative measure of dispersion including range, quartile deviation, mean deviation, variance, standard deviation and coefficient of variation for grouped and ungrouped data. Definition and interpretation of variance and standard deviation, computation of variance and standard deviation, properties of standard deviation and variance; skewness: Karl Pearson's coefficient of skewness, Bowley's coefficient of skewness; kurtosis and chebyshev's theorem. Covariance and correlation analysis.

### **Basic Probability Concepts**

A survey of probability concepts: classical probability, empirical concept, subjective probability. Some rules of probability: rules of addition, rules of multiplication, conditional probability, counting rules: the multiplication formula, the permutation formula, the combination formula.

### **Recommended Books:**

- Anderson, D. R., Sweeney, D. J., Williams, T. A., Camm, J. D., & Cochran, J. J. (2014). Essentials of statistics for business and economics. Cengage Learning.
- Lind, Douglas A., Marshal, William G. and Mason, Robert D., (2015) Statistical Techniques in Business and Economics (16th edition). Boston: McGraw Hill, 2003.



- Robert Johnson and Patricia Kuby (Latest issue), “Elementary Statistics”, Thomson Brooks/COLE.

**Course Code: STAT-102****Title: Probability and Probability Distributions****Credit Hours: 03****Prerequisite: Elementary Statistics****Course Objectives:**

This course is designed to equip students with higher statistical tools and their application in economic analysis.

**Course Contents:****Random Variables and Discrete Probability Distribution**

Random variables, discrete random variable, continuous random variable, discrete probability distribution; The mean, variance and standard deviation of a probability distribution; binomial probability distribution, and its computation. Cumulative probability distributions, properties of binomial probability distribution.

**Continuous Probability Distribution**

Continuous probability distribution: the normal probability distribution: properties of normal distribution, Applications of the standard normal distribution, finding areas under the normal curve.

**Sampling and Sampling Distributions**

What is sampling? Defining population, determining sampling frame, sampling design (probability versus non-probability sampling) and appropriate sample size. Issues of precision and confidence in determining a sample size. Sampling with and without replacement, sampling and non-sampling error, sampling bias; sampling distribution of the mean; The central limit theorem; sampling distribution of differences between means; sampling distribution of sample proportion; sampling distribution of differences between proportions.

**Estimation and Confidence Intervals**

Point estimates and confidence intervals; estimation by confidence interval: confidence interval estimate of a population mean (known variance), confidence interval estimate of a population mean (unknown variance) confidence interval for differences of means,

confidence interval for differences of means; confidence interval for population proportion, confidence interval for differences between proportions.

### **Hypothesis Testing**

One sample test of hypothesis; one sample; one tail and two tails tests of significance; testing for a population mean with a known population standard deviation: two-tailed test, one-tailed test; P-value in hypothesis testing; testing for a population mean: large sample, population standard deviation unknown.

### **Chi Square Applications**

Introduction; goodness-of-fit test: equal expected frequencies; goodness-of-fit test: Unequal expected frequencies; limitations of Chi square.

### **Analysis of Variance**

Introduction, the F-distribution; comparing two population variances; ANOVA assumptions; ANOVA test; inferences about pairs of treatment means; two-way analysis of variance.

### **Recommended Books:**

- Anderson, D. R., Sweeney, D. J., Williams, T. A., Camm, J. D., & Cochran, J. J. (2014). Essentials of statistics for business and economics. Cengage Learning.
- Anderson, D. R., Williams, T. A., & Sweeney, D. J. (2011). Statistics for Business and Economics. 12th. Cengage Learning.
- Lind, Douglas A., Marshal, William G. and Mason, Robert D., (2015) Statistical Techniques in Business and Economics (16th edition). Boston: McGraw Hill, 2003.

## **Course Code: CAL-111**

**Title: Calculus-I**

**Credit Hours: 03**

**Prerequisite: Mathematics at Secondary Level**

### **Course Objectives:**

To prepare the students with the essential tools of algebra/calculus to apply the concepts and the situations in Economics.

### **Course Contents**

#### **Preliminaries**

Demystifying Mathematics and Mathematical Economics, Mathematics vs. Non-Mathematical Economics, Mathematical Economics versus Econometrics. Ingredients of a Mathematical Model: Variables and Economic Variables, Constants and Parameters. A Few Aspects of Logic: Proportions, Implications and Necessary and Sufficient Conditions. The Real-Number System. Use of Sets in Economics. Cartesian Coordinates.

### **Functions and Equations:**

Functions, Domain and Range, Relations. Schultz Demand Function, Laffer Curve. Population and Consumption Functions, Quadratic Cost Function and Profit Function of a Monopoly, Quadratic Production Possibilities Frontier, Cubic Cost Functions, Rectangular Hyperbolic Functions, Population Growth using General Exponential Functions, Population Growth using Natural Exponential Functions, Equations and Identities, Definitional, Behavioral and Conditional Equations with Economic examples. Structural and Reduced Form Equations. Partial Linear Market Equilibrium and effects of Tax, shifts in Demand and supply curves. A Nonlinear Model, Numerical Solution of Two Good Case. National Income Equilibrium, extension of Induced and Autonomous Tax, Proportion of Government Expenditure.

### **Matrices**

Matrices and Vectors, Matrix Operations. Commutative, Associative and Distributive Laws. Vector Operations, Transpose, Cofactors, Adjoint, Determinant of a Matrix, Inverse of a Matrix, Minors and Cofactors. Market Model Analysis using Matrix Inversion Method. National Income Analysis using Matrix Inversion Method. Market Model and National Income Determination using Cramer's Rule.

### **Homogeneous Functions**

Homogeneous Functions, Homogeneity of CES Production Function, Cobb-Douglas Production Function and Returns to Scale. Expansion Path using First Order Condition. Homothetic Functions, Homotheticity of Cobb-Douglas Production Function.

### **Differentiation**

The Need and Nature of Comparative Statics, Concept of Limit and Continuity, Rate of Change, Slope & Derivative, Differentiation Rules for Single Variable Functions: Constant Function Rule & Power Function Rule, Sum-Difference Rule of Differentiation, Numerical Analysis of Cost Function, Product Rule of Differentiation, Quotient Rule of Differentiation, Marginal Cost, Marginal Revenue and Marginal Product Analysis. Chain Rule and Marginal Revenue Product of Labor (MRPL) Analysis.

### **Recommended Books:**

- Budnick, Frank, Applied Mathematics for Business, Economics and Social Sciences. - Latest Edition.
- Chiang A.C. Fundamental Methods of Mathematical Economics McGraw Hill –Latest Edition.
- Dowling Edward T. Mathematics for Economics Schaum Series, Latest Edition.
- Weber E. Jean, Mathematical Analysis, Business and Economic Application (latest edition), Harper and Row Publishers, New York.

## **Course Code: CAL-211**

**Title: Calculus-II**

**Credit Hours: 03**

**Prerequisite: Calculus-I**

### **Course Objectives:**

To prepare the students with the essential tools of calculus to apply the concepts and the techniques in Economics.

### **Course Contents**

#### **Partial & Total Differentiation**

Partial Differentiation. Marginal Physical Product of Labor and Capital. Marginal Utility Functions. Money Market Analysis, Demand for Money Function Analysis using Partial Derivatives. Income Elasticity of Demand and Cross Price Elasticity of Demand using Partial Derivatives.

Second and Higher Order Derivatives. Profit Maximization Condition using Second Derivative. Young's Theorem. Partial Derivatives Application on Three Input Production Function

Differentials versus Derivatives. Point Elasticity using Differentials, Income and Price Elasticity of Demand using Differentials. Concept of Total Differentials. Savings Function and Total Differentials. Specific Utility Function and Total Differentials.

Concept of Total Derivatives. Specific Production Function with time-dependent Labor and Capital. Exponential Functions and Growth. Rules of Differentiation of Exponential and Logarithmic Functions. Optimal Timing: A Problem of Wine Storage, A Problem of Timber Cutting, A Problem of Land Purchase for Speculation.

Finding the Rate of Growth using Exponential and Logarithmic Functions. Point Elasticity. Rate of Growth of Per Capita Employment. Rate of Growth of Profit.

#### **Optimization**

Concept of Optimization: Calculus Approach to Optimization, 1<sup>st</sup> Order and 2<sup>nd</sup> Order Conditions. Average Cost Minimization Analysis. Matrix Approach to Optimization (2<sup>nd</sup> Order) Test: Hessian. Profit Maximization Analysis, Profit Maximization of Technically Related Goods, Monopolistic Firm Producing Related Goods, Firm Producing Substitute Goods. Short Run Production Function Analysis.

Optimization of More than one Choice Variable, Economic Application on Multi-product Firm, Multi-plant Firm, Price Discrimination by Monopoly, Price Discrimination by Monopsony, Input Decision of a Firm, Profit Maximization of Two-Product Firm.

Comparative-Static Aspects of Optimization, Rationale for Constrained Optimization. Finding Stationary Values using Substitution-Elimination Method. Finding Stationary Values Method of Lagrange Multiplier. Interpretation of the Lagrange Multiplier. Second Order Condition: The Bordered Hessian. Convexity and Concavity using Second Order Derivative. Economic Applications: Utility Maximization, Law of Equi-marginal Utility using Lagrangian Multiplier, Production Function Maximization, Logarithmically Transformed Production Function.

### **Integral Calculus**

Dynamics and Integration. The Nature of Indefinite Integrals. Basic Rules of Integration. The Substitution Rule and the Rule of Integration by Parts. Economic Applications of Integrals: Finding Total Functions from Marginal Functions, Investment & Capital Formation, Utility Function. Definite Integrals. Major Properties of Definite Integral. A Definite Integral as an Area Under a Curve. Improper Integrals.

### **Consumer Analysis using Integrals**

Applications of Definite Integrals: Consumer's Reservation Price using Differential Equations, Consumer Willingness to Pay using Integrals, Consumer Surplus using Integrals, Differential of Consumer Surplus using Integrals, Producer & Social Surplus using Integrals, Producer Surplus with Tax using Integrals.

### **Firm Related Microeconomic Applications of Integrals**

Wine Storage Problem with Storage Costs. Breakeven Analysis of a Firm. Scale of Production Analysis. Mobile Production Analysis. Firm's Average Profit Analysis. Firm's Average Cost Analysis. Average Production Analysis. Average Inventory Analysis. Advertisement Analysis. Profit over Useful Life of Machine. Oil Extraction Analysis, Dynamics of Labour Cost.

### **Miscellaneous Microeconomic Applications of Integrals**

Efficiency Analysis using Integrals. Cost Effectiveness Decision using Differential Equations. Depreciation Analysis using Integrals. Retirement Annuity Analysis using Integrals. Wage Differential using Integrals. Present Value of Cash Flow. Present Value of a Perpetual Flow.

## Macroeconomic Applications of Integrals

Gini Index using Integrals. Depletion of Energy Resources Analysis using Integrals. Present Value of Gold Mine using Integrals. Balance of Trade Analysis using Integrals. Demand Analysis of Oil using Integrals. Demand Analysis of Oil using Integrals. Domar Growth Model: Framework, Solution and Numerical.

### Recommended Books:

- Chiang A.C. Fundamental Methods of Mathematical Economics McGraw Hill –Latest Edition.
- Weber E. Jean, Mathematical Analysis, Business and Economic Application (latest edition).
- Bradley, T. Essential mathematics for economics and business. Latest edition. John Wiley & Sons.
- Dowling, E. T. Theory and Problems of Introduction to Mathematical Economics. Latest Edition, McGraw-Hill, New York.
- Hoffman, L. D., & Bradley, G. L. Calculus for business, economics, and the social and life sciences. McGraw-Hill. Latest edition.
- Kolman, B., & Beck, R. E. (1995). Elementary linear programming with applications. Academic Press.
- Sydsæter, K. and Hammond, P. Essential Mathematics for Economic Analysis. Latest edition., Pearson Education Limited, England.

| Code    | Subject Title                                                                                                                                                                                                                                                                                                                      | Cr. Hrs | Semester |
|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|----------|
| ENG-111 | English-I (Language in Use)                                                                                                                                                                                                                                                                                                        | 3       | I        |
| Year    | Discipline                                                                                                                                                                                                                                                                                                                         |         |          |
| 1       | Botany, Zoology, Mathematics-I,II, Statistics-I,II,III, Chemistry-I,II, Applied Psychology, Business Administration, Commerce, Economics, English, Sociology & Sociocultural Studies, Social Work, Political Science, Physics, Mass Communication, Islamic Education, History, Education (Elementary), Education (Secondary), Urdu |         |          |

### Recommended Reading:

Eastwood, John. Oxford Practice Grammar. UK: Oxford University Press, 1992.

- Unit No. 76, 77,78,79,80,81,82,83,84,85,86,87,88,89,90. (ARTICLES)
- Unit No. 98, 99, 100, 101, 102 103. (PRONOUNS)
- Unit No. 104, 105, 106, 107, 110, 111, 112. (ADJECTIVES AND ADVERBS)
- Unit No. 118, 119, 120, 121, 122, 123, 124, 125, 126, 127. (PREPOSITIONS)
- Unit No. 150, 151, 152, 153. (LINKING WORDS)
- Appendix 3 Page No. 372

### Reading Comprehension and Summarizing Skills Recommended Reading:

Ellison, Peter and Gould, Mike. Focus on C o m p r e h e n s i o n . Volume 4. Singapore: Learner's Publishing Pte, 2009.

- Section 1: Unit No. 1.
- Section 2: Unit No. 2, 3, 4, 5. Practice Unit 1.
- Section 3: Unit No. 6, 7, 8, 9, 10, 11. Practice Unit 2.

### Paragraph Writing I

The students are required to know basics of Paragraph Writing with an emphasis on Topic sentences and Supporting sentences and a possible Concluding sentence. (Word Limit Up to 120 words)

### Recommended Reading:

Arnaudet, Martin L. and Barrett, MaryEllen. Paragraph Development: A Guide for Students of English as a Second Language. Prentice-Hall, Englewood Cliff, N.J. 1981 (Page 1

-32).



## Listening and Speaking Skills

This unit will lead up to the teaching and evaluation of “Oral Presentation Skills” in the following semesters also.

### Recommended Reading:

Eastwood, John. Oxford Practice Grammar. UK: Oxford University Press, 1992. Unit No. 34, 35, 36, 38, 39.

## Vocabulary Building Skills

GAT HIGH FREQUENCY WORDS (GAT Word List) Page No. 143-152

### Recommended Reading:

Idrees, Muhammad. Guide for GAT General Test. SMART BRAIN. GRE (General, Local) 2010-2011 Ed. Lahore: Dogar Brother Publishers, 2010.

**Note: The concepts listed in the syllabus contents may be acquired from sources other than those recommended.**

| Code    | Subject Title                                                                                                                                                                                                                                                                                                                      | Cr. Hrs | Semester |
|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|----------|
| ENG-112 | English-II (Academic Reading and Writing)                                                                                                                                                                                                                                                                                          | 3       | II       |
| Year    | Discipline                                                                                                                                                                                                                                                                                                                         |         |          |
| 1       | Botany, Zoology, Mathematics-I,II, Statistics-I,II,III, Chemistry-I,II, Applied Psychology, Business Administration, Commerce, Economics, English, Sociology & Sociocultural Studies, Social Work, Political Science, Physics, Mass Communication, Islamic Education, History, Education (Elementary), Education (Secondary), Urdu |         |          |

## Basics of Grammar II

### Recommended Reading:

Eastwood, John. Oxford Practice Grammar. UK: Oxford University Press, 1992. Unit No. 4, 5, 8, 9, 11, 12, 13, 16, 18, 20, 22, 23, 24, 26, 28, 29, 30 (Tenses) Unit No. 128, 129, 130, 131 (Phrasal Verbs).

## Reading and Critical Thinking

### Recommended Reading:

Ellison, Peter and Gould, Mike. Focus on Comprehension. Volume 4. Singapore: Learner's Publishing Pte, 2009. Section 4: Unit No. 12, 13, 14, 15. Practice Unit 3.

Section 5: Unit No 16, 17, 18.

Section 6: Unit No. 19. Practice Unit 4.

### **Paragraph Writing II**

The students are required to know basics of Paragraph Writing with an emphasis on Topic sentences and Supporting sentences and a possible Concluding sentences. (Word Limit Up to 120 words)

#### **Recommended Reading:**

Arnaudet, Martin L. and Barrett, MaryEllen. Paragraph Development: A Guide for Students of English as a Second Language. Prentice-Hall, Englewood Cliff, N.J. 1981. (Page 179-185).

#### **Study Skills**

The students are expected to be proficient in Reading Skills like Skimming, Scanning, Speed Reading and avoiding Faulty Reading Habits.

#### **Recommended Reading:**

Aziz, Nadeem. English Language Communication Skills for B.Ed. Pakistan: Majeed Book Publishers, (Page 139-159).

#### **Vocabulary Building Skills**

WORD ROOT METHOD Unit 1-6. Page No. 88-103.

#### **Recommended Reading:**

Idrees, Muhammad. Guide for GAT General Test. SMART BRAIN. GRE (General, Local) 2010-2011 Ed. Lahore: Dogar Brother Publishers, 2010.

**Note: The concepts listed in the syllabus contents may be acquired from sources other than those recommended.**

| Code    | Subject Title                                                                                                                                                                                                                                                                          | Cr. Hrs | Semester |
|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|----------|
| ENG-211 | English-III (Communication Skills)                                                                                                                                                                                                                                                     | 3       | III      |
| Year    | Discipline                                                                                                                                                                                                                                                                             |         |          |
| 2       | Botany, Zoology, Mathematics-I,II, Statistics-I,II,III, Chemistry-I,II, Applied Psychology, Economics, Sociology & Sociocultural Studies, Social Work, Political Science, Physics, Mass Communication, Islamic Education, History, Education (Secondary), Urdu, Education (Elementary) |         |          |

### **Formal Letters**

The students are expected to be proficient in formal letter writing like Letters to the editor, public officials (WAPDA, WASA etc.)

#### **Advanced Reading and Comprehension I**

The students are required to read the given prose critically and answer the questions.

#### **Recommended Reading:**

Rise. B. Axelrod. and Cooper, Charles R. *The St. Martin's Guide to Writing* New York: St. Martin's Press, 1985.

(Page 18-19, 26-34, 49-55, 66-67, 77-80, 88-94, 104-105, 110-115, 129-137).

### Oral Presentations

The students must be taught strategies for oral presentations. This will prepare them for formal presentations. In order to make them proficient, the teacher should assign them presentations in class.

### Recommended Reading:

Murphy, Herta A. *Effective Business Communication*. 7th Ed. New Delhi: Tata McGraw-Hill Publishing Company Limited, 2009. (Page 384-399).

### Connected Paragraph Writing and Picture Description

The students are required to practice paragraph writing with an emphasis on Topic sentence and Supporting sentences. The students are supposed to write **at least 3** connected paragraphs on a single theme (word limit: 350 words). The students are required to learn how to analyze and describe pictures in correct English.

### Recommended Reading:

Arnaudet, Martin L. and Barrett, Mary Ellen. *Paragraph Development: A Guide for Students of English as a Second Language*. Prentice-Hall, Englewood Cliff, N.J., 1981. (Page 179-185).

### Vocabulary Building Skills

WORD ROOT METHOD Unit 7-11. Page No. 103-116.

### Recommended Reading:

Idrees, Muhammad. *Guide for GAT General Test. Smart Brain GRE (General, Local)*. 2010-2011 ed. Lahore: Dogar Brother Publishers, 2010.

| Code    | Subject Title                                                                                                                                                                                                                                       | Cr. Hrs | Semester |
|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|----------|
| ENG-212 | English-IV (English for Practical Aims)                                                                                                                                                                                                             | 3       | IV       |
| Year    | Discipline                                                                                                                                                                                                                                          |         |          |
| 2       | Botany, Zoology, Mathematics-I,II, Statistics-I,II,III, Chemistry-I,II, Applied Psychology, Economics, Social Work, Political Science, Physics, Mass Communication, Islamic Education, History, Education (Secondary), Urdu, Education (Elementary) |         |          |

### Professional Correspondence

CV and covering letter.

Follow up messages after the job interview.

### Recommended Reading:

Murphy, Herta A. *Effective Business Communication*. 7th Ed. New Delhi: Tata McGraw-Hill Publishing Company Limited, 2009 (Page 504-529, 540-548).

### Advanced Reading and Comprehension II

The students are required to read the given prose critically and answer the questions.

### Recommended Reading:

(Rise. B. Axelrod. and Cooper, Charles R. *The St. Martin's Guide to Writing* New York:

St. Martin's Press, 1985. Page 146-147, 152-155, 158-172).

### **Job Interviews**

The students should learn to handle job interviews through “mock interviews”.

### **Recommended Reading:**

Murphy, Herta A. *Effective Business Communication*. 7th Ed. New Delhi: Tata McGraw-Hill Publishing Company Limited, 2009. (Page 539-539).

### **Essay Writing**

The students should be able to compose essays of 4 to 6 paragraphs relying on what they have learnt in the previous semesters about paragraph writing. (Word Limit about 500 words).

### **Vocabulary Building Skills**

WORD ROOT METHOD Unit 12-17. Page No. 116-131.

### **Recommended Reading:**

Idrees, Muhammad. *Guide for GAT General Test. Smart Brain GRE (General, Local)*. 2010-2011 ed. Lahore: Dogar Brother Publishers, 2010.

| Code     | Subject Title                                                                                                                                                                      | Cr. Hrs | Semester |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|----------|
| Math-111 | Elementary Mathematics I (Algebra)                                                                                                                                                 | 3       | I        |
| Year     | Discipline                                                                                                                                                                         |         |          |
| 1        | Business Administration, Economics, History, Botany, Zoology, Chemistry I, Social Work, Political Science, Mass Communication, Education (Elementary), Education (Secondary), Urdu |         |          |

### **Course Objectives:**

Specific Objectives of the Course: To prepare the students, not majoring in mathematics, with the essential tools of algebra to apply the concepts and the techniques in their respective disciplines.

### **Course Outline:**

Preliminaries: Real-number system, complex numbers, introduction to sets, set operations, functions, types of functions. Matrices: Introduction to matrices, types, matrix inverse, determinants, system of linear equations, Cramer's rule.

Quadratic Equations: Solution of quadratic equations, qualitative analysis of roots of a quadratic equation, equations reducible to quadratic equations, cube roots of unity, relation between roots and coefficients of quadratic equations.

Sequences and Series: Arithmetic progression, geometric progression, harmonic progression. Binomial Theorem: Introduction to mathematical induction, binomial theorem with rational and irrational indices. Trigonometry: Fundamentals of trigonometry, trigonometric identities.

### Recommended Books:

- Dolciani MP, Wooton W, Beckenback EF, Sharron S, Algebra 2 and Trigonometry, 1978, Houghton & Mifflin,
- Kaufmann JE, College Algebra, and Trigonometry, 1987, PWS-Kent Company, Boston
- Swokowski EW, Fundamentals of Algebra and Trigonometry (6th edition), 1986, PWS-Kent Company, Boston
- 

| Code     | Subject Title                                                                                                                                                                                                                                                               | Cr. Hrs | Semester |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|----------|
| COMP-111 | Computer (Introduction & Applications)                                                                                                                                                                                                                                      | 3       | I        |
| Year     | Discipline                                                                                                                                                                                                                                                                  |         |          |
| 1        | Botany, Zoology, Mathematics-I,II, Statistics-I,III, Chemistry-I,II, Business Administration, Economics, English, Sociology & Sociocultural Studies, Social Work, Political Science, mass Communication, History, Education (Secondary), Urdu, Education (Elementary), Urdu |         |          |

### Course Contents

Introduction to computers, Characteristics of computers, Components of computers, Categories of computers, Computer applications/uses in society, Components of system unit; Presenting the Internet, Communicating through the Internet; Interacting with the computer (Input devices), Using the Keyboard and Mouse; Seeing, Hearing, and Printing Data (Output devices), Video and Sound, Printing; Processing Data, Transforming Data into Information; Storing Data, Types of Storage Devices; Operating Systems, Operating System Basics, Survey of PC and Network Operating Systems; Working with Application Software, Productivity Software, Graphics and Multimedia Software; Networks, Types of computer networks, Data communication; Database Management, Database Management Systems, Survey of Database Systems; Development of Information Systems, Basics of Information Systems; Information Assurance: Security, Privacy, and Ethics.

Microsoft Office: introduction to word, creating, saving, closing and opening a file, difference between save and save, moving through the word document, selecting, cut, copy and paste through shortcuts and edit menu, undo and repeat, word menu bar and drop down menu, tool bar and status bar. Formatting using menu and formatting bar, paragraph formatting,

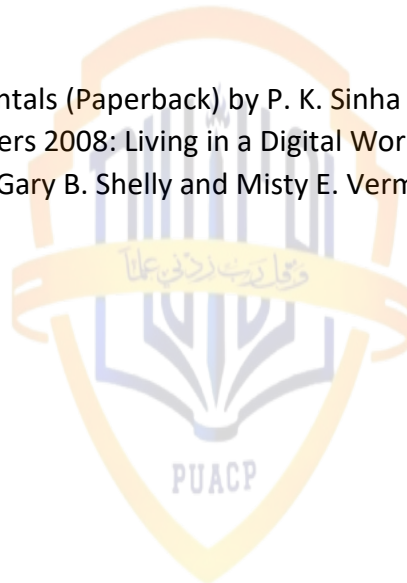
borders and shading, spell checking, creating and manipulating table; introduction to Excel, creating, saving closing and opening a worksheet, entering data, calculations using operators , using Functions, IF statement, creating and manipulating charts; Introduction to Power Point, presentations, creating a new blank presentation, presentation type, different view icons, new, outline, slide and slide show, slide animations, animation effect, slide timings, picture and backgrounds, drawing tools, selecting object, grouping and ungrouping, resizing and rotating objects.

**Text Books:**

- Peter Norton's Introduction to Computers, 6th Edition
- Microsoft Office 2003: The Complete Reference by Jennifer Kettell, Guy Hart-Davis and Curt Simmons

**Recommended Books:**

- Computer Fundamentals (Paperback) by P. K. Sinha
- Discovering Computers 2008: Living in a Digital World, Fundamentals (Shelly Cashman Series) by Gary B. Shelly and Misty E. Vermaat



## **Optional Courses: (Non-Economics)**

**Code: SOC-211**

**Title: Introduction to Sociology**

**Credit Hours: 03**

### **Course Objectives:**

The objective of this course is to introduce the students with the fundamentals of the subject of Sociology

### **Course Contents**

#### **Introduction**

What is sociology?

Sociology and other social sciences

Sociology and common sense / Stereotypes

Key terms – Social Problems / Sociological Issues

Sociological Perspective

#### **Structuralism**

Structural Functionalism

Social Conflict

#### **Social Action**

Symbolic Interactionism

#### **Culture**

What is culture?

Elements of Culture

Cultural Diversity

#### **Socialization**

Socialization and its importance

Agents of Socialization

Socialization through the life course

(i) C. H. Cooley

(ii) George Herbert Mead





## **Social Structure**

(i) Status

(ii) Status Set

(iii) Achieved and Ascribed

(iv) Master Status

(v) Role

(vi) Role Set

(vii) Role Conflict

(viii) Role Strain

Theories of Social Interaction (Brief)

(i) Social Construction of Reality

(ii) Ethno Methodology

## **Groups and Organizations**

Types of Groups

Studies of Group Behaviour

Bureaucracy and its Characteristics

## **Deviance**

Deviance, Crime and Social Control

Types of Crime

## **Recommended Books:**

- Anderson, Margaret and Howard F. Taylor. (2001) Sociology the Essentials. Australia: Wadsworth.
- Brown, Ken. (2004). Sociology. UK: Polity Press
- Gidden, Anthony (2002). Introduction to Sociology. UK: Polity Press. Macionis, John J. (2006). Sociology. 10th ed. New Jersey: Prentice-Hall
- Tischler, Henry L. (2002). Introduction to Sociology. 7th ed. New York: The Harcourt Press.

**Code: EDU-111**

**Title: Introduction to Education**

**Credit Hours: 03**

**Course Objectives:**

This course will focus on the ideological, philosophical, psychological, socio-economic and historical foundations of education. The major focus will be on developing an understanding of the students, how different philosophical theories affect education. The course will also include historical development of education of the Pakistan. Emphasize will be given on analyzing various sociological, political, economic and ideological forces that influence the process of education in our culture. The students will be able to: Explain the important features of foundation of education, Specify the role of educational thinkers in education, Discuss the modes of education, Evaluate the issues and problems of education.

**Course Contents**

**Ideological Foundations of Education**

Islamic Foundations, Islamic concept of Peace, Other religions and Islam

Ideology and teachers

**Philosophical Foundations of Education**

Philosophy and Education

Main Philosophical Thoughts

Idealism

Realism

Pragmatism

Re-constructionism

**Psychological Foundations of Education**

Learning and Maturation

Individual Differences

Self-Concept

Academic Aptitude

Instructional Strategies and Psychology

**Socio-Economic Foundations of Education**

Concept of Society and Culture

Social Conditions and Education

Economic Conditions and Education

Politics and Education

### **Historical Foundations of Education in Pakistan**

Pre-Pakistan Period (712 A.D. to onward)

Period from 1947-1958

Period From 1959-1971

Period from 1972-1979

Period from 1980 -1991

Period from 1992 – to date

### **Aims of Education**

Aims, Goals and Objectives

Taxonomies of Objectives

Aims and Objectives of Education in Pakistan

### **Problems and Issues in Education in Pakistan**

Universalization of Primary Education, Literacy, Medium of Instruction, Diversification of Education, Environmental Education, Gender and Education

Islamization of Education

Special Education

Health Education / Drug Education

### **Recommended Books:**

- Canestrari, A. (2009). Foundations of Education. New York: Sage Publications.
- Eugene, F.P. (2005). Critical issues in education: Anthology of reading. New York: Sage Publications.
- Goldblatt, P.F., & Smith, D. (2005). Cases for teacher development. New York: Sage Publications.
- Holt, L.C. (2005). Instructional patterns: Strategies for maximizing students learning.
- Murphy, D. (2005). Foundations/Introduction to Teaching. USA: Allyn & Bacon, Inc. New York: Sage Publications.
- Semel, S. F. (2010). Foundations of education: The essential texts. USA: Routledge.

**Code: APSY-111A****Title: Introduction to Psychology****Credit Hours: 03****Course Objectives:**

The main aim is to familiarize students with history, main concepts, methods, and theoretical

frameworks in psychology.

The course will help students appreciate the human nature and its related concepts, thereby will gain insight into human behavior and human relationships.

**Course Outcome**

After successful completion of this course the students will be able to:

Have a grasp over basic concepts and theoretical perspectives explaining human behavior. They will be able to appreciate the complexity of human behavior and relationships.

They will be able to understand Psychology as science and empirical methods used for understanding different aspects of human behavior.

**Course Contents****Understanding Psychology**

Psychology: Scientific perspective

Historical perspective

Schools of psychology

Methods of psychology

Ethical issues

Fields of psychology and their application

**Biological Basis of Behavior**

Neuron and its function

Central nervous system

Peripheral nervous system

Endocrine system

**Sensation and Perception**

Senses: Vision, audition, smell, taste and kinesthetic

Introduction to perception

Gestalt principles

Binocular and monocular cues

Illusions and extra sensory perception

**Learning**

Definition of learning

Types of learning: Classical and operant conditioning

Latent and observational learning

**Memory**

Definition and types of memory

Processes and techniques of improving memory

Forgetting: Nature and causes

**Cognition and Language**

Concept of cognition  
Problem solving  
Judgment and decision making  
Language development  
Language and cognition  
Language and culture

### **Intelligence and Creativity**

Concept of intelligence  
Theories of intelligence  
Assessment of intelligence  
Mental retardation  
Concept of creativity and its stages

### **Motivation and Emotion**

Introduction to motivation  
Factors affecting motivation  
Introduction to emotions  
Types of emotions  
Physiology and emotion  
Theories of emotion

### **Personality**

Defining personality  
Theories of personality  
Personality assessment

### **Social Thinking and Social Influence**

Social facilitation  
Attribution theory  
Crowd behavior  
Conformity, Obedience  
Helping behavior

### **Recommended Books:**

Atkinson R. C., & Smith, E. E. (2000). *Introduction to psychology* (13th ed.). NY: Harcourt Brace College Publishers.

Coon, D., & Mitterer, J. (2008). *Introduction to psychology: Gateways to mind and behavior* (12th ed.). USA: Wadsworth Cengage Learning.

Fredrickson, B., Nolen-Hoeksema, S., Loftus, G., & Wagenaar, W. (2009). *Atkinson & Hilgard's introduction to psychology* (15th ed.). USA: Wadsworth.

Glassman, W.E. (2000). *Approaches to psychology*. Open University Press.

Hayes, N. (2000). *Foundation of psychology* (3rd ed.). UK: Thomson Learning.

Kalat, J. W. (2010). *Introduction to psychology*. USA: Cengage Learning, Inc.

Lahey, B. B. (2004). *Psychology: An introduction* (8th ed.). UK: McGraw-Hill Companies, Inc.

Leahey, T. H. (1992). *A history of psychology: Main currents in psychological thought*. New Jersey: Prentice-Hall International, Inc.

Myers, D. G. (2011). *Psychology* (10th ed.). USA: Wadsworth Publishers.

Ormord, J. E. (1995). *Educational psychology: Developing learners*. USA: Prentice Hall, Inc.

Rathus, S. (2011). *Psychology: Concepts and connections* (10th ed.). USA: Wadsworth Cengage Learning.



**Code: HIS-111**

**Title: Introduction to History**

**Credit Hours: 03**

**Course Objectives:**

The objective of this course is to introduce the students with the fundamentals of the subject of History.

**Course Contents**

**History of Pakistan Movement 1858-1947**

Failure of the War of Independence and its effects, Condition of the Muslims. Government of India Act 1858. Indian Councils Act 1861. Sir Syed Ahmad Khan and the Aligarh Movement, His political and educational services. Indian National Congress. Syed Ameer Ali and the Central National Muhammadan Association. Indian Councils Act 1882, limitations and Impact on Muslims. Urdu-Hind controversy, Mohsin ul Mulk and formation of Mohammadan Political Organization. Partition of Bengal. Hindu reaction and annulment. Simla Deputation, Separate Electorate. Formation of All India Muslim League, its objects and early history, role of Nawab Saleem Ullah Khan, Waqar-ul-Mulk and Sir Agha Khan Minto-Morily Reforms of 1909 Lucknow Pact, its importance and drawbacks. The Government of India Act, 1919. Dyarchy and its failure. The Rowlatt Act Jalianwala Bagh Tragedy. Khilafat Movement. Ali Brothers and their role. The Constitution Problems of '1920's, Simon Commission, Nehru Report, Quaid-i-Azam's Fourteen Points, Round Table Conferences, Communal Award & Poona Pact. Government of India Act 1935. Congress Ministries and their attitude towards Muslims, Pirpur Report, Evolution of the idea of a separate Muslim state, Iqbal's Allahbad Address. The ideology of Pakistan. Struggle for the Pakistan; Lahore Resolution 1940, Different Proposals for the Partition of the India, Cabinet Mission Plan; Elections of 1945-46, Third June Plan, Simla Conference Indian Independence Act 1947, Radcliffe Award. Services of Quaid-e-Azam Mohammad Ali Jinnah for the cause of Pakistan

**Recommended Books:**

- Shtiaq Hussain Qureshi: Struggle for Pakistan, Urdu translation.
- Jamil-ud-Din Ahmad: Early Phase of struggle for Pakistan. Middle Phase of Struggle for Pakistan. Final Phase of Struggle for Pakistan.
- Ch. Mohammad Ali: Emergence of Pakistan, Urdu

**Code: GEOG-211**

**Title: Introduction to Geography**

**Credit Hours: 03**

**Course Objectives:**

The objective of this course is to introduce the students with the fundamentals of the subject of Geography

**Course Contents**

**Introduction and Definitions of Geography**

- (a) Themes of Geography
- (b) Roots of the discipline and basic geographic concepts
- (c) The evolution of geography from ancient to modern period
- (d) Branches of Geography and its relations with other disciplines

**The Universe**

- (a) Galaxies and Solar system
- (b) Origin of the Earth

**Earth as a planet and celestial positions, its Shape and size**

- (a) Rotation and revolution and related phenomena
- (b) Earths' Satellite Moon
- (c) Lunar and solar Eclipses

**Positions on Map and Globe**

- (a) Geographical coordinates and its characteristics
- (b) World time zones standard and local time

**Geological time scale**

- (a) Internal Structure of the Earth & Rocks.
- (b) Distribution of land and water,
- (c) Major Landforms (Mountain, Plateau and Plain)

**Atmospheric Structure and Composition**

Elements of Weather and Climate

**Ocean and its Configuration**

Movement of Oceans

**Biomes**

**Recommended Books:**



1. Christopherson, R.W. (2000), Geo-systems, Prentice-Hall, Inc, USA.
2. De Blij, H. J and Muller, P.O. (1996), Physical Geography of the global Environment, USA, John Wiley and Sons Inc.
3. Diwan A.P. & D.K. Arora (1995), Origin of the Ocean, Anmol Publisher, Delhi.
4. Gabler, R.E, Sager, R.J and Wise, D.L. (1997), Essentials of Physical Geography, Saunders College Publishing, New York.
5. Kendrew, (1961), Climates of the continents, Longman London/New York
6. King, CAM (1980), Physical Geography, Oxford, Basil Blackwell
7. Mcliveen, J.F.R. (1992), Fundamentals of Weather and climate, Prentice Hall New Jersey
8. Miller, E.W. (1985), Physical Geography, Columbus, Charles E. Merrill
9. Miller, G.T. (1996), Living in the Environment, Principles, connections and solutions, Wadsworth
10. Monkhouse, F.J. (1996), Principles of Physical Geography, Hodder & Stoughton, London
11. Rathor, A. Hamid (1996), Tabhi Geographia, Islamabad Muqtadra Qaumi Zaban
12. Scott, R.C. (1996), Introduction to physical geography, West Publishing Co, New York.
13. Small, R. J. (1989), Geomorphology and Hydrology, London, Longman.
14. Strahlar, A.N., Strahlar, A.H. (2004), Physical Environment, John Wiley, New York
15. Stringer, E.T. (2004), Modern Physical Geography, New York: John Wiley.
16. Taylor, J. (1993), Integral Physical Geography, London Longman
17. Thompson, R.D. et. Al. (1986), Process in Physical Geography, London, Longman.
18. Thornbury, W.D. (1969), Principles of Geomorphology, John Willy & Sons. New York.
19. Thurman, H.V. & Mexrill (1996), Essentials of Oceanography, Menson, London
20. Shepherd (latest edition) Oceanography
21. Pakistan Geographical Review (Lahore)
22. Pakistan Journal of Geography (Peshawar)
23. Journal Geography (Karachi)
24. HEC Digital Library

## **Code: PHIL-211A**

**Title: Introduction to Philosophy**

**Credit Hours: 03**

### **Course Objectives:**

The objective of this course is to introduce the students with the fundamentals of the subject of Introduction to Philosophy.

### **Course Contents**

- What is Philosophy
- o The subject matter of Philosophy
- o The divisions of Philosophy

- o Historical Periods of Philosophy
- o The methods of Philosophy
- o The elements of Critical Thinking
- o The Value of Philosophy

- Theory of Knowledge
  - o Rationalism
  - o Empiricism

- Moral Philosophy
  - o Kantian Ethics
  - o Utilitarianism
  - o Relativism

- Political Philosophy
  - o Equality
    - i. Money
    - ii. Employment
    - iii. Democracy
  - o Freedom
    - i. Positive
    - ii. Negative

- Philosophy of Art
  - o Significant from Theory
  - o Idealist Theory
  - o Institutional Theory



#### Recommended Books

1. Thomas F. Wall. Thinking Critically about Philosophical Problems. Wadsworth Publishers.
2. Julian Baggini. Philosophy key themes. Palgrave macmilian publishers.
3. Nigel Warburton, 4th Edition. The basics philosophy. Routledge Publishers.

**Code: URDU-111**

**Title: Functional Urdu**

**Credit Hours: 03**

**Course Objectives:**

The objective of this course is to introduce the students with the fundamentals of the subject of Urdu

## **Course Contents**

### **1. THE BASIC CONCEPT OF URDU LANGUAGE AND ALPHABETS**

- a. Names & voices of Urdu alphabets.
- b. Study of alphabets in Urdu, Persian, Arabic
- c. Origin of Urdu language (short study)

### **2. ART OF TRANSLATION**

- a. Art of translation and its importance.
- b. Short history of translation in Urdu from English
- c. Principles of translation.
- d. Practice of translation (from English into Urdu)

### **3. TECHNICAL TERMS (ISTELAHAT)**

- a. Problems of technical terms in Urdu.
- b. How to form technical terms?
- c. To prepare list of some literary and scientific technical terms with than English terms

### **4. INTERVIEW AND DIALOGUE WRITING**

- a. Art of interview writing.
- b. Through interviews 3 to 5 persons from different
- c. Art of dialogue writing. Examples by drama books.
- d. Dialogue writing on different topics

### **5. GRAMMER**

- Idioms, proverbs, prefixes, suffixes, synonyms, opposite words, syntax kinds of sentences and common errors in writing
- Figures of Speech: Laf-o-Nasar, Husn-e-Talil, Talmeeh, Iham, Tashbeeh, Isteara, Majaze-Mursal, Tajahile-Aarfana, Morratul-Nazeer, Tajnees & Mubalega.

## **Code: BLAW-111**

**Title: Introduction to Business Law**

**Credit Hours: 03**

### **Course Objectives:**

To familiarize the students with the different Mercantile Laws affecting the economic and business environment in Pakistan. To make students understand the important elements and aspects of business and industrial laws. To equip the students with the necessary skills and aptitude to deal tactfully with the legal situations arising out of business routine matters

## **Course Contents**

### **Introduction to Law of Contract 1872**

Definition of Contract, Essentials of a valid contract, Contract and its kinds, Offer and Acceptance, Revocation of offer, Consideration and Object, Exceptions to Considerations, Unlawful considerations, Capacity of Parties, Persons of unsound mind, Disqualified persons, Free consent, Coercion and Undue influence, Fraud and misrepresentation,

Mistake and its types, Void Agreements, Types and conditions , Contingent and Quasi Contracts, Performance of Contracts, Performance of joint promises, Reciprocal promises, Contracts not requiring performance, Discharge of Contract, Breach of Contract and its remedies, Indemnity and Guarantee, Bailment and Pledge, Essential features and parties, Rights and duties of parties, Termination of pledge and Bailment, Contract of agency, Creation of agency, Rights and duties of Principal and agent, Termination of agency

### **Contract of Sale of Goods**

Essentials of a sale, Agreement to sell, Fixation of Price, Conditions and Warranties, Definitions and Difference, Doctrine of Caveat Emptor, Transfer of Property, Sale by non-owner, Performance of Contract of Sale, Rights of Unpaid seller

### **Law of Negotiable Instruments**

Parties of Negotiable Instruments, Presentment of Negotiable Instruments, Negotiation of Negotiable Instruments, Dishonor of Negotiable Instruments, Banker and customer

### **Law of Carriage of Goods**

Carriage of Goods of Land, Carriage of Goods by Sea, Carriage of Goods by Air

### **Law of Industrial Relations**

Unfair Labor practices, Law of Labor in factories, Law of Industrial and Commercial Employment

### **Law of Partnership**

Introduction and scope of Partnership Act 1932, Formation of and relations in Partnership, Rights, Duties and liabilities of Partners, Dissolution of firm and partnership

### **Recommended Books:**

- Cheema, Khalid Mahmood, Business Law. (Latest Editions). Sayed Mobin Mahmud & Co. Lahore.
- Saeed, Khaja Amjad. Mercantile and Industrial Laws in Pakistan. (Latest Editions). Institute of Business Management, Lahore.
- Internet source: [www.Paksearch.com](http://www.Paksearch.com)

**Code: BUS-111**

**Title: Introduction to Business Administration**

**Credit Hours: 03**

**Course Objectives:**

To enhance students to full range of functions of business organization

**Course Contents**

**Business**

Scope, Importance and Functions, Enterpreneminal qualities of Businessman.

Business environment and social responsibility.

Role of business in economic progress of the country

Problems of business.

**Forms of Business Organization**

Sole proprietorships – Nature, Advantages and Disadvantages, Partnership, Classification, Advantages and Disadvantages, Rights, Duties and Liabilities of Partners, dissolution of partnership. Joint Stock Company, Types – Chartered Statuary and Registered Formation – Memorandum of Association, Articles of Association, Prospectus, Capital Management, Meetings and Winding up Features, Merits and Demerits, Management of the Companies

**Cooperative Ownership and Business Combinations**

Types of Cooperatives, Advantages and Disadvantages. Business Combination, Meaning, Types and Forms causes and purposes, Advantages and Disadvantages

**Finance**

Definition, Components, Importance, Kinds and Sources, Long Term and Short Term Financing. Some Important Financial Institutions. Stock Exchange and Commodity Exchange, Nature, Role and Significance. Risk & Insurance, Protection against Risks, Importance and Types of Insurance, Types of Risks

**Marketing**

Process of Marketing, Definition, Functions and Scope Approaches to the Study of Marketing.

Marketing Mix for Physical Goods and Service Products, Importance. Channels of Distribution for Consumer Good and Organizational Goods. Marketing Communication and Promotion, Advertising, Personal Selling, Sales Promotion, Public Relation, Packaging.

**Trade**

Wholesale and Retail, Meaning, Importance, Functions, Advantages and Disadvantages. Foreign Trade, Imports and Exports, Role in Economic Development of the Country.

### **Physical Factors**

Selecting Business Location, Factors Influencing Locations, Small Business, VS Large Business

Purchasing and Inventory Control Coordination of Purchasing with Sales, Organization of Purchasing Department, centralize vs. Decentralize Purchasing, Purchasing Policies and Procedure Inventory Control

### **The Personnel Department**

Employment Division, Recruiting, Placement Promotion. Wages and Salary Section, Analysis & Job Classification, Pay Scales. Training Program, Apprentice, Supervisory etc.

### **Recommended Books:**

- Rober C. Appleby, Modern Business Administration, Shoukat Ali, Introduction to Business Muhammad Irshad, Introduction to Business. Latest Edition. Naveed Publication Lahore.
- Khalid Mehmood Cheema and Syed Mobin Mahmood. Introduction to Business & Co Lahore Latest Edition.
- Theodore J. Sielaff & Belmont. Introduction to Business. Latest Edition. California.
- Askari Zaidi, Fundamentals of Business, Orient Publishers, Karachi 8. Ali M. H., Introduction to Business

### **Code: ACC-111**

**Title: Principles of Accounting**

**Credit Hours: 03**

### **Course Objectives:**

The objective of the course is to provide an introductory knowledge of accounting to first-year students from a wide range of disciplines. While a general overview of accounting principles relating to the preparation of financial and managerial reports will be presented, the primary focus is to illuminate how accounting information is utilized by a variety of stakeholders in planning, controlling and investing decisions.

### **Course Contents**

#### **Introduction to Principles of Accounting**

Definition of terms such as business, accounting, transaction, bookkeeping, capital, assets, liabilities, income, expenses and profit, Types of business activities: trading, manufacturing and provision of services, Types of business organizations: -sole proprietors such as vendors, small to medium enterprises -partnerships -limited companies, Importance of accounting, Users of accounting information such as households, managers, banks and proprietors.

## **Accounting Cycle**

Accounting cycle: transactions, source documents, subsidiary books, ledger, trial balance, financial statements

## **Data Processing Methods**

Manual method, Electronic method

## **Source Documents**

Source documents and their significance: receipts, cheques, invoices, debit notes, credit notes, vouchers, bank statements, statements of account

## **Subsidiary Books**

Subsidiary books: 2 column cash book, sales journal, purchases journal, sales returns journal, purchases return journal, interpretation of entries in the subsidiary books, Use of books of original entry: 3 column cash book, general journal, petty cash book, Interpretation of entries in subsidiary books, Use of the general journal, Interpretation of entries in the general journal

## **Ledger**

Classification of entries, posting and balancing ledger accounts, interpreting ledger accounts, Classification, posting and balancing ledger accounts, Use of folio numbers, Types of ledgers and the ledger accounts, Interpretation of ledger accounts

## **Trial Balance and Errors**

Trial balance: purpose and preparation, Trial balance and errors: purpose, preparation, and errors, Correction of errors, Suspense account, Correction of the trial balance, Effect of errors on profit, Adjustments on gross and net profit, Revised statements of - Profit - financial position (extracts)

## **Accounting Concepts**

Concepts: matching/ accrual, prudence, going concern, consistency, historical cost, double entry

## **End of Year Financial Statements**

Income statement, Statement of financial position, Interpretation of financial statements

## **End of Year Adjustments**

Prepayments and accruals: for income and expenses, Depreciation: causes and methods, reducing balance method, straight line method, revaluation method, Closing entries, Disposal of non-current assets, Provisions for: doubtful debts, discounts allowable

## **Recommended Books:**

- Jerry J. Weygandt, Donald E. Kieso, Paul D. Kimmel. (2007). Accounting Principles, 8th Edition Published by Wiley.
- Leiwy, D. and R. Perks (2013). Accounting: Understanding and Practice (Maidenhead: McGraw-Hill) Fourth Edition.



- **Tyler, J. Godwin, N. and Alderman, C. W. ACCT2 Financial, 2nd Asia-Pacific Edition, Cengage Learning, Australia.**
- Wood, F., Sangster, A. (2008). Business Accounting (11th ed.). Edinburgh: Prentice Hall.
- Wan, M. W. I., Palil, M. R. (2014). Fundamentals of Business Accounting (2nd ed.). Oxford University Press

## **Code: POL-111**

**Title: Introduction to Political Science**

**Credit Hours: 03**

### **Course Objectives:**

The objective of this course is to introduce the students with the fundamentals of the subject of Political Science

### **Course Contents**

- Definition, Nature, Scope and Sub-fields of Political Science
- Relationship of Political Science with other social sciences
- Approaches to the study of Political Science: Traditional and Modern
- Basic concepts of Political Science: Power, Authority, Legitimacy, Nation and Sovereignty
- State: its origin and evolution; Western and Islamic concepts of State; Forms of State: Unitary, Federation, Confederation
- Organs of Government: Legislature, Executive, Judiciary
- Forms of Government: Parliamentary, Presidential, Authoritarian

### **Recommended Books:**

- Choudhry Ahmad Shafi. (1996). Usul-e-Siyasiyat (Urdu), Lahore: Standard Book Depot.
- Shaikh Bashir Ahmad. (1984). Riyasat Jo Ilm (Sindhi meaning Science of State), Jamshoro: Institute of Sindhalogy, University of Sindh.
- Mazhar ul Haq. (1996). Political Science: Theory and Practice, Lahore: Bookland.
- Michael G. Roskin, Robert L. Cord, James A. (2014). Medeiros and Walter S. Jones, Political Science: An Introduction, 12th edition. Pearson Education.

**Code: IR-111****Title: International Relations****Credit Hours: 03****Course Objectives:**

The objective of this course is to introduce the students with the fundamentals of the subject of International Relations.

**Course Contents**

- 1) Introduction, Nature and Scope of International Relations;
- 2) Approaches to the Study of International Relations:
  - a) Realism
  - b) Idealism (Liberalism)
  - c) Behaviouralism.
- 3) 3. Modern State System and Sovereignty;
- 4) 4. Doctrine of Power in International Relations:
  - a) Elements of Power
  - b) Balance of Power.
- 5) National Interests in International Relations
- 6) Concept of Nationalism
- 7) Concept of Diplomacy
  - a) Status of diplomacy during cold war;
  - b) Status of diplomacy in the post cold war era;
  - c) Status of diplomacy after 9/11.
- 8) Ideologies in the 21st century:
  - a) Clash of civilization (Islam vs the West);
  - b) End of History.

**Recommended Books**

1. Bruce Russett, Grasping the Democratic Peace: Principles for post Cold War World, Princeton, Princeton University Press, 1993.
2. Chris Brown, Understanding International Relations, London, Palgrave, 2005.
3. Hans J. Morgenthau, Politics Among Nations, New York, McGraw Hill, 1993.
4. J. Steans and L. Pettiford, International Relations: Perspectives and Themes, Harlow, Pearson Education Press, 2005.
5. James E. Dougherty and Robert L. Pfaltzraff Jr. Contending Theories of International Relations: Comprehensive Survey, Ed (New York), Harper and Row Publishers, 1981
6. John Baylis and Steve Smith, The Globalization of World Politics: An introduction to International Relations, Oxford University Press, London, 2005
7. John T. Rourke, International Politics on the World Stage, Boston, Boston University Press, 2004.
8. Karen A. Mingst, Essentials of International Relations, London, W.W. Norton & Company, 2004.
9. Robert Jervis and Art Robert, International Politics: Enduring Concepts and Contemporary Issues, New York, Addison Wesley, 2003.
10. Robert Jervis, Perception and Misperception in International Politics, Princeton, Princeton University Press, 1976.

## **Code: BSCS-111**

**Title: Mass Communication**

**Credit Hours: 03**

### **Communication**

- ☐ Definitions, types and significance of Communication
- ☐ Process of Communication: source; message; channel, noise; destination; encoding; decoding; and feedback
- ☐ Essentials of effective communication
- ☐ Role of Mass media in agenda setting
- ☐ Nature and functions of mass communication (Information, Education, Opinion Formation, Entertainment and Development.)

### **Media of Mass Communication**

- ☐ Print Media: newspapers, magazines, periodicals, books, pamphlets, etc.
- ☐ Broadcast media: radio, TV, film.
- ☐ Comparative study of mass media, their components, functions and effects.

### **Recommended Readings:**

1. Dominick, J. R. (2006). Dynamics of Mass Communication (8<sup>th</sup> ed.). New York; McGraw-Hill.
2. Straubhaar, LaRose.(2002). Media Now: Communication Media in the Information Age (3<sup>rd</sup> ed.). USA; Wadsworth.
3. Understanding Mass Communication, Defleur Dennis, Houghton Mufflin Co. Bostan. 2000
4. Introduction to Mass Communication, Edward J. Whetmore, Wadsworth Publishing Co. California 1995.
5. Journalism for All, Mehdi Hassan and Abdus Salam Khurshid, Aziz Book Depot, Lahore. 6<sup>th</sup> ed.; 1997.
6. Iblagh Kay Nazriyat, Muqaddara Shamsuddin, M. (Nation Language Authority, Islamabad 1990.
7. Dynamics of Mass Communication Roger Dominic 6<sup>th</sup> ed, McGraw Hill New York, London 1999.
8. Ta'araf-e-iblagh-e-Amma, Matin-ur-Rehman Murtaza, Department of Communication, University of Karachi 2000.
9. Introducing Mass Communication Michael W. Gambol, McGraw Hill, 1996.
10. Media of Mass Communication, John Vivian 5<sup>th</sup> ED. Allyn and Bacon, London, 2000.

**Code: ARB-111****Title: Arabic Language - I****Credit Hours: 03****Course Objectives:**

The aim of this course is to provide students with a basic knowledge of Arabic, which is common to the entire Arabic-speaking part of the world and not specifically associated with any given country or region. The course covers grammar and essential vocabulary and focuses on the four language skills (listening, speaking, reading and writing), aiming to enable students to communicate in everyday situations

**Course Contents**

- Introduction to the Arabic language
- Alphabet and pronunciation
- Introducing oneself: name, nationality, profession
- Forming simple sentences
- Male and female forms
- Talking about one's family
- Feminine words
- Demonstratives (What's this?)
- Talking about one's work
- Plural forms
- Describing things
- Everyday objects (Whose is this?)
- Possessive endings
- Making polite requests
- The indefinite and definite articles
- 'Sun and moon letters'
- Describing places, time, days, months, colors etc.
- Asking questions (Where? How? What? etc.)
- Prepositions
- Introduction to dialect features
- The idafa construction
- Forming more complex sentences
- Describing one's town
- Writing and listening activities
- Overview and progress test

**Recommended Books:**

- Wightwick, J. and Gaafar, M. (2014). Mastering Arabic 1 (3rd ed.). London: Palgrave Macmillan.
- Alif Baa with DVDs. Introduction to Arabic Letters and Sounds, 2nd ed. By Brustad et al.

- Al-Kitaab fii Taallum al-Arabiyya with DVDsc. A Textbook for Beg Arabic: Part 1, 2nd ed. by Brustad et al.
- Understanding Arabs: A Contemporary Guide to Arab Society, 5th edn., by Margaret Nydell; Nicholas Brealey Publishing, 2012

### **Code: FRN-111**

**Title: French Language**

**Credit Hours: 03**

#### **Course Objectives:**

The objective of this course is to introduce the students with the fundamentals of the subject of French

#### **Course Contents**

##### **Written**

##### **Grammar**

- Translation: French into English (Seen)
- Translation: English into French (Unseen)
- Simple Composition

#### **Recommended Books:**

- More Rapid French, Part I by W.F.H. Whitmarsh, Longmans, Green & Co., Ltd., London).
- More Rapid French, Part H (First ten lessons only) by W.F.H. Whitmarsh (Longmans, Green & Co., Ltd., London)

### **Code: SPN-111**

**Title: Introduction to Spanish**

**Credit Hours: 03**

#### **Course Objectives:**

The objective of this course is to introduce the students with the fundamentals of the subject of Introduction to Spanish

#### **Course Contents**

##### **Written**

##### **Grammar**

- Translation from Spanish into English (based on the text)
- Locuelonce, Cuestionarios, Electricios, Including Cosal Utile and Proverbics contained

in the Text

- Composition

**Recommended Books:**

- New Spanish Reader by L.A. Wikins, Published by Harrap and Co., Ltd., London.
- Grammar shall include the regular and irregular verbs contained in “Apendicede

**Code: CHI-111**

**Title: Introduction to Chinese**

**Credit Hours: 03**

**Course Objectives:**

The purpose of this course is to train students to communicate in Chinese in an authentic setting

**Course Contents**

- At the beginner’s level, students will learn writing skills like basic strokes, character components and single-component characters. They will also learn up to 260 Chinese characters, 40 grammatical points and 30 commonly-used sentences.
- Recognize and use simple words and phrases in general social situations (e.g., to greet, introduce, thank, apologize, compliment, express pleasure or regret).
- Grammar, Apply basic Chinese grammar rules and syntax in writing sentences on a given topic and in exercises.
- Vocabulary Talk about daily activities using basic vocabulary, grammar patterns, and standard usage with clear pronunciation and correct tones.
- Dialogues, Demonstrate comprehension of short reading selections (e.g., articles, dialogues, or short stories) by correctly responding to multiple choice or short-answer questions.
- Chinese pronunciation
- Tones
- Courtesy Expressions
- Pinyin
- Chinese writing system
- Numbers
- Radicals
- Family
- Dates and Time
- Hobbies
- Visiting Friends
- Making Appointments

- Idiomatic expressions
- Chinese culture
- Character writing
- Composing short paragraph using Microsoft Word
- Complete both oral and written tests on vocabulary, grammar patterns, and common usage throughout the semester.

#### **Recommended Books:**

- Yao, Taochung et al. Integrated Chinese 中文听说读写. Simplified Character Edition. Level 1, Part 1. 2 nd ed. Boston: Cheng & Tsui Company, 2005.
- Tao-chung Yao and Yuehua Liu. (2005). Integrated Chinese: Simplified Character Edition. 2nd ed. Cheng and Tsui: Boston.
- Tao-chung Yao and Yuehua Liu. (2005). Workbook, Integrated Chinese: Simplified Character Edition. 2nd ed. Cheng and Tsui: Boston.
- Tao-chung Yao and Yuehua Liu. (2005). Character Workbook, Integrated Chinese: Simplified Character Edition. 2nd ed. Cheng and Tsui: Boston.
- Chinese Made Easy. 2nd Edition, simplified characters version. By Yamin Ma and Xinying Li.
- The teacher may also sometimes use other texts. Whenever this happens, photocopies of the pages will be distributed.

#### **Code: GER-111**

**Title: Introduction to German**

**Credit Hours: 03**

#### **Course Objectives:**

The objective of this course is to introduce the students with the fundamentals of the subject of German.

#### **Course Contents**

##### **Written**

##### **Grammar**

- Translation: German into English (Seen)
- Translation: German into English (Unseen)
- Simple Composition

#### **Recommended Books:**

- Schulz Griesbach: Deutsche sprachlehre fuer Auslaendcr, (available from German Department, Punjab University, one Volume edition) upto Chapter 15



**Code: ITL-111****Title: Introduction to Italian****Credit Hours: 03****Course Objectives:**

The objective of this course is to introduce the students with the fundamentals of the subject of Introduction to Italian.

**Course Contents****Written****Grammar**

- Translation from Italian into English (Seen)
- Translation Horn English into Italian (Unseen)
- Simple Composition

**Recommended Books:**

- V. CIOFFARI. Beginning Italian Grammar, Heath Boston 1965 (First 19 Lessons i.e., pp. 1 to 150).

**Code: TUR-111****Title: Introduction to Turkish****Credit Hours: 03****Course Objectives:**

The objective of this course is to introduce the students with the fundamentals of the subject of Introduction to Turkish

**Course Contents****Written****Grammar**

- Translation from Turkish into English (Unseen)
- Translation Urdu into Turkish (unseen)
- Poetry
- Composition

**Recommended Books:**

- A Grammar of Modern Turkish, Part II. by Dr. Abdul Hafiz Akmut

**Prose and Poetry**

- Okuma Kitobi, art II. by Nevide Gokoydin

**Code: PER-111**

**Title: Introduction to Persian**

**Credit Hours: 03**

**Course Objectives:**

The objective of this course is to introduce the students with the fundamentals of the subject of Introduction to Persian.

**Course Contents****Written****Grammar**

- Translation from Persian into English (Unseen)
- Translation Urdu into Persian (unseen)
- Poetry
- Composition

**Recommended Books:**

- Abdul Ghani, (1929). A History of Persian Literature at the Mughal Court, Allahbad.
- Levy, R. (1969). Persian Literature, London.
- Nabi Hadi. (1995). A History of Indo-Persian Literature, New Delhi.
- Syces, P.M. (1915). A History of Persian Literature, London.
- Aryanpour, Minuchehr. (1973). A History of Persian Literature, Tehran.
- Browne, E.G. (1906). A literary History of Persia, New York.
- Yarshater, Ehsan. (1962). Some Characteristics of Persian Poetry and Arts, Studia Islamica, pp 61-72

## Checklist for a New Academic Program

| Parameters                                          |                          |
|-----------------------------------------------------|--------------------------|
|                                                     |                          |
|                                                     |                          |
| 1. Department Mission and Introduction              | <input type="checkbox"/> |
| 2. Program Introduction                             | <input type="checkbox"/> |
| 3. Program Alignment with University Mission        | <input type="checkbox"/> |
| 4. Program Objectives                               | <input type="checkbox"/> |
| 5. Market Need/ Rationale                           | <input type="checkbox"/> |
| 6. Admission Eligibility Criteria                   | <input type="checkbox"/> |
| 7. Duration of the Program                          | <input type="checkbox"/> |
| 8. Assessment Criteria                              | <input type="checkbox"/> |
| 9. Courses Categorization as per HEC Recommendation | <input type="checkbox"/> |
| 10. Curriculum Difference                           | <input type="checkbox"/> |
| 11. Study Scheme / Semester-wise Workload           | <input type="checkbox"/> |
| 12. Award of Degree                                 | <input type="checkbox"/> |
| 13. Faculty Strength                                | <input type="checkbox"/> |
| 14. NOC from Professional Councils (if applicable)  | <input type="checkbox"/> |

\_\_\_\_\_  
Program Coordinator

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Chairperson